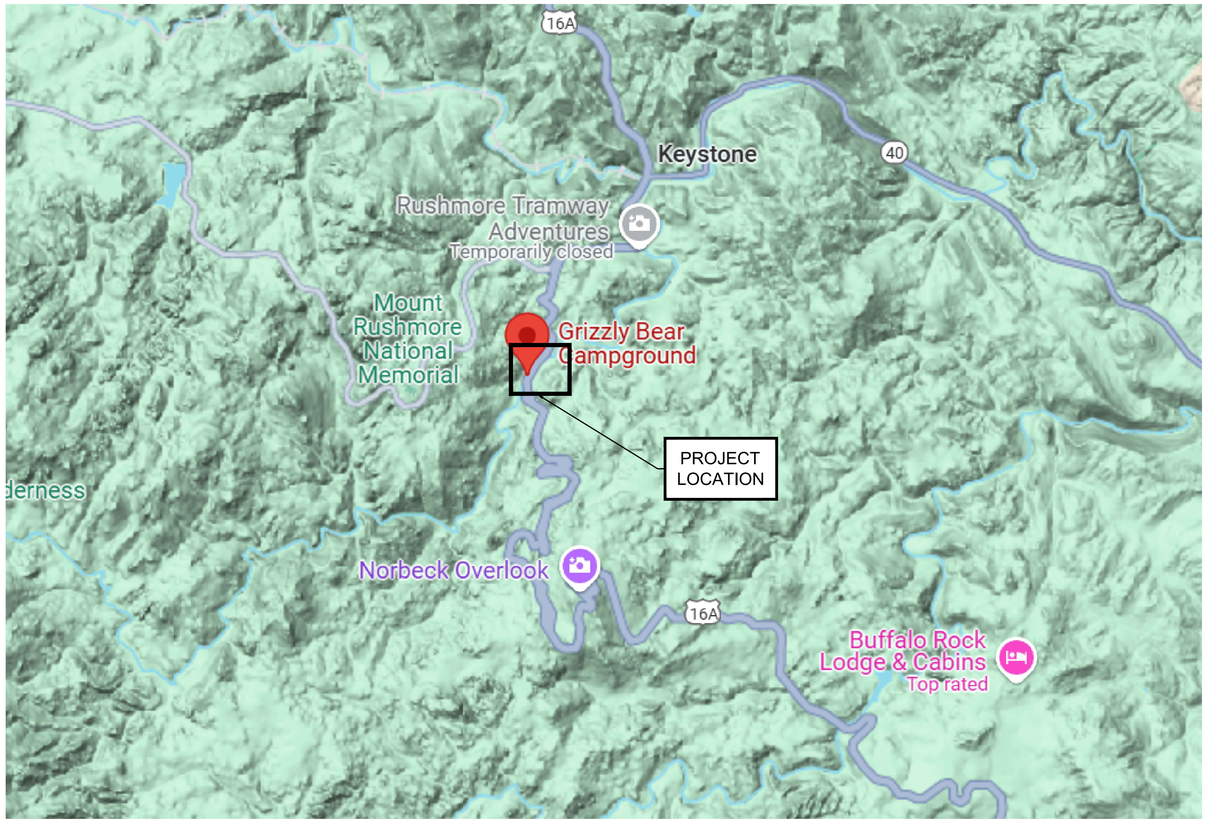


USDA FOREST SERVICE

BLACK HILLS NATIONAL FOREST
GRIZZLY BEAR CAMPGROUND
KEYSTONE, SOUTH DAKOTA

GAOA - GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

SHEET INDEX	
SHEET NUMBER	SHEET NAME
GENERAL	
G1.0	TITLE SHEET
CIVIL	
C1.0	GENERAL LAYOUT
C1.1	DEMOLITION PLAN
C2.0	OVERALL SITE PLAN
C2.1	SOUTH SITE PLAN
C2.2	NORTH SITE PLAN
C3.0	OVERALL GRADING PLAN
C3.1	SOUTH GRADING PLAN
C3.2	NORTH GRADING PLAN
C4.0	TYPICAL SECTIONS & ALIGNMENT DATA
C4.1	GRIZZLY ROADWAY PLAN & PROFILE
C4.2	GRIZZLY ROADWAY PLAN & PROFILE
C5.0	EROSION CONTROL PLAN
C5.1	EROSION CONTROL DETAILS
C6.0	DETAILS
C6.1	DETAILS
C6.2	DETAILS
C6.3	DETAILS
C6.4	DETAILS
C7.0	BID OPTION PLAN
C8.0	SIGNAGE PLAN
C8.1	SIGNAGE DETAILS
ARCHITECTURAL	
A1.1	DOUBLE TOILET FLOOR PLAN, ELEVATIONS, MISC
A1.2	DOUBLE TOILET DETAILS, MISC
A1.3	DOUBLE TOILET DETAILS, MISC
A2.1	PUMP HOUSE FLOOR PLAN, ELEVATIONS, SECTION
STRUCTURAL	
S0.1	GENERAL CONCRETE NOTES
S1.1	STRUCTURAL SITE PLAN
S2.0	TYPICAL CONCRETE DETAILS
MECHANICAL	
MD1.1	MECHANICAL/PLUMBING SITE DEMOLITION PLAN
M1.1	MECHANICAL/PLUMBING SITE PLAN
M1.2	MECHANICAL/PLUMBING ENLARGED SITE PLAN
M1.3	MECHANICAL/PLUMBING ENLARGED SITE PLAN
M5.0	MECHANICAL DETAILS EXISTING SEWAGE VAULT
M5.1	MECHANICAL DETAILS PUMP HOUSE WATER METER AND DISINFECTION PIPING
M5.2	MECHANICAL DETAILS HYDRANT DETAILS
M5.3	MECHANICAL DETAILS STORAGE TANK DETAIL
M5.4	MECHANICAL DETAILS STORAGE TANK INSTALLATION DETAILS
M5.5	MECHANICAL DETAILS MISCELLANEOUS UTILITY DETAILS
M6.0	MECHANICAL/PLUMBING BID OPTION SITE PLAN
M6.1	OPTIONAL MECHANICAL DETAILS NEW SEWAGE VAULT
ELECTRICAL	
E0.1	GENERAL NOTES, ABBREVIATIONS
ED1.1	ELECTRICAL SITE DEMOLITION PLAN
E1.1	ELECTRICAL SITE PLAN
E5.0	ELECTRICAL DETAILS ONE-LINE DIAGRAM
E5.1	ELECTRICAL DETAILS PUMP HOUSE ELECTRICAL LAYOUT
E5.2	ELECTRICAL DETAILS
E6.0	ELECTRICAL SCHEDULES PANEL SCHEDULES
E6.1	ELECTRICAL SCHEDULES SHORT CIRCUIT

I HAVE REVIEWED THE TECHNICAL ASPECTS OF THESE DRAWINGS:

PROJECT MANAGER — DATE

FOREST ENGINEER — DATE

RECOMMENDED BY:

ENGINEERING/REGIONAL SPECIALIST [FSP MANAGER] — DATE

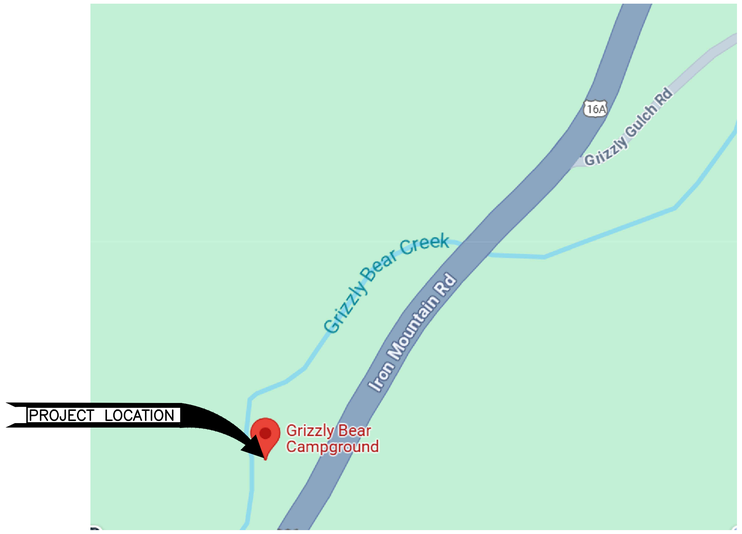
THESE DRAWINGS COMPLY WITH THE FOREST PLAN AND PROJECT SPECIFIC NEPA DOCUMENTATION:

DISTRICT RANGER — DATE

FOREST SUPERVISOR — DATE

APPROVED BY:

REGIONAL ENGINEER — DATE



PROJECT LOCATION

TRAVEL DIRECTIONS

GRIZZLY CAMPGROUND IS IN THE NE1/4 OF SECTION 18, T2S, R6E, BHM, PENNINGTON COUNTY, SOUTH DAKOTA. IT IS LOCATED ON US HIGHWAY 16A, APPROXIMATELY 1.5 MILES SOUTHWEST OF KEYSTONE, SOUTH DAKOTA. LAT: 43°52'36.89"N, LONG: 103°26'22.93"W

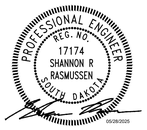
NOTES:

1. SURVEY INFORMATION PROVIDED BY USFS. CONTRACTOR TO COORDINATE WITH USFS FOR SURVEY BENCHMARKS AND CONTROL POINTS.

SHEETS COVERED BY THIS SEAL INCLUDE C1.0, C1.1, C2.0, C2.1, C2.2, C3.0, C3.1, C3.2, C4.0, C4.1, C4.2, C5.0, C5.1, C6.0, C6.1, C6.2, C6.3, C6.4, C7.0, C8.0 AND C8.1 ONLY.

SHEETS COVERED BY THIS SEAL INCLUDE A1.1, A1.2, A1.3, AND A2.1 ONLY.

SHEETS COVERED BY THIS SEAL INCLUDE S0.1, S1.1, AND S2.0 ONLY.



SHEETS COVERED BY THIS SEAL INCLUDE MD1.1, M1.1, M1.2, M1.3, M5.0, M5.1, M5.2, M5.3, M5.4, M5.5, M6.0, AND M6.1 ONLY.

SHEETS COVERED BY THIS SEAL INCLUDE E0.1, ED1.1, E1.1, E5.0, E5.1, E5.2, E6.0, AND E6.1 ONLY.

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ROCKY MOUNTAIN REGION

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NO.	REVISION / ISSUE	DATE
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PROJECT NAME
GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

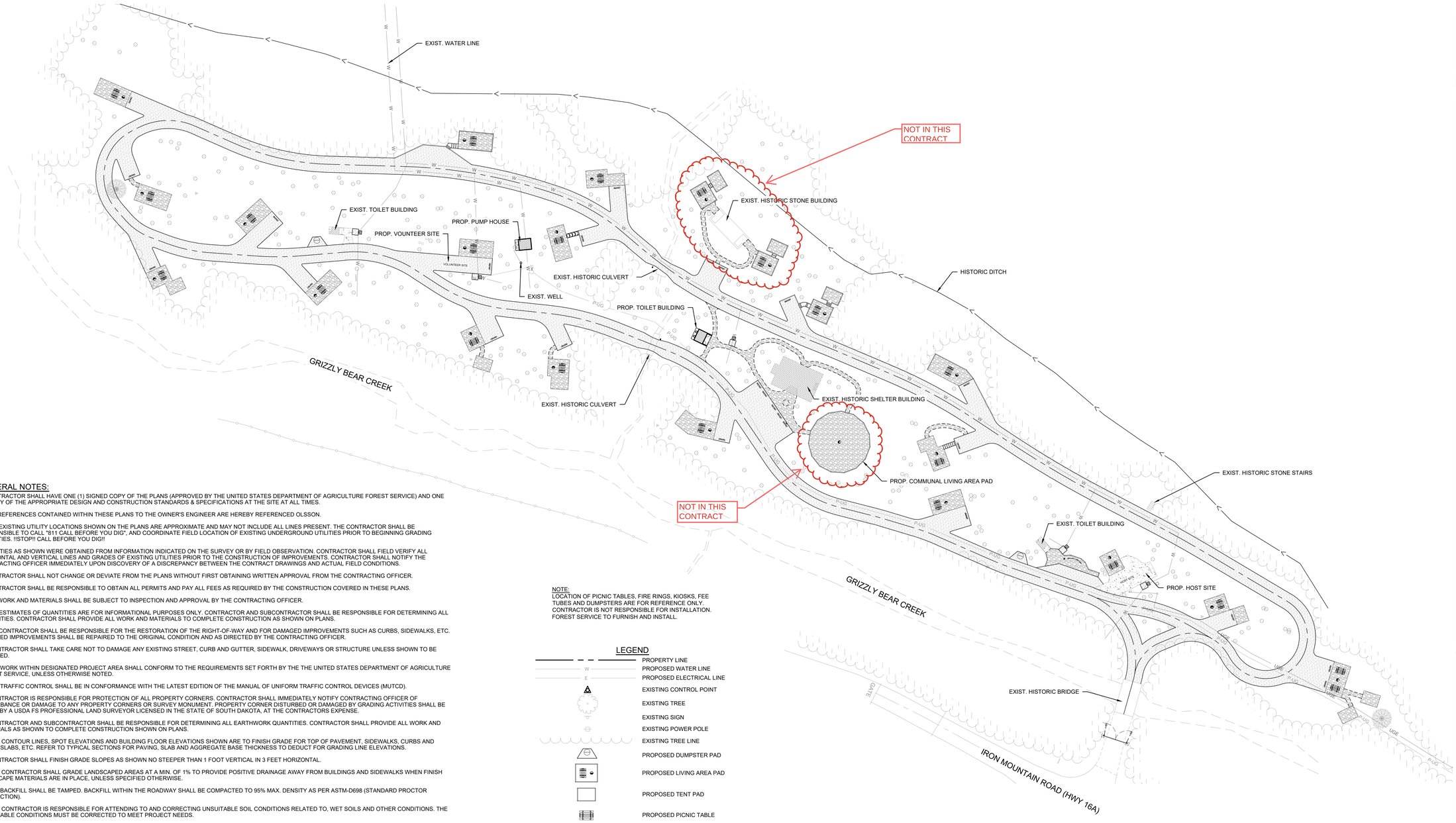
HELL CANYON RANGER DISTRICT

DRAWING TITLE
TITLE SHEET

ARCHIVE NO.

DESIGNER: JR	SUB SHEET NO. G1.0
CHECKED: SR	
DATE: 05.28.2025	SHEET 1 OF 48

5/28/25 18:59 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\STRUCTURAL\STRUCTURAL\GRIZZLY\G1.0.DWG;



NOTE:
LOCATION OF PICNIC TABLES, FIRE RINGS, KIOSKS, FEE TUBES AND DUMPSTERS ARE FOR REFERENCE ONLY. CONTRACTOR IS NOT RESPONSIBLE FOR INSTALLATION. FOREST SERVICE TO FURNISH AND INSTALL.

LEGEND

- PROPERTY LINE
- PROPOSED WATER LINE
- PROPOSED ELECTRICAL LINE
- EXISTING CONTROL POINT
- EXISTING TREE
- EXISTING SIGN
- EXISTING POWER POLE
- EXISTING TREE LINE
- PROPOSED DUMPSTER PAD
- PROPOSED LIVING AREA PAD
- PROPOSED TENT PAD
- PROPOSED PICNIC TABLE
- PROPOSED FIRE RING

GENERAL NOTES:

1. CONTRACTOR SHALL HAVE ONE (1) SIGNED COPY OF THE PLANS (APPROVED BY THE UNITED STATES DEPARTMENT OF AGRICULTURE FOREST SERVICE) AND ONE (1) COPY OF THE APPROPRIATE DESIGN AND CONSTRUCTION STANDARDS & SPECIFICATIONS AT THE SITE AT ALL TIMES.

2. ALL REFERENCES CONTAINED WITHIN THESE PLANS TO THE OWNER'S ENGINEER ARE HEREBY REFERENCED OLSSON.

3. THE EXISTING UTILITY LOCATIONS SHOWN ON THE PLANS ARE APPROXIMATE AND MAY NOT INCLUDE ALL LINES PRESENT. THE CONTRACTOR SHALL BE RESPONSIBLE TO CALL "811 CALL BEFORE YOU DIG", AND COORDINATE FIELD LOCATION OF EXISTING UNDERGROUND UTILITIES PRIOR TO BEGINNING GRADING ACTIVITIES. !!STOP!! CALL BEFORE YOU DIG!!

4. UTILITIES AS SHOWN WERE OBTAINED FROM INFORMATION INDICATED ON THE SURVEY OR BY FIELD OBSERVATION. CONTRACTOR SHALL FIELD VERIFY ALL HORIZONTAL AND VERTICAL LINES AND GRADES OF EXISTING UTILITIES PRIOR TO THE CONSTRUCTION OF IMPROVEMENTS. CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER IMMEDIATELY UPON DISCOVERY OF A DISCREPANCY BETWEEN THE CONTRACT DRAWINGS AND ACTUAL FIELD CONDITIONS.

5. CONTRACTOR SHALL NOT CHANGE OR DEVIATE FROM THE PLANS WITHOUT FIRST OBTAINING WRITTEN APPROVAL FROM THE CONTRACTING OFFICER.

6. CONTRACTOR SHALL BE RESPONSIBLE TO OBTAIN ALL PERMITS AND PAY ALL FEES AS REQUIRED BY THE CONSTRUCTION COVERED IN THESE PLANS.

7. ALL WORK AND MATERIALS SHALL BE SUBJECT TO INSPECTION AND APPROVAL BY THE CONTRACTING OFFICER.

8. ANY ESTIMATES OF QUANTITIES ARE FOR INFORMATIONAL PURPOSES ONLY. CONTRACTOR AND SUBCONTRACTOR SHALL BE RESPONSIBLE FOR DETERMINING ALL QUANTITIES. CONTRACTOR SHALL PROVIDE ALL WORK AND MATERIALS TO COMPLETE CONSTRUCTION AS SHOWN ON PLANS.

9. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE RESTORATION OF THE RIGHT-OF-WAY AND FOR DAMAGED IMPROVEMENTS SUCH AS CURBS, SIDEWALKS, ETC. DAMAGED IMPROVEMENTS SHALL BE REPAIRED TO THE ORIGINAL CONDITION AND AS DIRECTED BY THE CONTRACTING OFFICER.

10. CONTRACTOR SHALL TAKE CARE NOT TO DAMAGE ANY EXISTING STREET, CURB AND GUTTER, SIDEWALK, DRIVEWAYS OR STRUCTURE UNLESS SHOWN TO BE REMOVED.

11. ALL WORK WITHIN DESIGNATED PROJECT AREA SHALL CONFORM TO THE REQUIREMENTS SET FORTH BY THE THE UNITED STATES DEPARTMENT OF AGRICULTURE FOREST SERVICE, UNLESS OTHERWISE NOTED.

12. ALL TRAFFIC CONTROL SHALL BE IN CONFORMANCE WITH THE LATEST EDITION OF THE MANUAL OF UNIFORM TRAFFIC CONTROL DEVICES (MUTCD).

13. CONTRACTOR IS RESPONSIBLE FOR PROTECTION OF ALL PROPERTY CORNERS. CONTRACTOR SHALL IMMEDIATELY NOTIFY CONTRACTING OFFICER OF DISTURBANCE OR DAMAGE TO ANY PROPERTY CORNERS OR SURVEY MONUMENT. PROPERTY CORNER DISTURBED OR DAMAGED BY GRADING ACTIVITIES SHALL BE RESET BY A USDA FS PROFESSIONAL LAND SURVEYOR LICENSED IN THE STATE OF SOUTH DAKOTA, AT THE CONTRACTORS EXPENSE.

14. CONTRACTOR AND SUBCONTRACTOR SHALL BE RESPONSIBLE FOR DETERMINING ALL EARTHWORK QUANTITIES. CONTRACTOR SHALL PROVIDE ALL WORK AND MATERIALS AS SHOWN TO COMPLETE CONSTRUCTION SHOWN ON PLANS.

15. THE CONTOUR LINES, SPOT ELEVATIONS AND BUILDING FLOOR ELEVATIONS SHOWN ARE TO FINISH GRADE FOR TOP OF PAVEMENT, SIDEWALKS, CURBS AND FLOOR SLABS, ETC. REFER TO TYPICAL SECTIONS FOR PAVING, SLAB AND AGGREGATE BASE THICKNESS TO DEDUCT FOR GRADING LINE ELEVATIONS.

16. CONTRACTOR SHALL FINISH GRADE SLOPES AS SHOWN NO STEEPER THAN 1 FOOT VERTICAL IN 3 FEET HORIZONTAL.

17. THE CONTRACTOR SHALL GRADE LANDSCAPED AREAS AT A MIN. OF 1% TO PROVIDE POSITIVE DRAINAGE AWAY FROM BUILDINGS AND SIDEWALKS WHEN FINISH LANDSCAPE MATERIALS ARE IN PLACE, UNLESS SPECIFIED OTHERWISE.

18. ALL BACKFILL SHALL BE TAMPED. BACKFILL WITHIN THE ROADWAY SHALL BE COMPACTED TO 95% MAX. DENSITY AS PER ASTM-D698 (STANDARD PROCTOR COMPACTION).

19. THE CONTRACTOR IS RESPONSIBLE FOR ATTENDING TO AND CORRECTING UNSUITABLE SOIL CONDITIONS RELATED TO, WET SOILS AND OTHER CONDITIONS. THE UNSUITABLE CONDITIONS MUST BE CORRECTED TO MEET PROJECT NEEDS.

20. ALL EXTERIOR CONCRETE SHALL HAVE A MINIMUM 28-DAY COMPRESSIVE STRENGTH OF 4000 PSI AND BE AIR ENTRAINED.

21. CONTRACTOR SHALL BE RESPONSIBLE FOR KEEPING THE PUBLIC STREETS IN THE VICINITY OF THE JOB SITE CLEAN AND FREE OF ROCKS, SOIL/MUD AND DEBRIS DURING ALL PHASES OF CONSTRUCTION.

22. CONTRACTOR SHALL CLEAN OUT ALL EXISTING AND PROPOSED INLETS, PIPES AND MANHOLES OF DEBRIS AND SEDIMENTATION AT COMPLETION OF SITEWORK. THIS WORK SHALL BE DONE TO THE SATISFACTION OF THE UNITED STATES DEPARTMENT OF AGRICULTURE FOREST SERVICE.

23. CONTRACTOR SHALL RETAIN FLOATABLE WIND BLOWN MATERIALS ON SITE BY STORING ALL TRASH AND BUILDING MATERIAL WASTE IN ENCLOSURES UNTIL PROPER DISPOSAL AT AN OFF-SITE FACILITY. CONTRACTOR TO CHECK ADJACENT AREAS DAILY AND PICK UP CONSTRUCTION WASTE MATERIALS AND DEBRIS THAT HAVE BLOWN OR WASHED OFF-SITE.

24. CONTRACTOR IS RESPONSIBLE FOR PROVIDING BERMS, SILT FENCES, OR OTHER MEANS TO PREVENT ERODED MATERIALS FROM REACHING THE PUBLIC RIGHT-OF-WAY AND ADJACENT PROPERTIES. IN THE EVENT THE PREVENTION MEASURES ARE NOT EFFECTIVE, THE CONTRACTOR SHALL REMOVE ANY DEBRIS, SILT, OR MUD AND RESTORE THE RIGHT-OF-WAY TO ORIGINAL OR BETTER CONDITION.

25. CONTRACTOR SHALL BE RESPONSIBLE TO CONTROL DOWNSTREAM EROSION AND SILTATION DURING ALL PHASES OF CONSTRUCTION. EROSION CONTROL PROCEDURES SHALL BE IN PLACE PRIOR TO BEGINNING GRADING ACTIVITIES.

26. CONTRACTOR SHALL BE RESPONSIBLE FOR SECURING ACCESS TO THE PROJECT SITE AT ALL TIMES. GATE SHALL BE LOCKED AT ALL TIMES TO PREVENT PUBLIC ENTRY.

27. CONSTRUCTION ACCESS TO THE SITE SHALL BE LIMITED TO THE APPROVED TEMPORARY CONSTRUCTION ENTRANCE AS SHOWN ON SHEET C5.0.

28. TEMPORARY CONSTRUCTION ENTRANCE SHALL HAVE SHOT ROCK FOR ITS SURFACE.

29. EROSION CONTROL DEVICES SHALL BE MAINTAINED DURING THE WHOLE CONSTRUCTION PERIOD BY THE CONTRACTOR.

30. CONTRACTOR SHALL PROTECT ANY STORM INLETS FROM SEDIMENT THAT TAKE STORM WATER FROM THE AREA OF CONSTRUCTION.

31. ALL DISTURBED AREAS SHALL BE STABILIZED IN ACCORDANCE WITH THE APPROVED EROSION CONTROL PLANS.

NOTE:
OLSSON DID NOT PERFORM ANY BOUNDARY OR TOPOGRAPHIC SURVEY. ALL EXISTING SURVEY INFORMATION WAS PROVIDED BY USFS.

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

GENERAL LAYOUT

ARCHIVE NO.

DESIGNER CRW	SUB SHEET NO. C1.0
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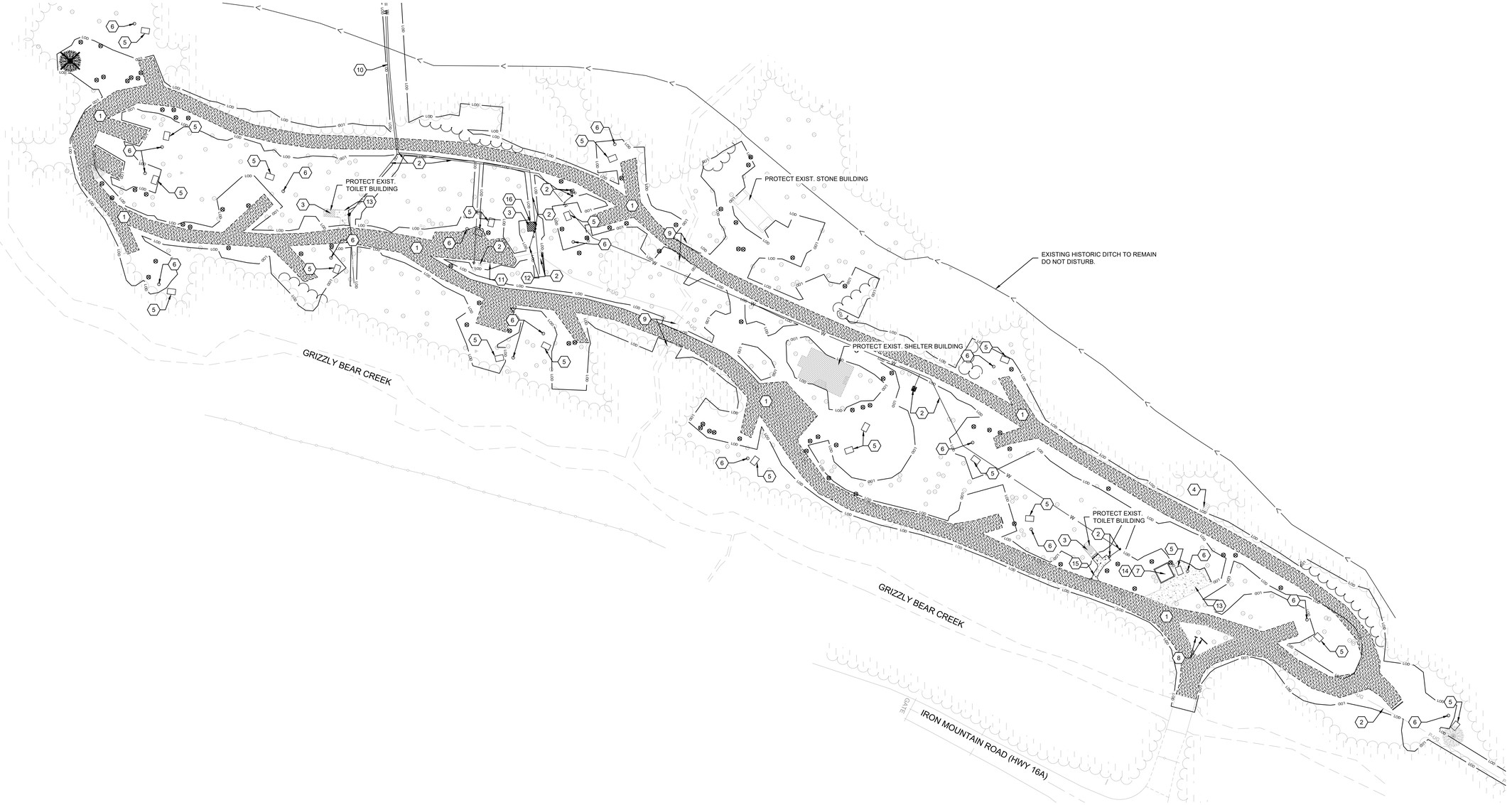
DATE 05/28/2025	SHEET 2 OF 48



DEMOLITION NOTES:

1. ALL DEMOLITION IS TO BE PERFORMED IN ACCORDANCE WITH ALL APPLICABLE CITY, COUNTY, STATE AND/OR GOVERNING BODY'S STANDARDS.

2. THE CONTRACTOR IS RESPONSIBLE FOR OBTAINING ALL PERMITS ASSOCIATED WITH THE DEMOLITION WORK AS DESCRIBED & SHOWN ON THESE PLANS.

3. THE EXISTING UTILITY LOCATIONS SHOWN ON THE PLANS ARE APPROXIMATE AND MAY NOT INCLUDE ALL LINES PRESENT. THE CONTRACTOR SHALL BE RESPONSIBLE TO CALL "811", AND COORDINATE FIELD LOCATION OF EXISTING UNDERGROUND UTILITIES PRIOR TO BEGINNING DEMOLITION ACTIVITIES. !!KNOW WHAT'S BELOW. CALL BEFORE YOU DIG!!

4. CONTRACTOR SHALL COORD. WITH ELECTRIC, TELEPHONE/FIBER, AND CABLE PROVIDERS TO TERMINATE ALL EXISTING SERVICES TO BUILDINGS MARKED FOR DEMOLITION ON THESE PLANS. CONTRACTOR SHALL VERIFY WITH UTILITY PROVIDERS THAT SERVICES ARE DISCONNECTED PRIOR TO ANY REMOVAL OF INTERNAL SERVICE LINES.

5. THE DEMOLITION PLAN AND EXISTING CONDITIONS SHOWN ARE APPROXIMATE. CONTRACTOR SHALL INSPECT AND FIELD VERIFY THE EXISTING SITE CONDITIONS TO BECOME ACQUAINTED WITH ALL EXISTING CONDITIONS WHICH EFFECT ANY AND ALL WORK ASSOCIATED WITH THE DEMOLITION PLANS.

6. ALL EXISTING BUILDING RELATED FEATURES NOT SHOWN ON THIS PLAN SHALL BE COORD. WITH ALL OTHER PLANS ASSOCIATED WITH THIS PROJECT. ALL CONCRETE OR ASPHALT PAVEMENT, CURB & GUTTER, RETAINING WALLS AND MISCELLANEOUS STRUCTURES INCLUDING BUT NOT LIMITED TO WATER HYDRANTS, FIRE RINGS, PICNIC TABLES, SITE NUMBERING POSTS, WHEELSTOPS, KIOSKS, INFO BOARDS, AND MISCELLANEOUS DEBRIS SHALL BE DEMOLISHED AND REMOVED FROM THE SITE AND SHALL BE DISPOSED OF OFF-SITE AT CONTRACTORS EXPENSE TO AN APPROVED LANDFILL, UNLESS OTHERWISE NOTED.

7. IN THE EVENT THE CONTRACTOR ENCOUNTERS ON THE SITE MATERIAL REASONABLY BELIEVED TO BE ASBESTOS, LEAD-BASED PAINT, OR ANY OTHER HAZARDOUS MATERIAL, CONTRACTOR SHALL IMMEDIATLEY NOTIFY THE CONTRACTING OFFICER

8. DEBRIS SHALL NOT BE BURIED ON SITE. ALL UNSUITABLE MATERIAL AND DEBRIS SHALL BE REMOVED FROM THE SITE AND PROPERLY DISPOSED OFF-SITE IN ACCORDANCE WITH ALL CITY, STATE, COUNTY & FEDERAL LAWS AND ORDINANCES. NO ON-SITE BURNING WILL BE PERMITTED.

9. IF EXISTING UN-CAPPED WATER WELLS ARE DISCOVERED ON-SITE, CONTRACTOR SHALL IMMEDIATELY NOTIFY THE CONTRACTING OFFICER.

10. CONTRACTOR SHALL COORDINATE WITH FOREST SERVICE TO PUMP SEPTIC VAULTS OF ALL WASTEWATER AND SLUDGE. CONTRACTOR SHALL REMOVE EXISTING VAULTS AND ANY ASSOCIATED APPURTENANCES IN ACCORDANCE WITH CITY, COUNTY, STATE, AND FEDERAL LAWS AND ORDINANCES.

11. ANY AREA DISTURBED BY DEMOLITION THAT WILL NOT BE DISTURBED IN PROPOSED GRADING ACTIVITIES SHOWN ON THESE PLANS SHALL HAVE TOPSOIL PLACED, SEEDED AND MULCHED ACCORDING TO THE CONTRACT SPECIFICATIONS.

14. OWNER HAS FIRST RIGHT OF REFUSAL OF ALL MATERIALS REMOVED FROM THE SITE.

DEMOLITION KEYNOTES:

1. EXIST. GRAVEL PAVEMENT TO BE RESHAPED AND INCORPORATED INTO THE SUBGRADE FOR PROPOSED LOOP ROAD. SEE SHEETS C2.1-C2.2 & C4.1-C4.2 FOR MORE INFORMATION
2. * CONTRACTOR TO USE CAUTION WHEN WORKING AROUND AND CROSSING EXIST. UTILITIES. REFER TO MEP PLANS FOR UTILITIES TO BE REMOVED AND REPLACED/INSTALLED
3. * CONTRACTOR TO USE CAUTION WHEN WORKING AROUND EXIST. BUILDINGS. REFER TO MEP & ARCH. PLANS FOR EXISTING BUILDINGS TO BE MODIFIED
4. EXIST. HISTORIC STONE STAIRS TO BE PROTECTED IN PLACE.
5. EXIST. PICNIC TABLE TO BE REMOVED AND DISPOSED.
6. EXIST. FIRE RING TO BE REMOVED AND DISPOSED.
7. EXIST. TENT PAD AND TIMBER TO BE REMOVED AND DISPOSED
8. EXIST. SITE BULLETIN BOARD AND FEE TUBE TO BE REMOVED. SEE SHEET C2.2 FOR NEW LOCATION.
9. EXIST. HISTORIC BRIDGE, CULVERTS, & STONE DRAINAGE CROSSINGS TO REMAIN AND BE PROTECTED IN PLACE. CONTRACTOR TO USE EXTREME CAUTION WHEN WORKING AROUND EXIST. CULVERTS.
10. EXIST. WATER LINE TO BE REPLACED. BURIED WATER TANK TO BE REMOVED AND REPLACED. REFER TO MEP PLANS FOR DETAILS
11. EXIST. SEWAGE VAULT TO REMAIN IN PLACE. SEE MEP PLANS FOR MORE INFORMATION.
12. EXIST. WELL. CONTRACTOR TO PROVIDE POSITIVE DRAINAGE AWAY FROM WELL CAP. REFER TO MEP PLANS FOR DETAILS.
13. EXIST. PAVEMENT TO BE PROTECTED IN PLACE.
14. REMOVE EXISTING SEWAGE VAULT, ELECTRIC HOOKUP, AND HYDRANT. COORD. WITH MEP PLANS FOR DETAILS.
15. EXIST. CONCRETE SIDEWALK TO BE REMOVED.
16. EXIST. PUMP HOUSE TO BE REMOVED AND DISPOSED. SEE MEP PLANS FOR MORE INFORMATION.

NOTE:
OLSSON DID NOT PERFORM ANY BOUNDARY OR TOPOGRAPHIC SURVEY. ALL EXISTING SURVEY INFORMATION WAS PROVIDED BY USFS.

ALL BARRIER BOULDERS ENCOUNTERED ON SITE SHALL BE SALVAGED AND RE-USED. CONTRACTOR SHALL COORDINATE WITH FOREST SERVICE.

* CONTRACTOR TO REFER TO MEP SHEETS FOR ALL UTILITIES AND ASSOCIATED UTILITY WORK TO BE REMOVED AND REPLACED

** ALL EXISTING TREES AROUND DISTURBED AREAS THAT ARE NOT MARKED TO BE REMOVED SHALL BE PROTECTED (SEE DETAIL ON SHEET C5.1). CONTRACTOR SHALL FLAG ALL TREES FOR REMOVAL AND FOREST SERVICE SHALL APPROVE PRIOR TO REMOVAL.

DEMOLITION LEGEND

- LOD — LIMITS OF CONSTRUCTION/DEMOLITION
- W — WATER PIPE TO BE REMOVED
- P-UG — UNDERGROUND ELECTRIC TO BE REMOVED
- EXIST. GRAVEL PAVEMENT TO BE REMOVED
- ** EXIST. TREES OR BUSHES TO BE REMOVED
- EXIST. TREES OR BUSHES TO REMAIN

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FOREST SERVICE

R2

ROCKY MOUNTAIN REGION

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

DEMOLITION PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C1.1
CHECKED: CRW	
DATE: 05/28/2025	SHEET 3 OF 48

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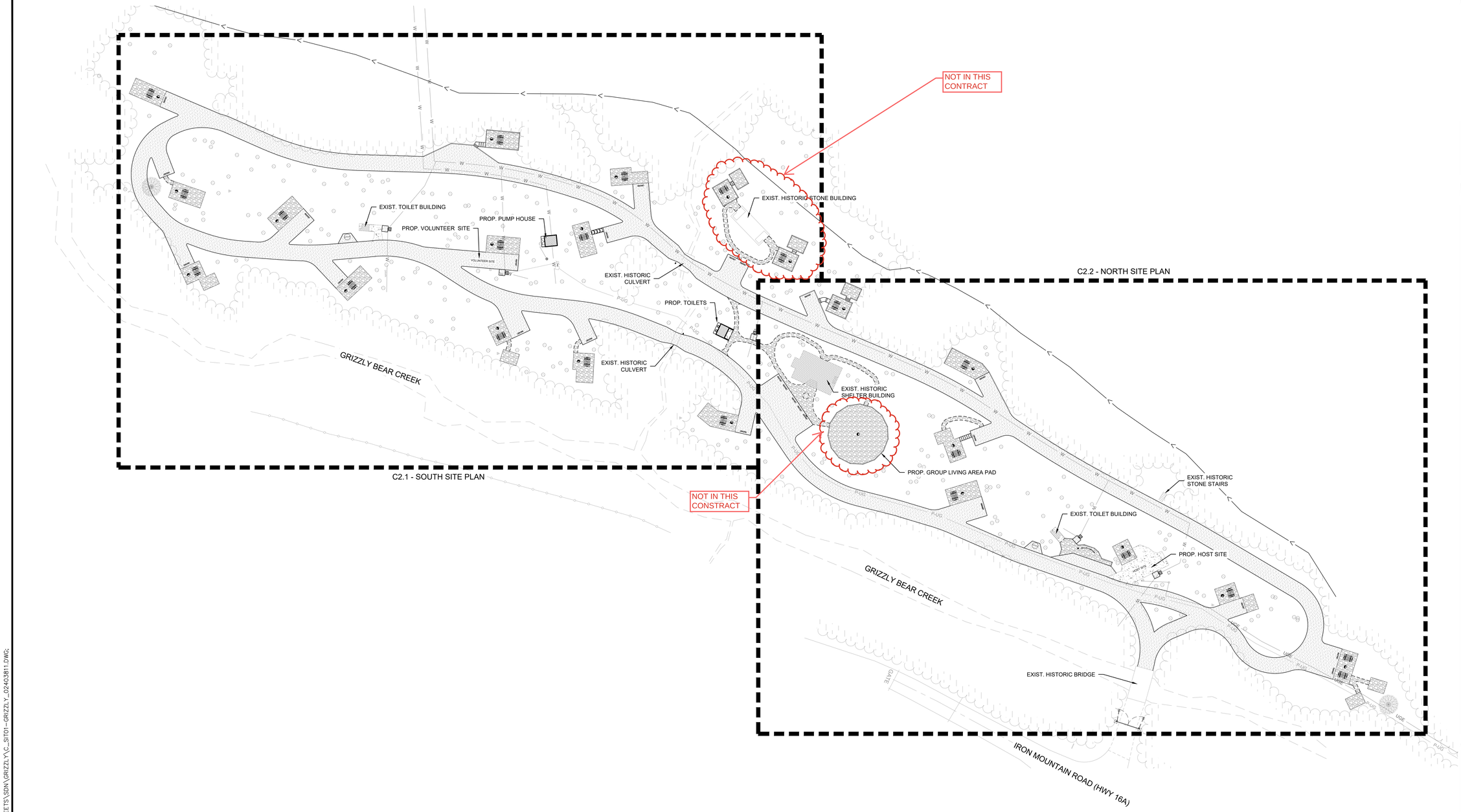

LEGEND

PROPERTY LINE
RIGHT-OF-WAY LINE
EASEMENT LINE

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SCALE IN FEET

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

OVERALL SITE PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C2.0
CHECKED: CRW	
DATE: 05/28/2025	SHEET 4 OF 48

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KEYNOTES:

- 1.1 INSTALL GRAVEL DRIVE SECTION (SEE DETAIL ON SHEET C6.0) (REFER TO SHEETS C4.1-C4.2 FOR ALIGNMENT AND DIMENSIONS)
- 1.2 INSTALL CONCRETE SIDEWALK/PAD (SEE STRUCTURAL PLANS FOR DETAILS)
- 1.3 INSTALL GRAVEL WALKWAY (SEE DETAIL ON SHEET C6.2)
- 2.1 INSTALL BACK-IN SPUR/PULL-OFF (SEE TYPICAL DETAIL ON SHEET C6.0)
- 2.2 INSTALL LIVING AREA AND OR TENT PAD (SEE TYPICAL DETAIL ON SHEET C6.2)
- 2.3 INSTALL GRAVEL DUMPSTER PAD
- 2.4 PROP. TOILET BUILDING. REFER TO ARCH/MEP PLANS FOR DETAILS.
- 2.5 INSTALL TIMBER RETAINING WALL AS NEEDED TO MEET PROP. GRADE (SEE DETAIL ON SHEET C6.1-C6.2) (SEE GRADING ON SHEET C3.0)
- 2.6 INSTALL TIMBER STAIRCASE (SEE DETAIL ON SHEET C6.1)
- 2.7 INSTALL TIMBER STAIRCASE W/ LANDING (SEE DETAIL ON SHEET C6.1)
- 2.8 INSTALL TIMBER FRAMING (SEE DETAIL ON SHEET C6.1-C6.2)
- 2.9 PROP. PUMP HOUSE. SEE ARCH. & MEP PLANS FOR DETAILS.
- 3.1 6' X 9' CUSTOM MADE WOOD PICNIC TABLES. FOREST SERVICE SHALL FURNISH AND INSTALL.
- 3.2 GOVERNMENT FURNISHED FS-30 ACCESSIBLE CAMPFIRE RING. FOREST SERVICE SHALL FURNISH AND INSTALL. (SEE DETAIL ON SHEET C6.1)
- 3.3 INSTALL BARRIER ROCK "WHEELSTOP". CONTRACTOR TO COORD. ROCK ORIENTATION W/ OWNER. (SEE DETAIL SHEET C6.0)
- 3.4 INSTALL ISA SIGNAGE (SEE DETAIL SHEET C6.3)
- 3.5 INSTALL YARD HYDRANT (SEE MEP PLANS FOR DETAIL)
- 3.6 PROP. 3-PANEL KIOSK. FOREST SERVICE SHALL FURNISH AND INSTALL. (SEE DETAIL ON SHEET C6.4)
- 3.7 INSTALL DUAL-ARM SECURITY GATES (SEE DETAIL ON SHEET C6.3)
- 3.8 ROCKART 03-560 FEE COLLECTION STATION. FOREST SERVICE SHALL FURNISH AND INSTALL.

NOTE:
LOCATION OF PICNIC TABLES, FIRE RINGS, KIOSKS, FEE TUBES AND DUMPSTERS ARE FOR REFERENCE ONLY. CONTRACTOR IS NOT RESPONSIBLE FOR INSTALLATION. FOREST SERVICE SHALL FURNISH AND INSTALL.

LEGEND

- PROP. ROCK PARKING BARRIER
- PROP. SITE SIGN
- PROP. FIRE RING
- PROP. PICNIC TABLE
- PROP. YARD HYDRANT
- PROP. DUMPSTER
- PROP. TIMBER STAIRCASE
- INSTALL GRAVEL WALK, LIVING PAD, OR TENT PAD SECTION (SEE DETAIL ON SHEET C6.2)

N

0' 10' 20' 40'
SCALE IN FEET

USDA FOREST SERVICE
UNITED STATES DEPARTMENT OF AGRICULTURE
FOREST SERVICE
R2
ROCKY MOUNTAIN REGION

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

SOUTH SITE PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C2.1
CHECKED: CRW	
DATE: 05/28/2025	SHEET 5 OF 48

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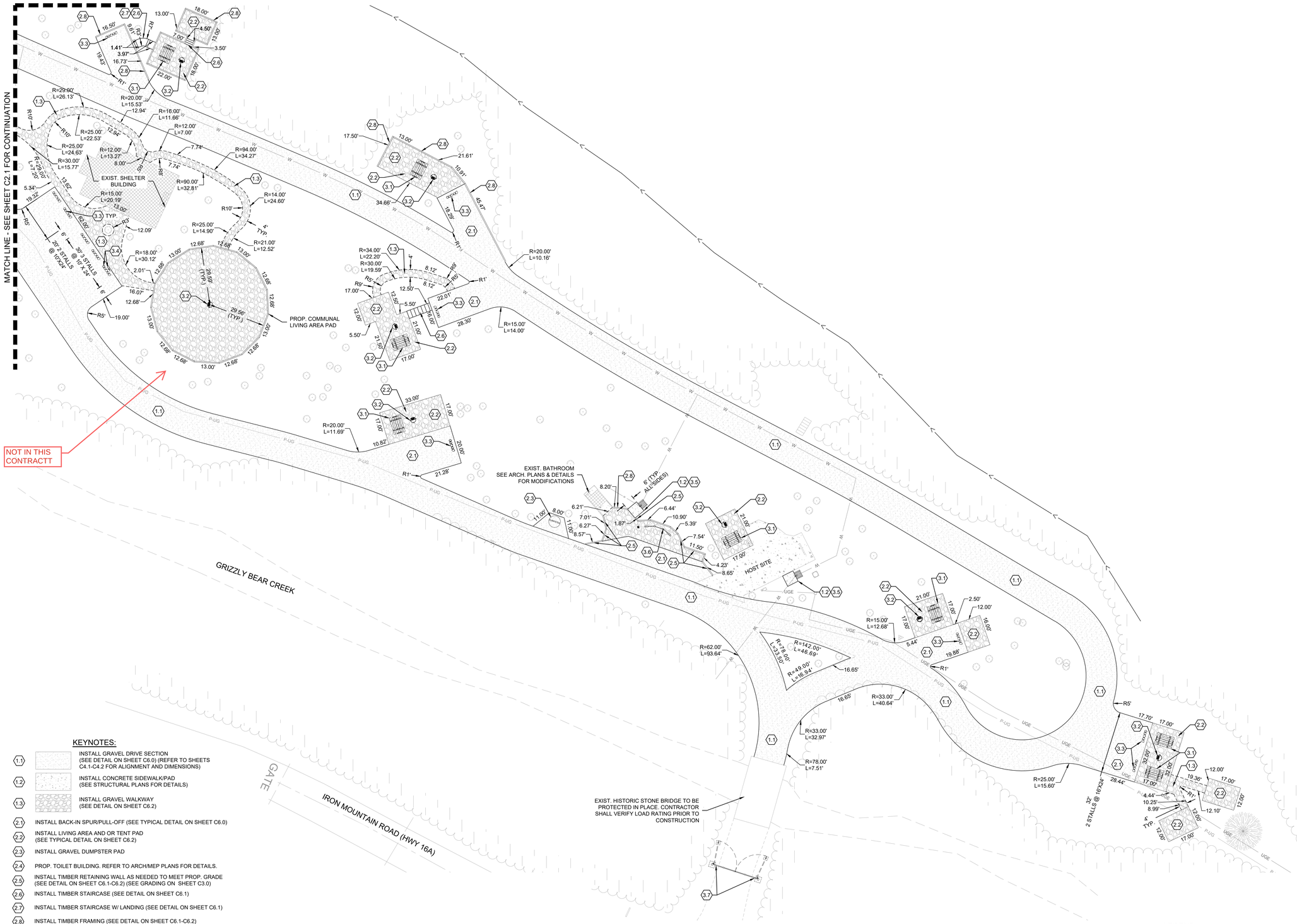

KEYNOTES:

- 1.1 INSTALL GRAVEL DRIVE SECTION (SEE DETAIL ON SHEET C6.0) (REFER TO SHEETS C4.1-C4.2 FOR ALIGNMENT AND DIMENSIONS)
- 1.2 INSTALL CONCRETE SIDEWALK/PAD (SEE STRUCTURAL PLANS FOR DETAILS)
- 1.3 INSTALL GRAVEL WALKWAY (SEE DETAIL ON SHEET C6.2)
- 2.1 INSTALL BACK-IN SPUR/PULL-OFF (SEE TYPICAL DETAIL ON SHEET C6.0)
- 2.2 INSTALL LIVING AREA AND OR TENT PAD (SEE TYPICAL DETAIL ON SHEET C6.2)
- 2.3 INSTALL GRAVEL DUMPSTER PAD
- 2.4 PROP. TOILET BUILDING. REFER TO ARCH/MEP PLANS FOR DETAILS.
- 2.5 INSTALL TIMBER RETAINING WALL AS NEEDED TO MEET PROP. GRADE (SEE DETAIL ON SHEET C6.1-C6.2) (SEE GRADING ON SHEET C3.0)
- 2.6 INSTALL TIMBER STAIRCASE (SEE DETAIL ON SHEET C6.1)
- 2.7 INSTALL TIMBER STAIRCASE W/ LANDING (SEE DETAIL ON SHEET C6.1)
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NOTE:
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LEGEND

- PROP. ROCK PARKING BARRIER
- PROP. SITE SIGN
- PROP. FIRE RING
- PROP. PICNIC TABLE
- PROP. YARD HYDRANT
- PROP. DUMPSTER
- PROP. TIMBER STAIRCASE
- INSTALL GRAVEL WALK, LIVING PAD, OR TENT PAD SECTION (SEE DETAIL ON SHEET C6.2)

N

0' 10' 20' 40'
SCALE IN FEET

USDA FOREST SERVICE
UNITED STATES DEPARTMENT OF AGRICULTURE
FOREST SERVICE
R2
ROCKY MOUNTAIN REGION

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

NORTH SITE PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C2.2
CHECKED: CRW	
DATE: 05/28/2025	SHEET 6 OF 48

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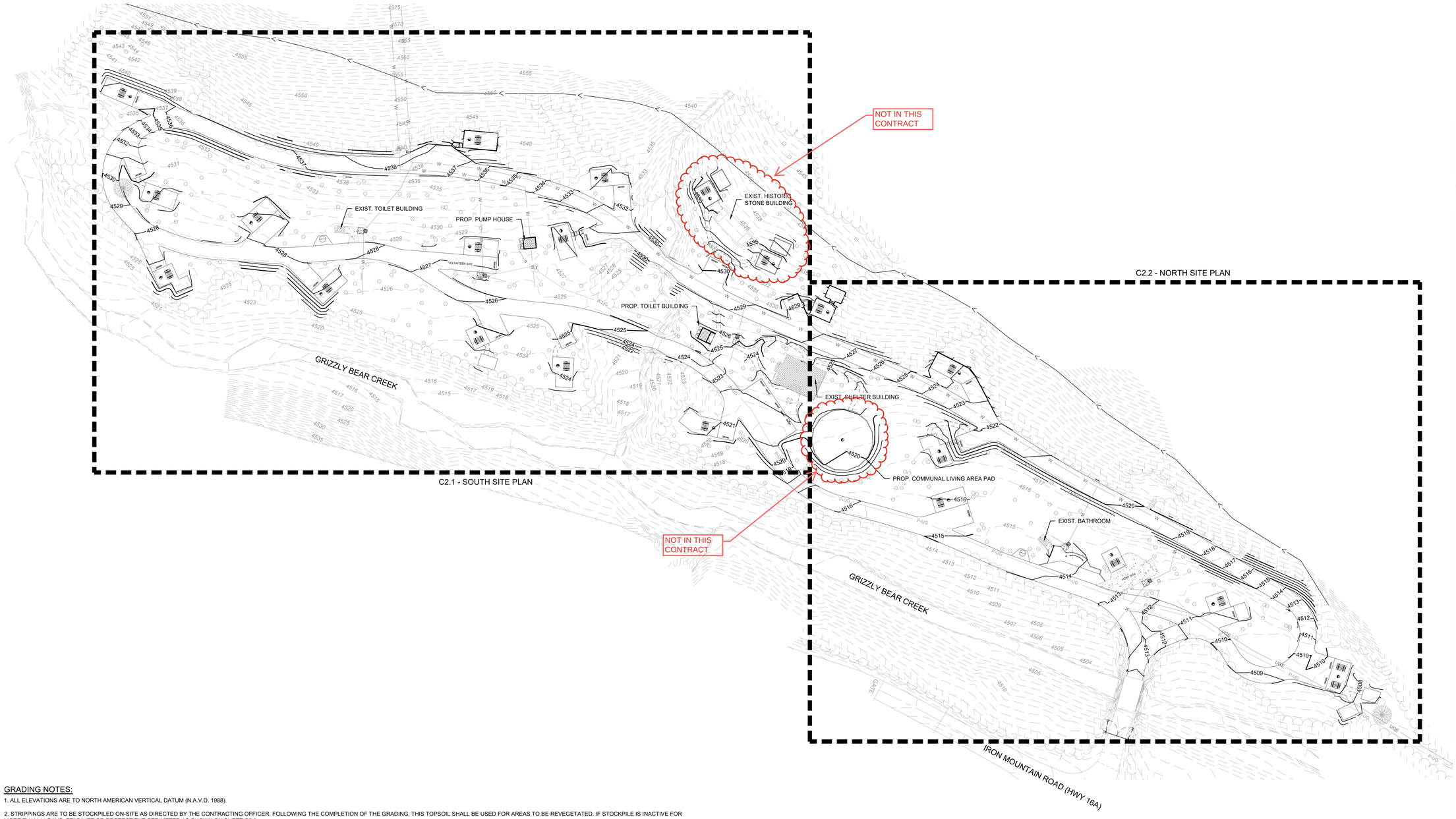

GRADING NOTES:

1. ALL ELEVATIONS ARE TO NORTH AMERICAN VERTICAL DATUM (N.A.V.D. 1988).

2. STRIPPINGS ARE TO BE STOCKPILED ON-SITE AS DIRECTED BY THE CONTRACTING OFFICER. FOLLOWING THE COMPLETION OF THE GRADING, THIS TOPSOIL SHALL BE USED FOR AREAS TO BE REVEGETATED. IF STOCKPILE IS INACTIVE FOR MORE THAN 14 DAYS, STABILIZE OR PROTECT THE PERIMETER AS SHOWN ON SHEET C5.0.

3. ALL HERBACEOUS VEGETATION SHALL BE REMOVED FROM WITHIN THE LIMITS OF THE GRADING AND REDISTRIBUTED WITH THE TOPSOIL AS OUTLINED IN NOTE #2.

4. CONTRACTOR SHALL USE CAUTION AROUND ANY EXISTING UTILITIES LOCATED ON-SITE. THEY SHALL BE RESPONSIBLE FOR THE REPAIRS OF SUCH STRUCTURES WHEN BROKEN OR OTHERWISE DAMAGED BY THE NEW CONSTRUCTION.

5. THE PROPOSED CONTOURS REPRESENT TOP OF SLAB IN PAVEMENT AREAS AND FINISHED GRADE IN ALL OTHER AREAS. QUANTITIES DO NOT TAKE INTO ACCOUNT CORE-OUT OR ANY OVER-EXCAVATION.

6. PRIOR TO MOVING OFF THE JOB THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER TO DO A FINAL WALK-THROUGH OF THE CONSTRUCTION SITE.

7. THE EXISTENCE AND LOCATION OF UNDERGROUND UTILITIES, PIPES AND STRUCTURES SHOWN ON THESE PLANS WERE OBTAINED BY A SEARCH OF AVAILABLE RECORDS AND TO THE BEST OF OUR KNOWLEDGE CONSTITUTES ALL KNOWN FACILITIES. HOWEVER, THE CONTRACTOR IS REQUIRED TO TAKE DUE PRECAUTIONARY MEASURES TO PROTECT ANY EXISTING UTILITIES OR STRUCTURES LOCATED AT THE WORK SITE. IT IS THE CONTRACTOR'S RESPONSIBILITY TO CONTACT AN UNDERGROUND LOCATE SERVICE IN ADVANCE OF ANY EXCAVATION FOR THE MARK-OUT OF THE LOCATION OF UTILITIES AND NOTIFICATION OF COMMENCEMENT OF WORK.

8. BEFORE EXCAVATING FOR THIS CONTRACT, THE CONTRACTOR SHALL FIELD VERIFY LOCATION OF UNDERGROUND UTILITIES. CONTRACTOR SHALL MAKE EXPLORATION EXCAVATIONS AND LOCATE EXISTING UNDERGROUND UTILITIES SUFFICIENTLY AHEAD OF CONSTRUCTION TO PERMIT REVISIONS TO PLANS IF REVISIONS ARE NECESSARY BECAUSE OF ACTUAL LOCATION OF EXISTING FACILITIES.

9. CONTRACTOR IS REQUIRED TO TAKE PRECAUTIONARY MEASURES TO PROTECT THE UTILITY LINES SHOWN AND ANY OTHER EXISTING LINES NOT OF RECORD OR NOT SHOWN ON THESE PLANS.

10. CONTRACTOR SHALL BE RESPONSIBLE FOR THE MAINTENANCE OF THE SEDIMENT CONTROL BARRIERS FOR A PERIOD OF 1 YEAR OR UNTIL THEY ARE RELEASED FROM THIS RESPONSIBILITY BY THE CONTRACTING OFFICER, WHICHEVER PERIOD IS SHORTER.

11. ALL ITEMS REMOVED SHALL BE DISPOSED OFF SITE BY THE CONTRACTOR AS PER THE UNITED STATES DEPARTMENT OF AGRICULTURE FOREST SERVICE REQUIREMENTS.

12. IN ALL LOCATIONS WHERE NEW PAVEMENT ABUTS EXISTING, SAW CUT EXISTING PAVEMENT AS NECESSARY TO PROVIDE A SMOOTH TRANSITION AT THE JOINT. MATCH EXISTING GRADES AT THE JOINT IN ALL LOCATIONS.

13. ALL WALKWAYS SHALL COMPLY WITH FOREST SERVICE ACCESSIBILITY GUIDELINES. COORDINATE WITH FOREST SERVICE FOR AREAS THAT MAY EXCEED GUIDELINES. ALL LANDINGS OUTSIDE OF EXTERIOR ENTRANCES/EXITS SHALL BE NO STEEPER THAN 2.00% FOR THE FIRST 5'-0".

14. CONTRACTOR SHALL FINISH GRADE SLOPES AS SHOWN NO STEEPER THAN 1 FOOT VERTICAL IN 3 FEET HORIZONTAL.

15. ALL FINISH GRADES AROUND BUILDING SHALL BE 6" BELOW FINISH FLOOR UNLESS SPECIFIED OTHERWISE. CONTRACTOR SHALL FINISH GRADE TO PROVIDE POSITIVE DRAINAGE AWAY FROM BUILDING.

LEGEND

- BREAK IN GRADE
- 4500 EXIST. CONTOUR
- 4500 FINISH GRADE CONTOUR

N

0' 20' 40' 80'
SCALE IN FEET

UNITED STATES
DEPARTMENT OF AGRICULTURE

FOREST SERVICE

R2

ROCKY MOUNTAIN REGION

STAMPS, LOGOS, AND SEALS

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

OVERALL GRADING PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C3.0
CHECKED: CRW	
DATE: 05/28/2025	SHEET 7 OF 48

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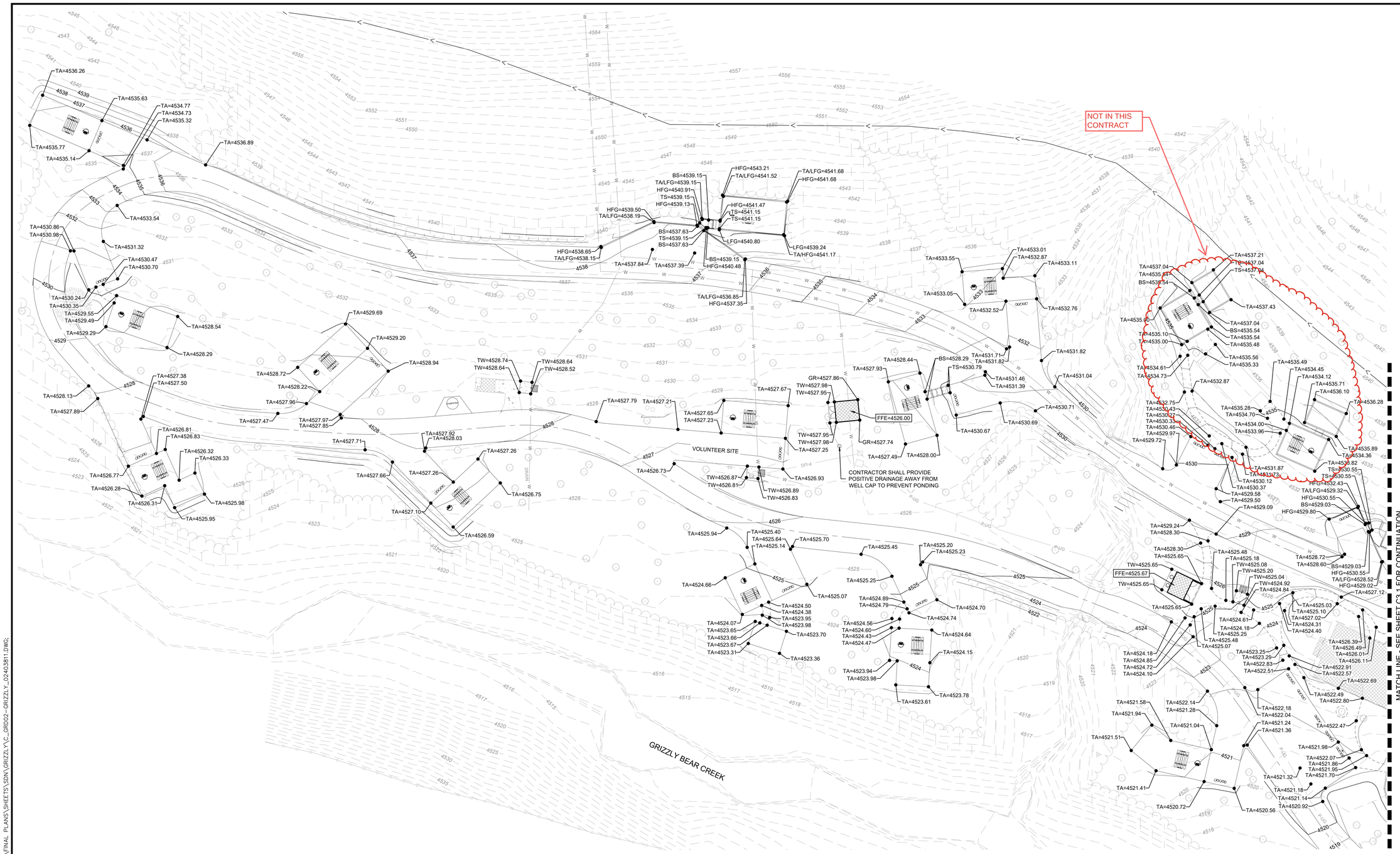

LEGEND

— · — · — · —	BREAK IN GRADE
4500	EXIST. CONTOUR
4500	FINISH GRADE CONTOUR
TA=4500.00	TOP OF AGGREGATE
TW=4500.00	TOP OF SIDEWALK
BS=4500.00	BOTTOM OF STAIR
TS=4500.00	TOP OF STAIR
LFG	WALL LOW FINISH GRADE
HFG	WALL HIGH FINISH GRADE
ME	MATCH EXISTING

N

0' 10' 20' 40'
SCALE IN FEET

USDA FOREST SERVICE
UNITED STATES DEPARTMENT OF AGRICULTURE
FOREST SERVICE
R2
ROCKY MOUNTAIN REGION

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

SOUTH GRADING PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C3.1
CHECKED: CRW	
DATE: 05/28/2025	SHEET 8 OF 48

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LEGEND

— · — · — · — · —	BREAK IN GRADE
4500	EXIST. CONTOUR
4500	FINISH GRADE CONTOUR
TA=4500.00	TOP OF AGGREGATE
TW=4500.00	TOP OF SIDEWALK
BS=4500.00	BOTTOM OF STAIR
TS=4500.00	TOP OF STAIR
LFG	WALL LOW FINISH GRADE
HFG	WALL HIGH FINISH GRADE
ME	MATCH EXISTING

N

0' 10' 20' 40'
SCALE IN FEET

USDA FOREST SERVICE

UNITED STATES
DEPARTMENT OF AGRICULTURE

FOREST SERVICE

R2

ROCKY MOUNTAIN REGION

STAMPS, LOGOS, AND SEALS

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

NORTH GRADING PLAN

ARCHIVE NO.

DESIGNER CRW	SUB SHEET NO. C3.2
CHECKED CRW	
DATE 05/28/2025	SHEET 9 OF 48

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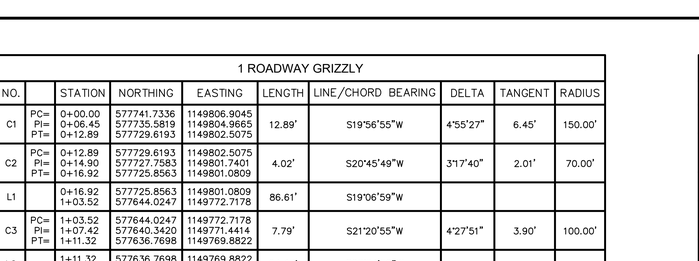
1 ROADWAY GRIZZLY									
NO.		STATION	NORTHING	EASTING	LENGTH	LINE/CHORD BEARING	DELTA	TANGENT	RADIUS
C1	PC= PI= PT=	0+00.00 0+06.45 0+12.89	577741.7336 577735.5819 577729.6193	1149806.9045 1149804.9665 1149802.5075	12.89'	S19°56'55"W	4°55'27"	6.45'	150.00'
C2	PC= PI= PT=	0+12.89 0+14.90 0+16.92	577729.6193 577727.7583 577725.8563	1149802.5075 1149801.7401 1149801.0809	4.02'	S20°45'49"W	3°17'40"	2.01'	70.00'
L1		0+16.92 1+03.52	577725.8563 577644.0247	1149801.0809 1149772.7178	86.61'	S19°06'59"W			
C3	PC= PI= PT=	1+03.52 1+07.42 1+11.32	577644.0247 577640.3420 577636.7698	1149772.7178 1149771.4414 1149769.8822	7.79'	S21°20'55"W	4°27'51"	3.90'	100.00'
L2		1+11.32 1+75.75	577636.7698 577577.7183	1149769.8822 1149744.1071	64.43'	S23°34'50"W			
C4	PC= PI= PT=	1+75.75 1+87.72 1+99.66	577577.7183 577566.7485 577555.2860	1149744.1071 1149739.3190 1149735.8733	23.91'	S20°09'21"W	6°50'59"	11.97'	200.00'
L3		1+99.66 2+33.00	577555.2860 577523.3543	1149735.8733 1149726.2746	33.34'	S16°43'51"W			
C5	PC= PI= PT=	2+33.00 2+37.39 2+41.78	577523.3543 577519.1506 577514.8954	1149726.2746 1149725.0110 1149723.9330	8.78'	S15°28'25"W	2°30'53"	4.39'	200.00'
L4		2+41.78 2+77.17	577514.8954 577480.5908	1149723.9330 1149715.2422	35.39'	S14°12'58"W			
C6	PC= PI= PT=	2+77.17 3+19.01 3+57.13	577480.5908 577440.0327 577416.5551	1149715.2422 1149704.9672 1149670.3357	79.96'	S35°02'27"W	41°38'58"	41.84'	110.00'
L5		3+57.13 4+27.42	577416.5551 577377.1088	1149670.3357 1149612.1489	70.30'	S55°51'56"W			
C7	PC= PI= PT=	4+27.42 4+68.96 5+08.01	577377.1088 577353.8037 577315.2055	1149612.1489 1149577.7718 1149562.4389	80.58'	S38°45'55"W	34°12'02"	41.53'	135.00'
L6		5+08.01 5+66.97	577315.2055 577260.4044	1149562.4389 1149540.6696	58.97'	S21°39'54"W			
C8	PC= PI= PT=	5+66.97 6+01.17 6+34.95	577260.4044 577228.6232 577194.6190	1149540.6696 1149528.0447 1149524.4185	67.97'	S13°52'34"W	15°34'41"	34.20'	250.00'
L7		6+34.95 6+77.11	577194.6190 577152.6895	1149524.4185 1149519.9471	42.17'	S6°05'13"W			
C9	PC= PI= PT=	6+77.11 7+12.92 7+47.42	577152.6895 577117.0822 577087.0299	1149519.9471 1149516.1499 1149496.6779	70.30'	S19°30'50"W	26°51'13"	35.81'	150.00'
L8		7+47.42 7+75.53	577087.0299 577063.4386	1149496.6779 1149481.3923	28.11'	S32°56'26"W			
C10	PC= PI= PT=	7+75.53 8+13.09 8+47.91	577063.4386 577031.9183 576994.4897	1149481.3923 1149460.9692 1149464.0880	72.39'	S14°05'19"W	37°42'14"	37.56'	110.00'
L9		8+47.91 8+90.06	576994.4897 576952.4882	1149464.0880 1149467.5878	42.15'	S4°45'48"E			
C11	PC= PI= PT=	8+90.06 9+13.39 9+36.64	576952.4882 576929.2344 576905.9607	1149467.5878 1149469.5255 1149467.8442	46.58'	S0°18'56"E	8°53'43"	23.33'	300.00'

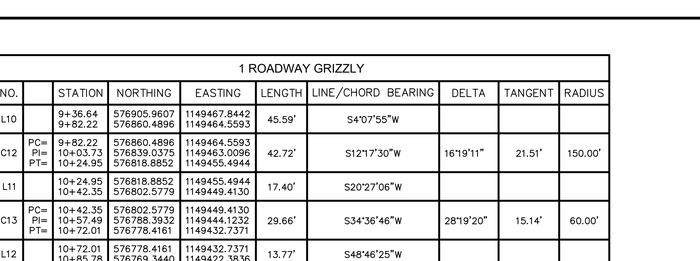
1 ROADWAY GRIZZLY									
NO.		STATION	NORTHING	EASTING	LENGTH	LINE/CHORD BEARING	DELTA	TANGENT	RADIUS
L10		9+36.64 9+82.22	576905.9607 576860.4896	1149467.8442 1149464.5593	45.59'	S4°07'55"W			
C12	PC= PI= PT=	9+82.22 10+03.73 10+24.95	576860.4896 576839.0375 576818.8852	1149464.5593 1149463.0096 1149455.4944	42.72'	S12°17'30"W	16°19'11"	21.51'	150.00'
L11		10+24.95 10+42.35	576818.8852 576802.5779	1149455.4944 1149449.4130	17.40'	S20°27'06"W			
C13	PC= PI= PT=	10+42.35 10+57.49 10+72.01	576802.5779 576788.3932 576778.4161	1149449.4130 1149444.1232 1149432.7371	29.66'	S34°36'46"W	28°19'20"	15.14'	60.00'
L12		10+72.01 10+85.78	576778.4161 576769.3440	1149432.7371 1149422.3836	13.77'	S48°46'25"W			
C14	PC= PI= PT=	10+85.78 11+49.90 11+76.63	576769.3440 576727.0857 576784.1582	1149422.3836 1149374.1570 1149344.9285	90.85'	N79°10'20"W	104°06'29"	64.12'	50.00'
L13		11+76.63 11+84.12	576784.1582 576790.8270	1149344.9285 1149341.5132	7.49'	N27°07'06"W			
C15	PC= PI= PT=	11+84.12 12+15.03 12+41.83	576790.8270 576818.3398 576846.6333	1149341.5132 1149327.4231 1149339.8718	57.71'	N1°41'05"W	50°52'02"	30.91'	65.00'
L14		12+41.83 12+82.96	576846.6333 576884.2834	1149339.8718 1149356.4374	41.13'	N23°44'56"E			
C16	PC= PI= PT=	12+82.96 13+31.70 13+79.23	576884.2834 576928.8926 576977.6078	1149356.4374 1149376.0649 1149377.4992	96.27'	N12°43'04"E	22°03'45"	48.74'	250.00'
L15		13+79.23 14+54.93	576977.6078 577053.2776	1149377.4992 1149379.7272	75.70'	N1°41'11"E			
C17	PC= PI= PT=	14+54.93 15+26.27 15+96.65	577053.2776 577124.5860 577192.4619	1149379.7272 1149381.8267 1149403.7849	141.72'	N9°48'24"E	16°14'25"	71.34'	500.00'
L16		15+96.65 16+03.76	577192.4619 577199.2290	1149403.7849 1149405.9741	7.11'	N17°55'36"E			
C18	PC= PI= PT=	16+03.76 16+43.80 16+82.78	577199.2290 577237.3159 577267.7264	1149405.9741 1149418.2954 1149444.3266	79.02'	N29°14'42"E	22°38'12"	40.03'	200.00'
L17		16+82.78 17+04.81	577267.7264 577284.4588	1149444.3266 1149458.6493	22.03'	N40°33'48"E			
C19	PC= PI= PT=	17+04.81 17+63.70 18+21.76	577284.4588 577329.2021 577383.0859	1149458.6493 1149496.9492 1149520.7269	116.95'	N32°11'13"E	16°45'08"	58.90'	400.00'
L18		18+21.76 18+77.24	577383.0859 577433.8462	1149520.7269 1149543.1264	55.48'	N23°48'39"E			
C20	PC= PI= PT=	18+77.24 18+85.77 18+94.30	577433.8462 577441.6486 577449.5638	1149543.1264 1149546.5694 1149549.7443	17.05'	N22°50'01"E	1°57'16"	8.53'	500.00'
L19		18+94.30 19+15.62	577449.5638 577469.3490	1149549.7443 1149557.6805	21.32'	N21°51'23"E			
C21	PC= PI= PT=	19+15.62 19+20.48 19+25.34	577469.3490 577473.8639 577478.3427	1149557.6805 1149559.4915 1149561.3900	9.73'	N22°24'50"E	1°06'53"	4.86'	500.00'

1 ROADWAY GRIZZLY									
NO.		STATION	NORTHING	EASTING	LENGTH	LINE/CHORD BEARING	DELTA	TANGENT	RADIUS
L20		19+25.34 19+82.34	577478.3427 577530.8147	1149561.3900 1149583.6321	56.99'	N22°58'17"E			
C22	PC= PI= PT=	19+82.34 20+27.88 20+73.17	577530.8147 577572.7472 577610.7782	1149583.6321 1149601.4066 1149626.4649	90.84'	N28°10'33"E	10°24'33"	45.54'	500.00'
L21		20+73.17 21+21.69	577610.7782 577651.2920	1149626.4649 1149653.1592	48.52'	N33°22'50"E			
C23	PC= PI= PT=	21+21.69 21+71.56 22+21.35	577651.2920 577692.9364 577737.1043	1149653.1592 1149680.5984 1149703.7580	99.66'	N30°31'32"E	5°42'36"	49.87'	1000.00'
L22		22+21.35 22+86.76	577737.1043 577795.0284	1149703.7580 1149734.1307	65.40'	N27°40'14"E			
C24	PC= PI= PT=	22+86.76 23+13.42 23+40.06	577795.0284 577818.6393 577841.5571	1149734.1307 1149746.5112 1149760.1321	53.31'	N29°11'51"E	3°03'15"	26.66'	1000.00'
L23		23+40.06 24+33.54	577841.5571 577921.9118	1149760.1321 1149807.8902	93.48'	N30°43'29"E			
C25	PC= PI= PT=	24+33.54 22+85.25 25+84.46	577921.9118 577794.4356 577864.8868	1149807.8902 1149732.1259 1149862.6136	150.92'	S43°49'12"E	210°54'37"	148.29'	41.00'
C26	PC= PI= PT=	25+84.46 25+90.47 25+96.42	577864.8868 577862.0320 577858.0056	1149862.6136 1149857.3260 1149852.8653	11.96'	S54°46'55"W	13°42'22"	6.01'	50.00'
C27	PC= PI= PT=	25+96.42 26+02.82 26+09.11	577858.0056 577853.7219 577848.1968	1149852.8653 1149848.1199 1149844.9038	12.68'	S39°03'56"W	17°43'28"	6.39'	41.00'
L24		26+09.11 26+34.75	577848.1968 577826.0354	1149844.9038 1149832.0038	25.64'	S30°12'12"W			
C28	PC= PI= PT=	26+34.75 26+65.63 26+95.66	577826.0354 577799.3461 577768.6905	1149832.0038 1149816.4681 1149812.7385	60.91'	S18°34'12"W	23°16'00"	30.88'	150.00'
C29	PC= PI= PT=	26+95.66 27+07.86 27+20.00	577768.6905 577756.5830 577744.8725	1149812.7385 1149811.2655 1149807.8558	24.34'	S11°35'07"W	9°17'50"	12.20'	150.00'



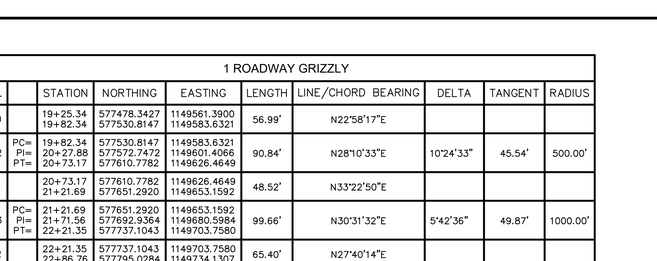
TYPICAL SECTION #1 - SINGLE LANE ROAD SECTION (LEFT)

NOT TO SCALE



TYPICAL SECTION #2 - SINGLE LANE ROAD SECTION (RIGHT)

NOT TO SCALE



TYPICAL SECTION #3 - SINGLE LANE ROAD SECTION

NOT TO SCALE

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

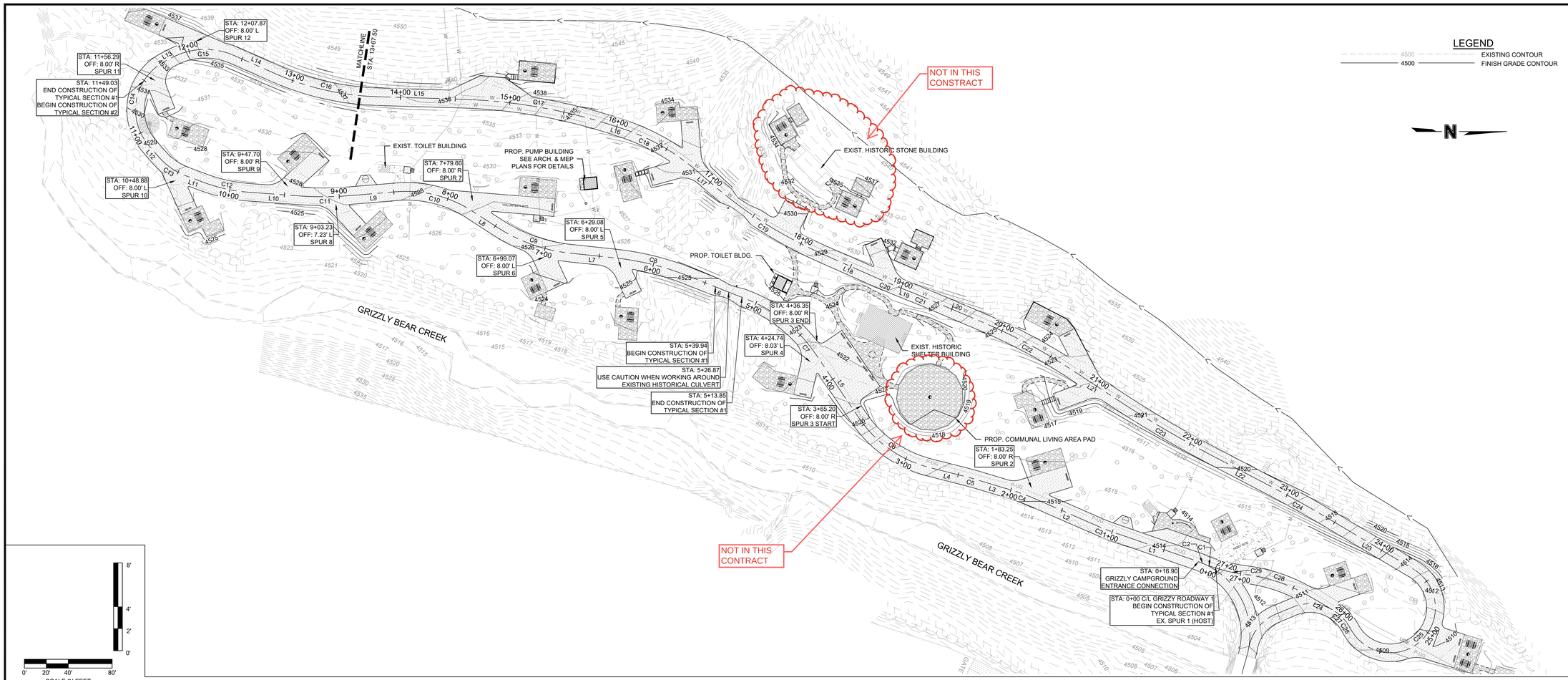
DRAWING TITLE

TYPICAL SECTIONS & ALIGNMENT DATA

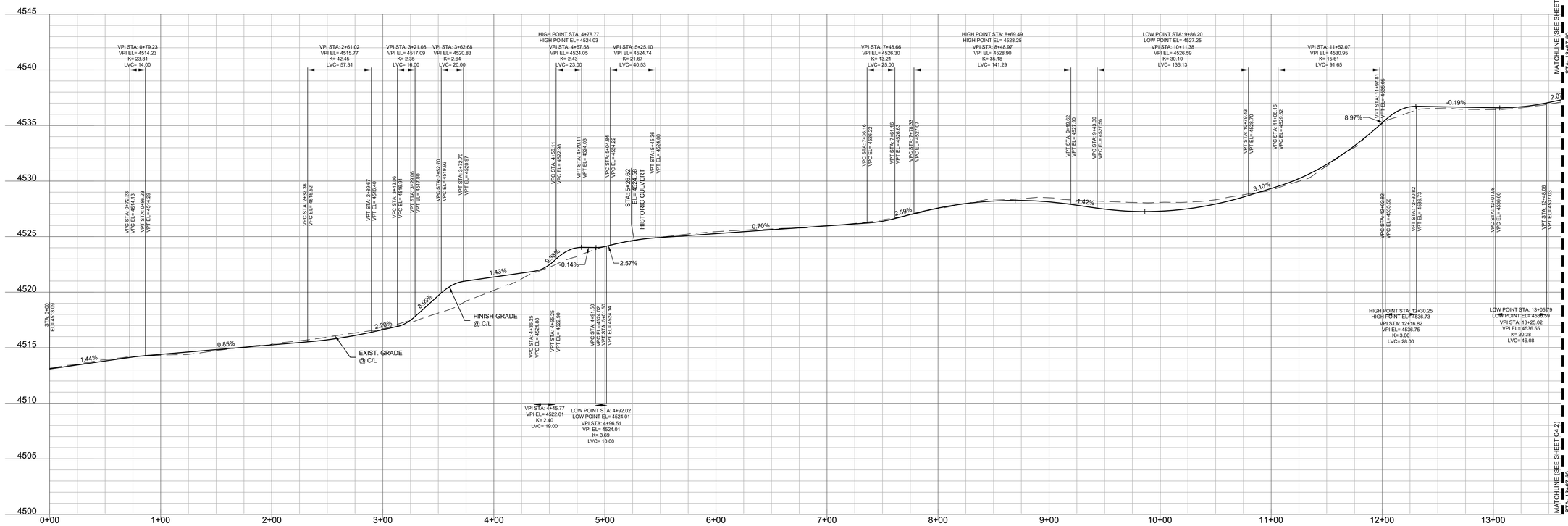
ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C4.0
CHECKED: CRW	
DATE: 05/28/2025	SHEET 10 OF 48

5/28/25 14:47 FBOYD F:\2024\03501-04000\024-03811\40-DESIGN\AUTOCAD\FINAL PLANS\SHEETS\SDN\GRIZZLY\C_ALN01-GRIZZLY_024038111.DWG;



GRIZZLY ROADWAY 1
STA: 0+00.00 - 13+67.50



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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

GRIZZLY ROADWAY PLAN AND PROFILE

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C4.1
CHECKED: CRW	
DATE: 05/28/2025	SHEET 11 OF 48

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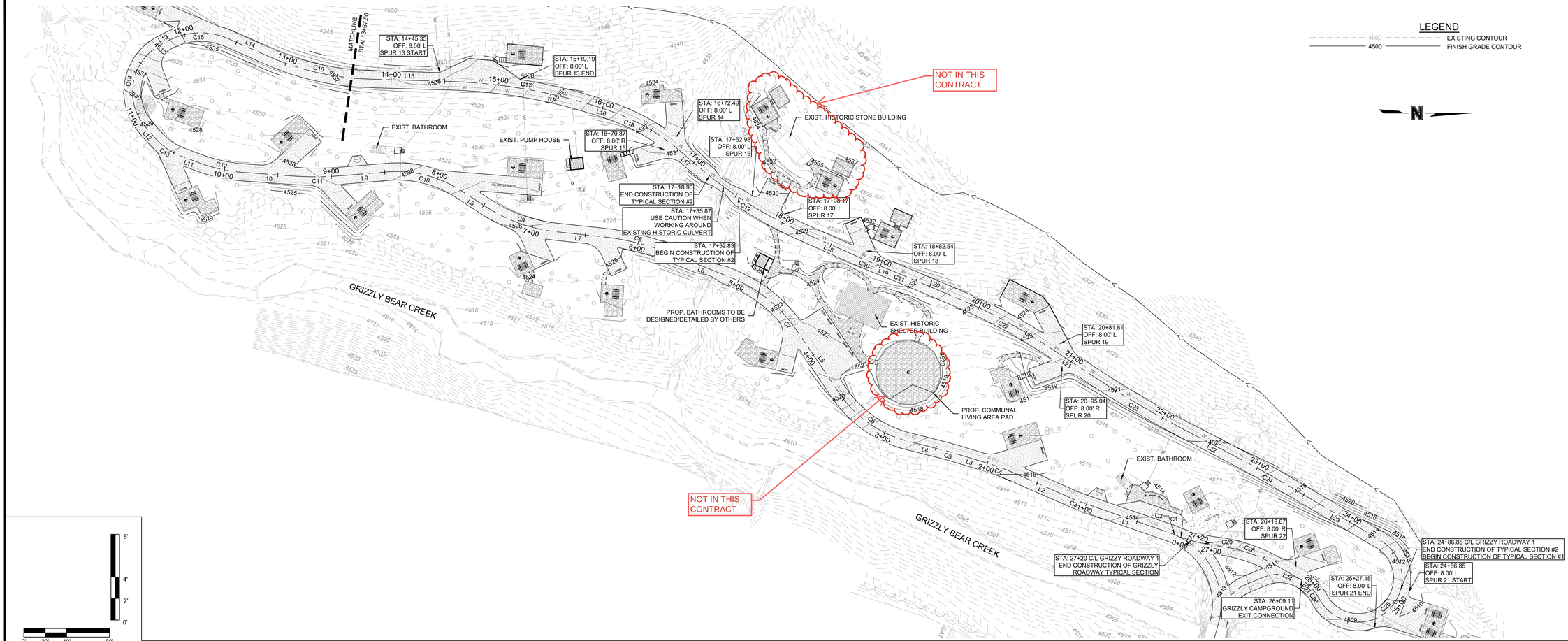
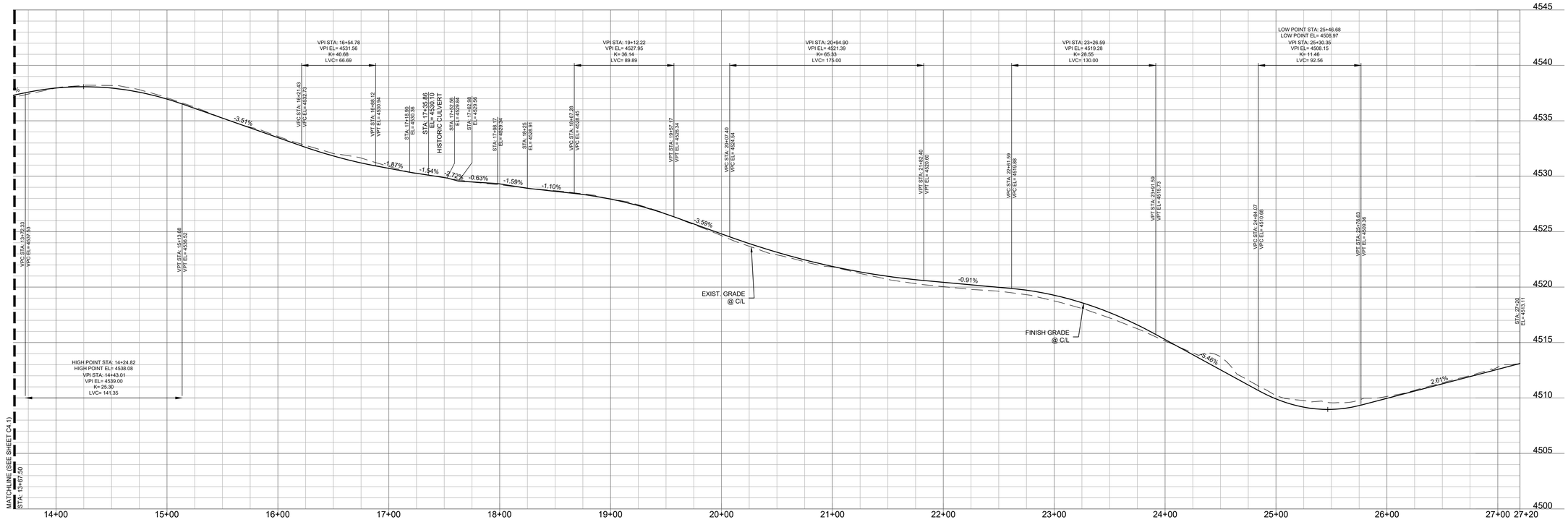


GRIZZLY ROADWAY 1
STA: 13+50.00 - 27+20.00



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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

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HELL CANYON RANGER DISTRICT

DRAWING TITLE

GRIZZLY ROADWAY PLAN AND PROFILE

ARCHIVE NO.

DESIGNER CRW	SUB SHEET NO. C4.2
CHECKED CRW	
DATE 05/28/2025	SHEET 12 OF 48

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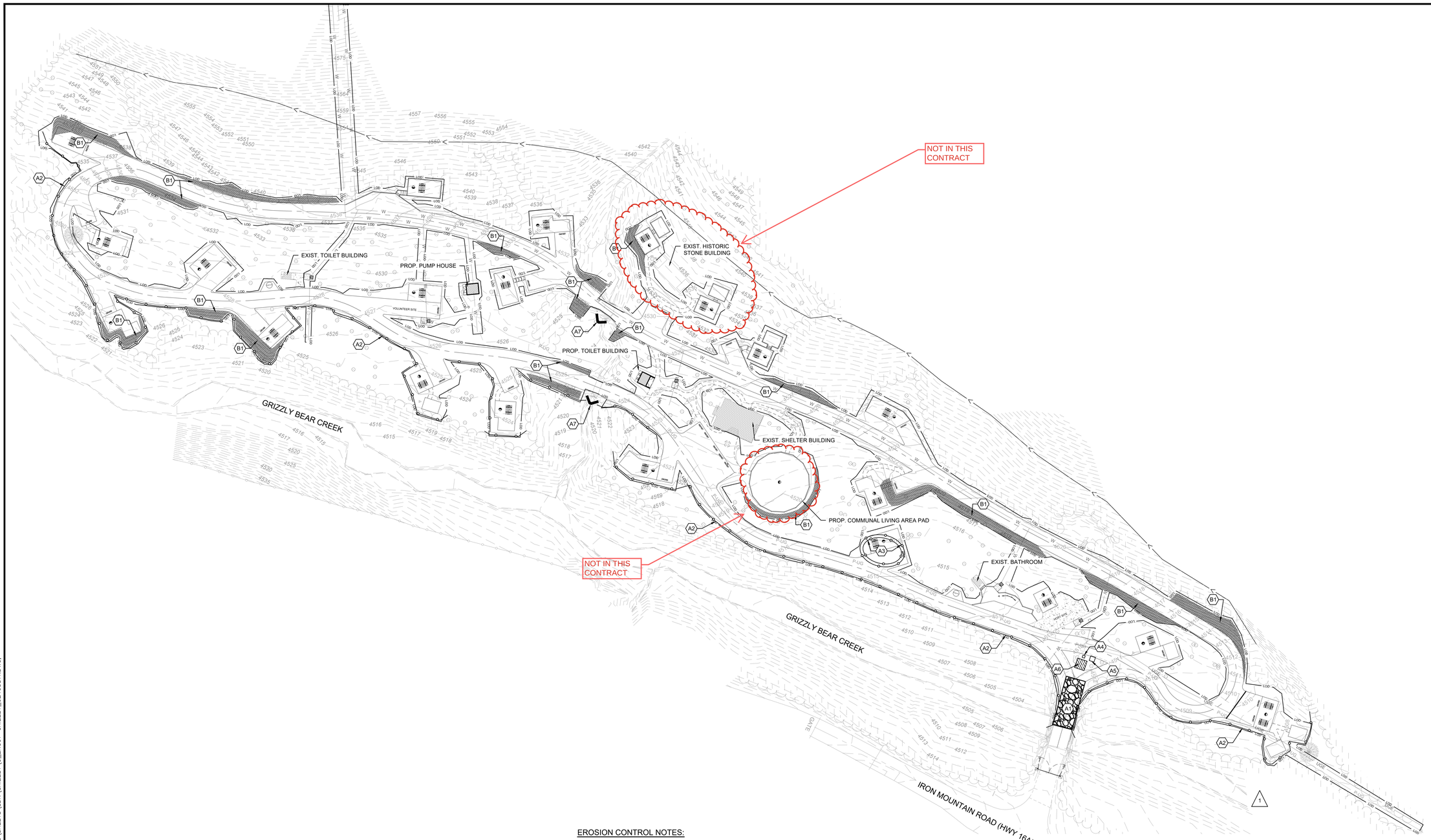

EROSION CONTROL PHASING CHART

PROJECT STAGE	BMP PLAN REF. #	BMP DESCRIPTION	REMOVE AFTER STAGE	NOTES:	INSTALLATION DATE / REMOVAL DATE
A - PRE-CONSTRUCTION	A1	CONSTRUCTION ENTRANCE/EXIT	C	SEE DETAIL SHEET C5.1	
	A2	TEMPORARY COMPOST FILTER SOCK	C	SEE DETAIL SHEET C5.1	
	A3	TEMPORARY STAGING/STOCKPILE AREA	C	SEE APPROX. LOCATION ON THIS SHEET	
	A4	INSTALL PORTA-POTTY(S)	C	CONTRACTOR TO MARK FINAL PLACEMENT ON ERC PLAN.	
	A5	INSTALL DUMPSTER, SITE SIGN & SPILL KIT	C	CONTRACTOR TO MARK FINAL PLACEMENT ON ERC PLAN.	
	A6	INSTALL CONCRETE WASHOUT	B	SEE DETAIL SHEET C5.1	
	A7	INSTALL FLARED END SECTION ROCK CHECK	C	SEE DETAIL SHEET C5.1	
B - MASS GRADING & ON-SITE IMPROVEMENTS		TEMPORARY SEEDING	N/A	PROVIDE TEMPORARY SEEDING (SEE EROSION CONTROL NOTES & TEMP. SEEDING NOTES ON SHEET C5.1)	
	B1	INSTALL SLOPE REINFORCEMENT	N/A	INSTALL PER MANUFACTURERS RECOMMENDATIONS	
		CONTRACTOR SHALL MAINTAIN EROSION CONTROL DEVICES THROUGHOUT CONSTRUCTION PHASE	N/A		
C - POST-CONSTRUCTION/ FINAL STABILIZATION	C1	PLACE TOPSOIL, INSTALL EROSION CONTROL MATTING SEED, MULCH, HYDROSEED, LANDSCAPE (AS INDICATED ON PLAN)	N/A	ESTABLISH PERENNIAL VEGETATION WITH A 70% DENSITY OVER 100% OF DISTURBED AREA. REFER TO SHEET C5.1 FOR ADDITIONAL INFORMATION.	

EROSION CONTROL NOTES:

1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTENANCE OF ALL EROSION & SEDIMENT CONTROL MEASURES AND PRACTICES THROUGHOUT THE PROJECT. ANY AND ALL FINES ASSOCIATED WITH EROSION CONTROL VIOLATIONS WILL BE THE CONTRACTOR'S RESPONSIBILITY.
2. EROSION CONTROL IS THE CONTRACTOR'S RESPONSIBILITY. THIS PLAN SHOULD BE USED AS A GUIDE AND REPRESENTS THE MINIMUM EROSION CONTROL DEVICES REQUIRED.
3. EROSION CONTROL MEASURES WILL BE MAINTAINED AT ALL TIMES. ADDITIONAL EROSION & SEDIMENT CONTROL MEASURES WILL BE INSTALLED IF DEEMED NECESSARY BY ON-SITE INSPECTION.
4. CONTRACTOR IS RESPONSIBLE FOR REPAIRING ALL EROSION & SEDIMENT CONTROL DEVICES AFTER EACH RAINFALL EVENT.
5. THE CONTRACTOR SHALL PROVIDE ANY FURTHER EROSION CONTROL MEASURES IN ADDITION TO THOSE LISTED TO ENSURE THAT SILT WILL NOT LEAVE THE PROJECT CONFINES.
6. THE CONTRACTOR IS RESPONSIBLE FOR REMOVING THE TEMPORARY EROSION & SEDIMENT CONTROL DEVICES AFTER COMPLETION OF CONSTRUCTION AND ONLY WHEN AREAS HAVE BEEN STABILIZED WITH A HEALTHY STAND OF PERMANENT VEGETATION.
7. THE CONTRACTOR IS RESPONSIBLE FOR REMOVING SILT FROM SITE IF NOT REUSABLE ON-SITE AND ASSURING PLAN ALIGNMENT AND GRADE IN ALL DITCHES AT COMPLETION OF CONSTRUCTION.
8. THE CONTRACTOR SHALL ENSURE THAT ALL DRAINAGE STRUCTURES, FLUMES, PIPES, ETC. ARE CLEANED OUT AND WORKING PROPERLY AT TIME OF ACCEPTANCE.
9. THE CONTRACTOR SHALL PROVIDE ANY TEMPORARY STABILIZATION AS REQUIRED.
10. THE CONTRACTOR SHALL PROVIDE A TEMPORARY CONSTRUCTION ENTRANCE FOR VEHICULAR TRAFFIC AT LOCATION(S) SHOWN.
11. ALL EROSION CONTROL DEVICES SHALL CONFORM TO THE APPLICABLE SECTIONS OF THE STANDARD SPECIFICATIONS AND DESIGN CRITERIA OF THE UNITED STATES DEPARTMENT OF AGRICULTURE FOREST SERVICE.
12. REFERENCE DETAILS ON SHEET C5.1 FOR TYPICAL EROSION CONTROL DEVICE INSTALLATION.
13. THE CONTRACTOR WILL BE REQUIRED TO CLEAN THE STREETS OF DEPOSITED MUD AS FREQUENTLY AS NEEDED AS DETERMINED BY THE ENGINEER IN ORDER TO KEEP THEM USABLE AND TO CONTROL DUST.
14. SEE TEMPORARY VEGETATION REQUIREMENT NOTES ON SHEET C5.1 FOR EXPOSED SOIL WHERE NO ACTIVITY WILL OCCUR FOR MORE THAN 14 DAYS.
15. CONTRACTOR IS RESPONSIBLE FOR PHASED INSTALLATION OF EROSION CONTROL BMP'S IN ORDER TO PREVENT SEDIMENT FROM BREACHING THE LIMITS OF DISTURBANCE.

LEGEND

- RIGHT-OF-WAY LINE
- PROPERTY LINE
- UTILITY EASEMENT
- 4500 EXIST. CONTOUR
- 4500 FINISH GRADE CONTOUR
- LOD LIMITS OF DISTURBANCE
- INSTALL COMPOST FILTER SOCK
- INSTALL FES OUTLET PROTECTION (SEE DETAIL SHEET C5.1)
- INSTALL GRAVEL CONSTRUCTION ENTRANCE/EXIT
- INSTALL CONCRETE WASHOUT
- INSTALL NORTH AMERICAN GREEN SC-150 EROSION CONTROL BLANKET PER MANUFACTURER'S RECOMMENDATIONS ON SLOPES 4:1 OR STEEPER. ECB SHALL BE INSTALLED INCREMENTALLY AS FILL IS ADDED. CONTRACTOR TO ALSO INSTALL PERMANENT SEEDING IN THIS AREA (SEE NOTES ON SHEET C5.1)
- X STORM OUTFALL

LAND DISTURBANCE:

TOTAL SITE AREA: 12.96± AC.

TOTAL DISTURBED AREA: 2.98± AC.

N

0' 20' 40' 80'

SCALE IN FEET

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

EROSION CONTROL PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C5.0
CHECKED: CRW	
DATE: 05/28/2025	SHEET 13 OF 48

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TEMPORARY & PERMANENT VEGETATION REQUIREMENTS:

TOPSOIL REQUIREMENTS:

PERMANENT AND TEMPORARY SEEDING -
LOOSEN COMPACTED SOILS TO A DEPTH OF 4 INCHES. IF RAINFALL CAUSES THE SURFACE TO BECOME SEALED OR CRUSTED, LOOSEN IT JUST PRIOR TO SEEDING. SLOPES STEEPER THAN 33 PERCENT (3:1) GRADE SHOULD BE GROOVED OR FURROWED ON THE CONTOUR BEFORE SEEDING. A GOOD SEEDBED IS WELL PULVERIZED, LOOSE, AND UNIFORM.

PERMANENT SEEDING -
A MINIMUM OF 4 INCHES OF LOOSE TOPSOIL SHOULD BE SPREAD ON AREAS TO BE SEEDED.

LIME REQUIREMENTS:

PERMANENT AND TEMPORARY SEEDING -
LIME SHOULD BE APPLIED ACCORDING TO SOIL TEST RECOMMENDATIONS. IF THE PH OF THE SOIL IS UNKNOWN, LIME SHALL BE INCORPORATED INTO THE TOP 4 INCHES OF SOIL AT THE RATE OF 1500 POUNDS EFFECTIVE NEUTRALIZING MATERIAL (ENM) PER ACRE. SOILS WITH A PH OF SIX OR HIGHER NEED NOT BE LIMED.

FERTILIZER REQUIREMENTS:

PERMANENT SEEDING -
FERTILIZER SHOULD BE APPLIED BASED ON SOIL TESTS. WHEN THESE ARE NOT POSSIBLE, A 13-13-13 GRADE FERTILIZER SHALL BE INCORPORATED INTO THE TOP 4 INCHES OF SOIL AT THE RATE OF 500 POUNDS PER ACRE.

TEMPORARY SEEDING -
FERTILIZER SHOULD BE APPLIED BASED ON SOIL TESTS. WHEN THESE ARE NOT POSSIBLE, A 10-10-10 GRADE FERTILIZER SHALL BE INCORPORATED INTO THE TOP 4 INCHES OF SOIL AT THE RATE OF 200 POUNDS PER ACRE.

SEED REQUIREMENTS:

PERMANENT SEEDING -
SEED MIX SHALL CONSIST OF NINETY PERCENT (90%) TURF TYPE TALL FESCUE AND TEN PERCENT (10%) ANNUAL RYEGRASS. SEED MIXTURE SHALL BE APPLIED AT A RATE OF 400 POUNDS PER ACRE. FINAL STABILIZATION WILL BE ACHEIVED ONCE VEGETATION COVERS AT LEAST 70% OVER ALL OF THE DISTURBED AREA.

TEMPORARY SEEDING - SEED SELECTION VARIES DEPENDING UPON THE TIME OF YEAR. SEED MIX SHALL CONSIST OF SPRING WHEAT (APRIL-JULY), WINTER WHEAT (AUGUST-NOVEMBER), OR OATS (APRIL-JULY). SEED MIXTURE SHALL BE APPLIED AT A RATE OF 56 POUNDS PER ACRE.

DORMANT SEASON SEEDING - SEED MIX SHALL CONSIST OF SPRING OATS. SEED MIXTURE SHALL BE APPLIED AT A RATE OF 100 POUNDS PER ACRE.

MULCH REQUIREMENTS:

PERMANENT AND TEMPORARY SEEDING -
MULCH IS REQUIRED AT THE RATE OF TWO TONS PER ACRE. MULCH SHALL MEET THE REQUIREMENTS OF SECTION 730 OF THE SOUTH DAKOTA DOT STANDARD SPECIFICATIONS FOR ROADS AND BRIDGES.

DATES FOR SEEDING:

PERMANENT SEEDING - MARCH 1 TO JUNE 1 AND SEPTEMBER 1 TO NOVEMBER 30
TEMPORARY SEEDING - APRIL TO NOVEMBER
DORMANT SEASON SEEDING - DECEMBER TO MARCH

POLLUTION PREVENTION PROCEDURE NOTES:

1. HANDLING AND DISPOSAL OF HAZARDOUS MATERIALS:

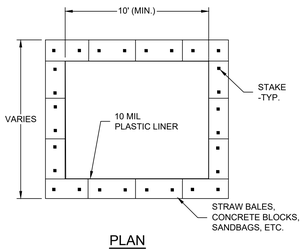
DO:
- PREVENT SPILLS
- USE PRODUCTS UP
- FOLLOW LABEL DIRECTIONS FOR DISPOSAL
- REMOVE LIDS FROM EMPTY BOTTLES AND CANS WHEN DISPOSING TRASH
- RECYCLE WASTES WHENEVER POSSIBLE

DON'T:
- DON'T POUR WASTE INTO SEWERS OR WATERWAYS ON THE GROUND
- DON'T POUR WASTE DOWN THE SINK, FLOOR DRAIN OR SEPTIC TANKS
- DON'T BURY CHEMICALS OR CONTAINERS, OR DISPOSE OF THEM WITH OTHER WASTE
- DON'T BURN OR MIX CHEMICALS OR CONTAINERS
- DON'T WASH SEDIMENT DOWN STORM SEWER INLETS

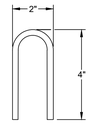
2. CONTAINERS SHALL BE PROVIDED FOR COLLECTION OF ALL WASTE MATERIAL INCLUDING CONSTRUCTION DEBRIS, TRASH, PETROLEUM PRODUCTS AND ANY HAZARDOUS MATERIALS TO BE USED ONSITE. ALL WASTE MATERIAL SHALL BE DISPOSED OF AT FACILITIES APPROVED FOR THAT MATERIAL.
3. NO WASTE MATERIALS SHALL BE BURIED ON-SITE.
4. MIXING, PUMPING, TRANSFERRING OR OTHERWISE HANDLING CONSTRUCTION CHEMICALS SUCH AS FERTILIZER, LIME, ASPHALT, CONCRETE DRYING COMPOUNDS, AND ALL OTHER POTENTIALLY HAZARDOUS MATERIALS SHALL BE PERFORMED IN AN AREA AWAY FROM ANY WATERCOURSE, DITCH OR STORM DRAIN.
5. EQUIPMENT FUELING AND MAINTENANCE, OIL CHANGING, ETC., SHALL BE PERFORMED ONLY IN AN AREA DESIGNATED FOR THAT PURPOSE. THE DESIGNATED AREA SHALL BE EQUIPPED FOR RECYCLING OIL AND CATCHING SPILLS.
6. CONCRETE WASH WATER SHALL NOT BE ALLOWED TO FLOW DIRECTLY TO STORM SEWERS, STREAMS, DITCHES, LAKES, ETC WITHOUT BEING TREATED. A CONCRETE WASHOUT AREA SHALL BE PROVIDED. SEE DETAIL ON THIS SHEET.
7. ALL PAINT, SOLVENTS, PETROLEUM PRODUCTS AND PETROLEUM WASTE PRODUCTS, AND STORAGE CONTAINERS (SUCH AS DRUMS, CANS, OR CARTONS) SHALL BE STORED ACCORDING TO BMPS. THE MATERIALS EXPOSED TO PRECIPITATION SHALL BE STORED IN WATERTIGHT, STRUCTURALLY SOUND, CLOSED CONTAINERS. ALL CONTAINERS SHALL BE INSPECTED FOR LEAKS OR SPILLAGE DURING THE ONCE PER WEEK INSPECTION OF BMPS. IF SUBSTANCES SUCH AS OIL, DIESEL FUEL, HYDRAULIC FLUID, ANTIFREEZE, ETC. ARE SPILLED, LEAKED, OR RELEASED ONTO SOIL, THE SOIL SHALL BE DUG UP AND PROPERLY DISPOSED OF. SPILLS ON PAVEMENT SHALL BE ABSORBED WITH SAWDUST, KITTY LITTER OR PRODUCT DESIGNED FOR THAT PURPOSED AND DISPOSED OF AT A LICENSED SANITARY LANDFILL. HAZARDOUS OR INDUSTRIAL WASTES SUCH AS MOST SOLVENTS, GASOLINE, OIL-BASED PAINTS, AND CEMENT CURING COMPOUNDS REQUIRE SPECIAL HANDLING. THESE MATERIALS WILL BE REMOVED FROM THE SITE AND RECYCLED OR DISPOSED OF IN ACCORDANCE WITH SOUTH DAKOTA DEPARTMENT OF AGRICULTURE AND NATURAL RESOURCES (DANR) REQUIREMENTS.
8. STATE LAW REQUIRES THE PARTY RESPONSIBLE FOR A PETROLEUM PRODUCT SPILL IN EXCESS OF 50 GALLONS TO REPORT THE SPILL TO SOUTH DAKOTA DANR (605-773-3296) AS SOON AS PRACTICAL AFTER DISCOVERY. FEDERAL LAW REQUIRES THE RESPONSIBLE PARTY TO REPORT ANY RELEASE OF OIL IF IT REACHES OR THREATENS A SEWER, LAKE, CREEK, STREAM, RIVER, GROUNDWATER, WETLAND, OR AREA, LIKE A ROAD DITCH, THAT DRAINS INTO ONE OF THE ABOVE.
9. SUFFICIENT TEMPORARY TOILET FACILITIES TO SERVE THE NUMBER OF WORKERS ON THE SITE SHALL BE PROVIDED. THE FACILITIES SHALL BE SERVICED FREQUENTLY TO MAINTAIN A SANITARY CONDITIONS.

NOTES:

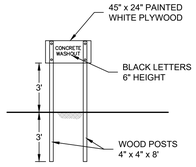
1. ACTUAL LAYOUT LOCATIONS & NUMBER OF WASH-OUTS TO BE DETERMINED BY CONTRACTOR IN FIELD.
2. THE CONCRETE WASH-OUT SIGN SHALL BE INSTALLED WITHIN 30 FT. OF THE TEMPORARY CONCRETE WASHOUT FACILITY.
3. WASH-OUTS SHALL BE LOCATED A MINIMUM OF 50 FT. FROM STORM DRAINS, OPEN DRAINAGE FACILITIES AND WATER COURSES. AVOID MIXING EXCESS AMOUNTS OF FRESH CONCRETE.
4. INSTALL BEFORE THE START OF ANY CONCRETE ACTIVITIES OR DELIVERIES.
5. INSPECT EVERY WEEK AND AFTER ½" STORM EVENT. REMOVE AND DISPOSE OF HARDENED CONCRETE AND RETURN THE FACILITY TO A FUNCTIONAL CONDITION. WASH-OUT FACILITIES MUST BE CLEANED, OR NEW FACILITIES MUST BE CONSTRUCTED AND READY FOR USE ONCE THE WASH-OUT IS 75% FULL.
6. WHEN TEMPORARY CONCRETE WASH-OUT FACILITIES ARE NO LONGER REQUIRED FOR THE WORK, THE HARDENED CONCRETE SHOULD BE REMOVED AND DISPOSED. MATERIALS USED TO CONSTRUCT TEMPORARY CONCRETE WASH-OUT FACILITIES SHALL BE REMOVED FROM THE SITE AND DISPOSED. HOLES, DEPRESSIONS OR OTHER GROUND DISTURBANCE CAUSED BY THE REMOVAL OF THE WASH-OUT SHALL BE BACKFILLED AND REPAIRED.



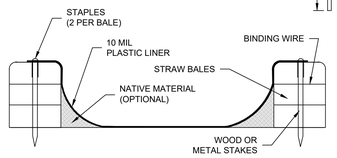
PLAN



STAPLE

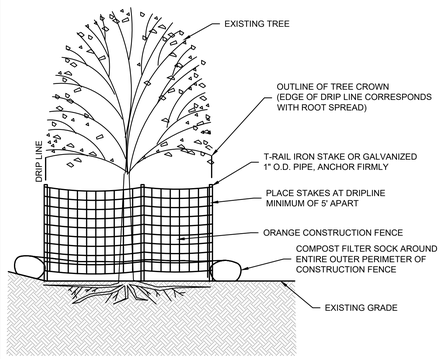


WASHOUT SIGN



SECTION

CONCRETE WASH-OUT DETAIL
NOT TO SCALE



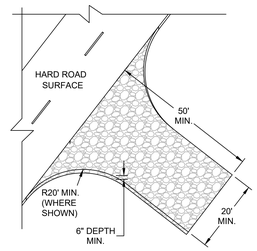
TREE PRESERVATION NOTES

1. GRADING AROUND TREES TO BE LIMITED TO OUTSIDE THE DRIPLINE UNLESS OTHERWISE NOTED ON THESE PLANS.
2. THERE SHALL BE NO GRADING OR FILLING WITHIN THE PROTECTED CRITICAL ROOT ZONES OF PROTECTED TREES UNLESS OTHERWISE NOTED ON THESE PLANS.
3. NO VEHICLES OR EQUIPMENT SHALL BE DRIVEN OR PARKED UNDER THE DRIPLINE OF PROTECTED TREES. NO MATERIALS SHALL BE STORED UNDER THE DRIPLINE OF PROTECTED TREES.
4. TREE FENCE AND SILT FENCE FOR ALL PROTECTED TREES MUST BE PROPERLY MAINTAINED THROUGHOUT CONSTRUCTION.
5. ALL REQUIRED TREE, SILT FENCING AND COMPOST FILTER SOCK SHALL REMAIN IN PLACE UNTIL ALL GRADING AND LANDSCAPING ACTIVITIES HAVE CEASED AND SHALL ONLY BE REMOVED UPON THE REVIEW AND APPROVAL BY THE LANDSCAPE ARCHITECT.
6. ATTACHMENTS (WIRES, FENCES, ETC.) OTHER THAN THOSE APPROVED FOR GUYING, BRACING OR WRAPPING MUST NOT BE ATTACHED TO PROTECTED TREES.

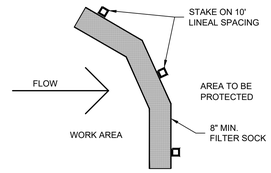
TREE PROTECTION DETAIL
NOT TO SCALE

NOTES:

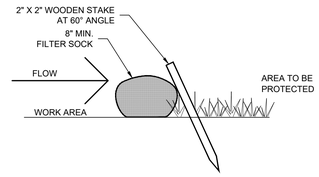
1. INSTALL AS SOON AS POSSIBLE BEFORE START OF GRADING.
2. PLACE FILTER FABRIC UNDER STONE
3. USE 3"-6" CLEAN CRUSHED LIMESTONE
4. REPLACE AS NEEDED TO MAINTAIN 6" DEPTH



GRAVEL CONSTRUCTION ENTRANCE / EXIT
NOT TO SCALE



PLAN VIEW



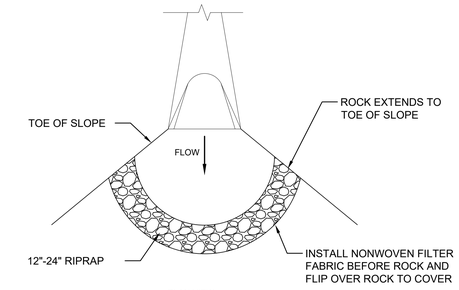
SECTION VIEW

NOTES:

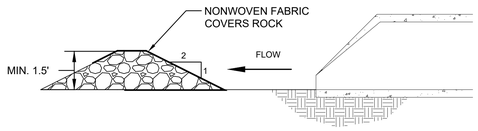
1. COMPOST FILTER SOCK SHALL MEET THE REQUIREMENTS OF SDDOT SECTION 734.2.B.1. CONTRACTOR SHALL SUBMIT SHOP DRAWINGS TO ENGINEER FOR APPROVAL BEFORE INSTALLING SOCK.
2. FILTER SOCK DEPICTED IS FOR MINIMUM SLOPES. GREATER SLOPES MAY REQUIRE LARGER SOCKS PER ENGINEER.
3. COMPOST MATERIAL TO BE DISPERSED ON SITE, AS DETERMINED BY ENGINEER.

COMPOST FILTER SOCK DETAILS
NOT TO SCALE

NOTES:
1. TEMPORARY ROCK CHECK FILTER SHALL BE INSTALLED AFTER FLARED END SECTION IS INSTALLED AND SHALL TRAP SEDIMENT DURING STORM PIPE INSTALLATION.
2. ROCK SIZE AND QUANTITY SHALL BE SIZED TO FILTER STORMWATER RUNOFF VOLUME.
3. NONWOVEN FILTER FABRIC SHALL BE INSTALLED FIRST. ONCE ROCK IS INSTALLED WRAP FILTER FABRIC OVER ROCK TO COVER.
4. ALL FLOW SHALL PASS THROUGH ROCK CHECK FILTER.
5. SEDIMENT ACCUMULATION SHALL BE REMOVED AFTER RAIN EVENTS AND SHALL NOT EXCEED 50%.
6. ROCK CHECK FILTER SHALL BE REMOVED ONCE AREA UPSTREAM OF PIPE IS PERMANENTLY STABILIZED.



PLAN VIEW



PROFILE VIEW

F.E.S. ROCK CHECK FILTER DETAILS
NOT TO SCALE

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

EROSION CONTROL DETAILS

ARCHIVE NO.

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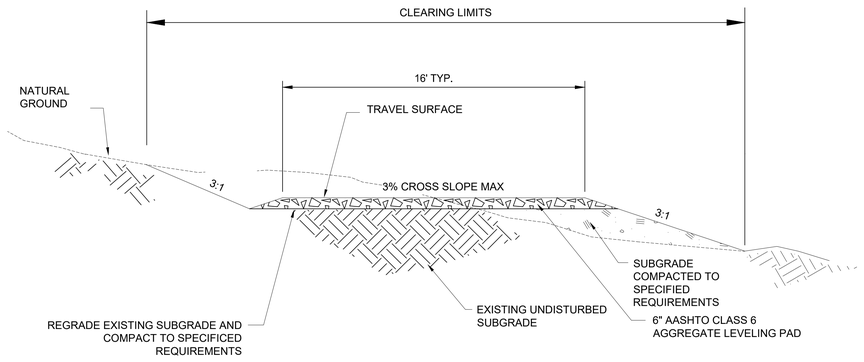
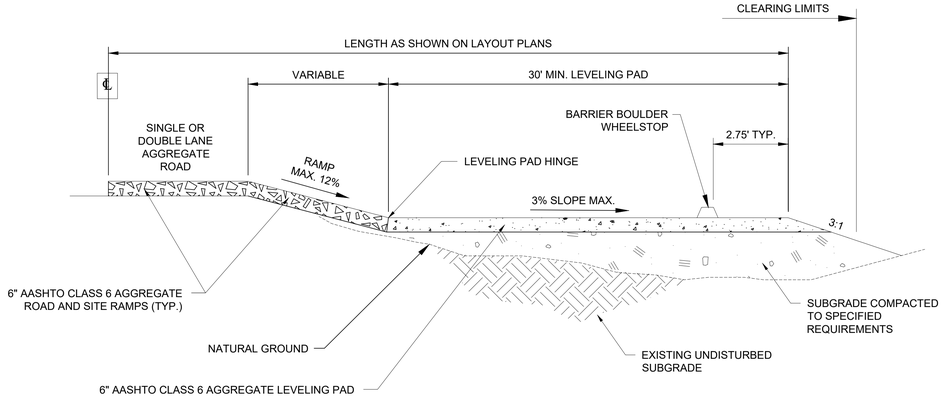


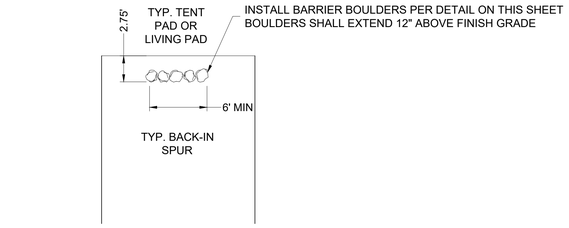
F C6.0 TYPICAL BACK-IN SPUR CROSS SECTION
NO SCALE



C C6.0 TYPICAL BACK-IN SPUR CENTERLINE FILL SECTIONS
NO SCALE

GENERAL RV CAMPGROUND SPUR NOTES

1. CAMPGROUND SPUR SITES CAN SHIFT TO AVOID EXISTING OR UNFORESEEN SITE CONDITIONS. APPROVAL BY CONTRACTING OFFICER IS REQUIRED.
2. INSTALL BARRIER BOULDERS AT END OF ALL BACK-IN SPURS, DUMPSTER SITES, AND THE HOST SITE. COORDINATE WITH CONTRACTING OFFICER TO PLACEMENT PRIOR TO INSTALLATION.
3. ALL SPUR SITES MUST MEET ABA / FSORG ACCESSIBILITY REQUIREMENTS.
4. ALL DISTURBED AREAS ARE TO BE SCARIFIED AND SEEDED PER SPECIFICATIONS.



E C6.0 BARRIER BOULDER WHEELSTOP
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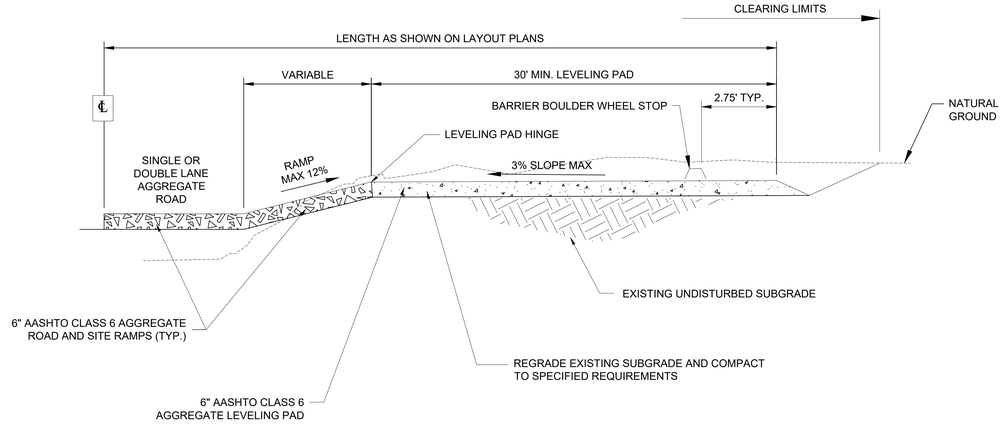
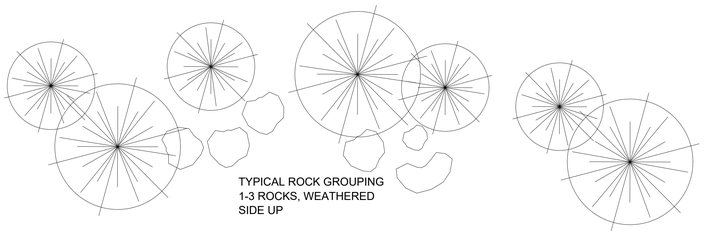


B C6.0 TYPICAL BACK-IN SPUR CENTERLINE CUT SECTIONS
NO SCALE



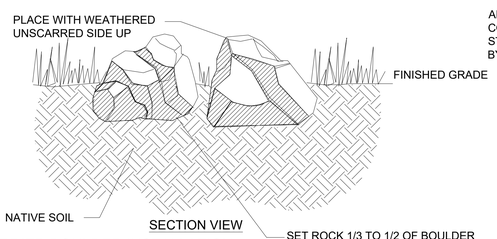
USE NATURAL CONTROLS AS MUCH AS POSSIBLE. EXISTING TREES, SHRUBS, ROCK OUTCROPPINGS, STEEP SLOPES AND CREEK CROSSINGS.

PLAN VIEW

NOTE: BARRIER ROCK LOCATIONS WILL BE STAKED BY THE CONTRACTING OFFICER. BARRIER ROCK SOURCE SHALL BE FROM RELOCATED/STOCKPILED ROCKS ON SITE. ADDITIONAL BOULDERS SHALL BE PROVIDED BY CONTRACTOR AS DETAILED IN SPECIFICATIONS

GENERAL BOULDER BARRIER NOTES

ALL BOULDERS SHOWN ON SITE PLANS ARE CONCEPTUAL IN NATURE. EXACT LOCATION TO BE STAKED IN FIELD BY CONTRACTOR AND APPROVED BY CONTRACTING OFFICER PRIOR TO PLACEMENT.



D C6.0 BARRIER BOULDER INSTALLATION
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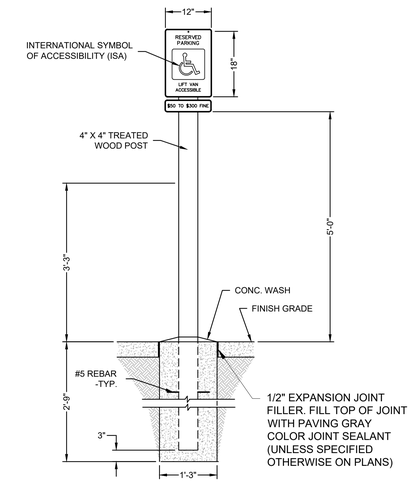

A C6.0 ISA STALL SIGN DETAIL
NO SCALE

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PROJECT NAME
GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
100% CDs
BLACK HILLS NATIONAL FOREST
HELL CANYON RANGER DISTRICT

DRAWING TITLE
DETAILS

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CRW	C6.0
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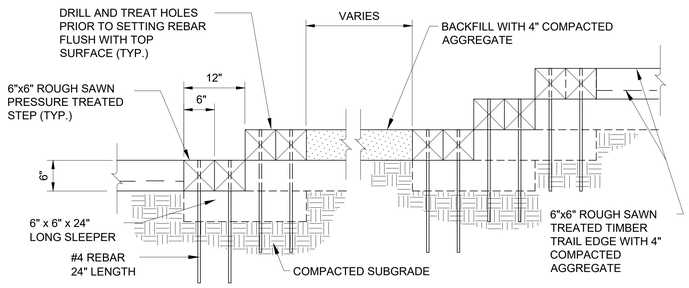
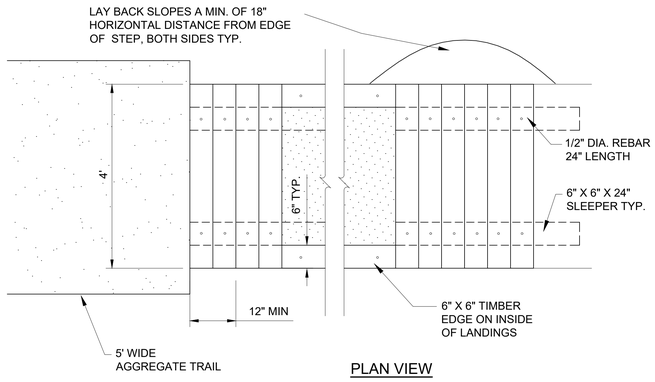


ELEVATION VIEW



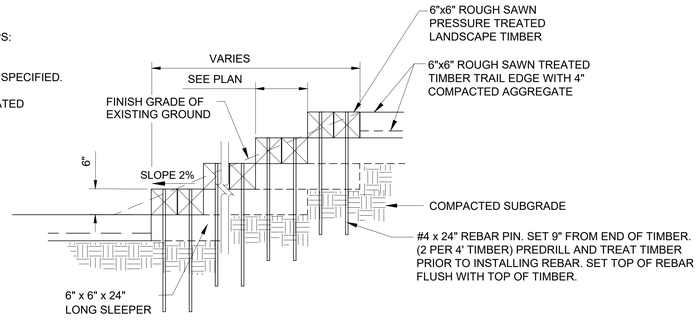
PLAN VIEW

D C6.1 TIMBER STEPS WITH LANDING DETAIL

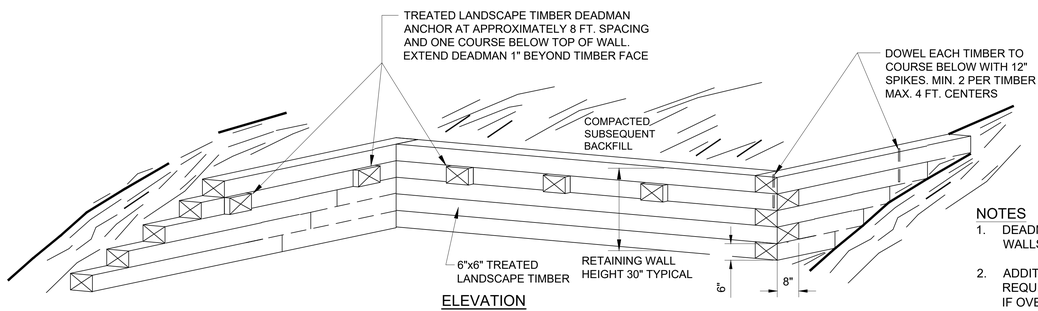
NO SCALE

NOTES

1. STEPS HAVE NO CROSS SLOPE.
2. NUMBER OF PINS ON DOUBLE TIMBER STEPS: 3'-0" TIMBER - 2 PINS PER TIMBER.
3. TREAT ALL CUT ENDS AND DRILL HOLLS AS SPECIFIED.
4. ALL TIMBER TO BE #2 GRADE TYPICAL TREATED LANDSCAPE TIMBER COLOR BROWN.



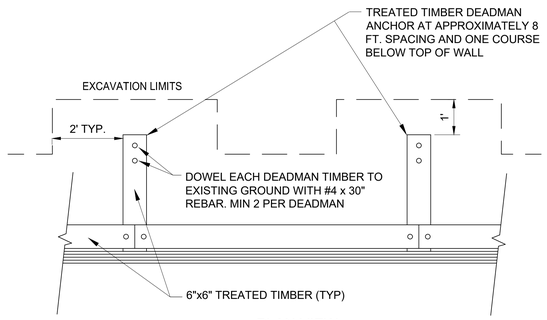
B C6.1 TIMBER STEPS DETAIL



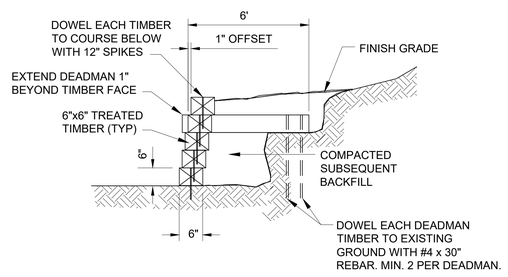
ELEVATION

NOTES

1. DEADMAN ANCHORS REQUIRED ON ANY WALLS OVER 3 COURSES HIGH
2. ADDITIONAL ROW(S) OF DEADMAN ANCHORS REQUIRED FOR EVERY 3 COURSES OF WALL IF OVER 30" HEIGHT
3. DEADMAN TIMBERS SHALL BE CONSIDERED INCIDENTAL TO THE RETAINING WALL BID PAY ITEM
4. REBAR AND SPIKES NOT SHOWN FOR CLARITY, TO BE SPACED 36" O.C.



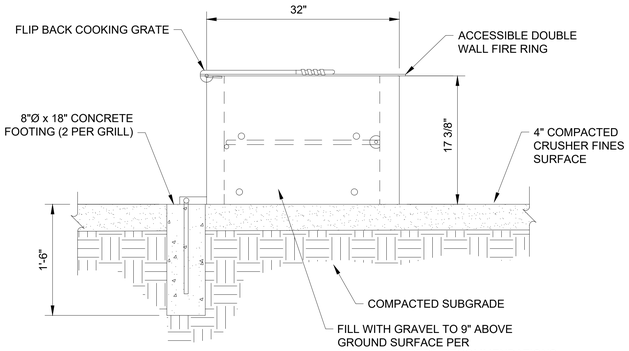
PLAN VIEW



SECTION

C C4.8 TIMBER RETAINING WALL DETAILS

NO SCALE



SECTION D

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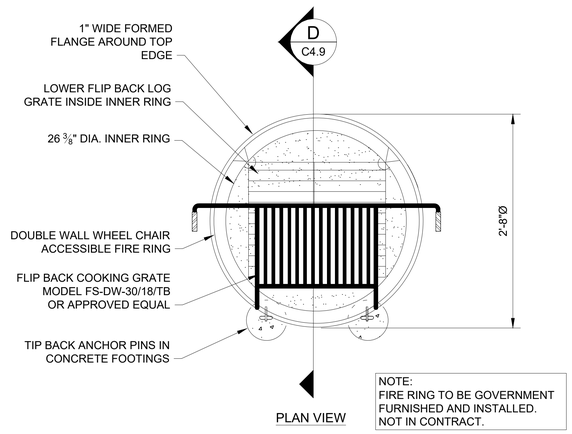

NOTE:
FIRE RING TO BE GOVERNMENT FURNISHED AND INSTALLED. NOT IN CONTRACT.

A C6.1 FIRE RING INSTALLATION DETAIL

NO SCALE

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

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ARCHIVE NO.

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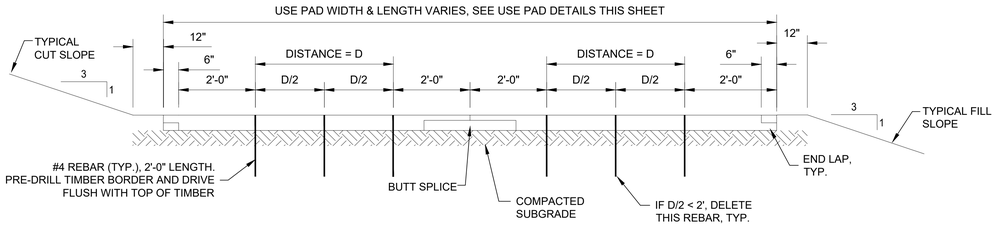
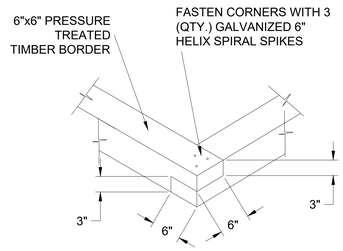


I C6.2 TIMBER PINNING INSTALL DETAIL

NO SCALE



F C6.2 TIMBER END LAP

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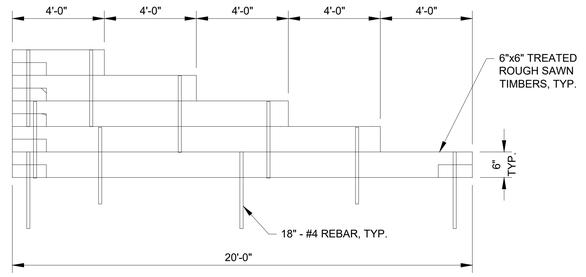
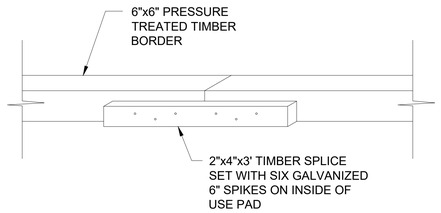
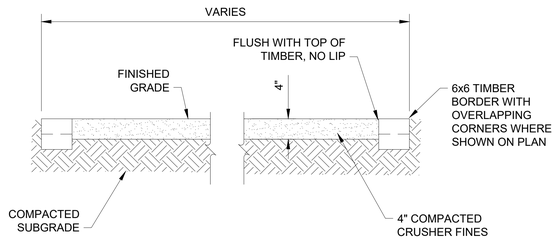


C C6.2 ADD-ON RETAINING WALL TO FACILITY / TENT PAD



H C6.2 TIMBER BUTT SPLICE

NO SCALE

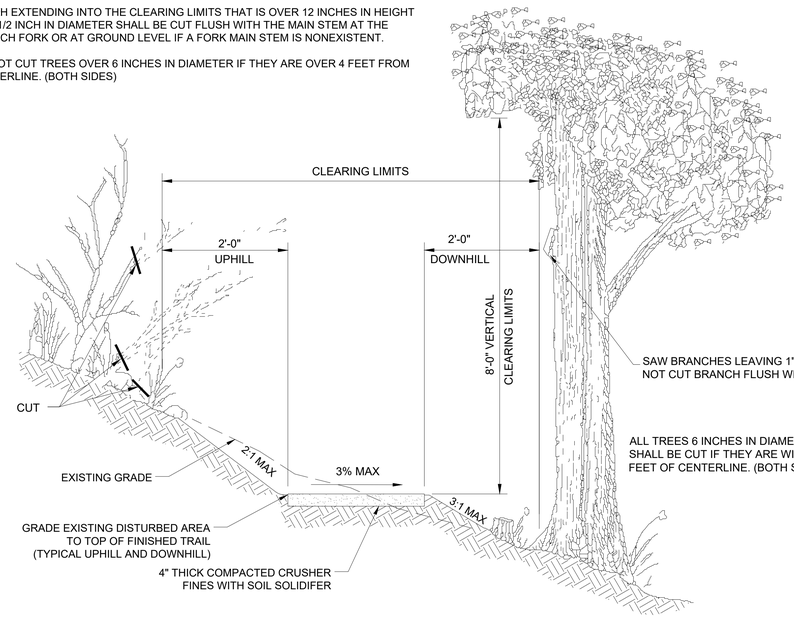


E C6.2 FACILITY PAD CROSS SECTION

NO SCALE

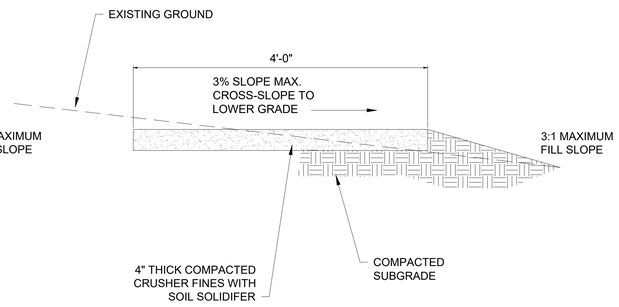
GENERAL NOTES

1. REFER TO TRAIL GRADE LIMITATIONS FOR MAXIMUM ALLOWABLE GRADES ALONG TRAIL IN DIRECTION OF TRAVEL.
2. MINIMIZE CUT AND FILL ON TRAIL, MAXIMUM 2:1 CUT SLOPE & 3:1 FILL SLOPE.
3. CONSTRUCT WITH A FULL BENCH WHENEVER POSSIBLE.
4. BRUSH EXTENDING INTO THE CLEARING LIMITS THAT IS OVER 12 INCHES IN HEIGHT AND 1/2 INCH IN DIAMETER SHALL BE CUT FLUSH WITH THE MAIN STEM AT THE BRANCH FORK OR AT GROUND LEVEL IF A FORK MAIN STEM IS NONEXISTENT.
5. DO NOT CUT TREES OVER 6 INCHES IN DIAMETER IF THEY ARE OVER 4 FEET FROM CENTERLINE. (BOTH SIDES)



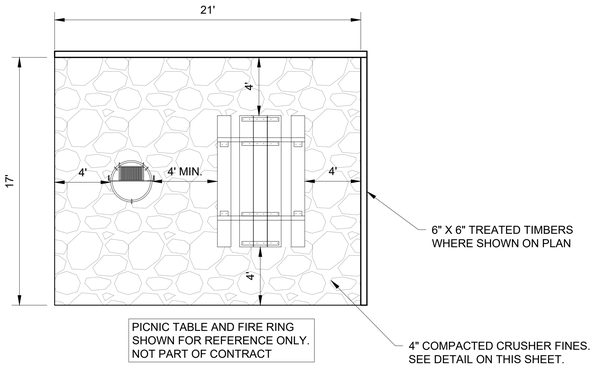
B C6.2 TYPICAL TRAIL CONSTRUCTION DETAIL TO WALK-IN SITES

NO SCALE



G C6.2 TYPICAL PATH SECTION INSIDE CAMPGROUND

NO SCALE



D C6.2 LIVING AREA PAD

NO SCALE

TRAIL GRADE	REMARKS
0% TO 5% [1:20]	TRAIL MEETS ACCESSIBILITY
> 5% [1:20] TO < 8.33% [1:12]	5'x5' LEVEL LANDING AREA* REQUIRED EVERY 50'
> 8.33% [1:12] TO < 10% [1:10]	5'x5' LEVEL LANDING AREA* REQUIRED EVERY 30'
> 10% [1:10] TO < 12.5% [1:8] MAX.	HORIZONTAL RUNS CANNOT EXCEED 10'
> 12.5% [1:8]	STEPS ARE REQUIRED (NOT ALLOWED ON ACCESSIBLE CONCRETE ROUTES)

(*) LEVEL LANDING AREA NOT TO EXCEED 2% IN ANY DIRECTION, LANDINGS LOCATIONS APPROVED OR FIELD LOCATED BY C.O.R.

A C6.2 TRAIL MAXIMUM GRADE LIMITATIONS

NO SCALE

CONSTRUCTION NOTES

1. FACILITY PAD AND TENT PAD LOCATIONS SHALL BE STAKED IN THE FIELD BY THE CONTRACTOR (WITH FOUR CORNER STAKES), AND APPROVED BY THE CONTRACTING OFFICER (C.O.) PRIOR TO CONSTRUCTION. FACILITY PAD AND TENT PAD LOCATIONS CAN SHIFT TO AVOID EXISTING OR UNFORESEEN SITE CONDITIONS. APPROVAL BY C.O. IS REQUIRED. APPROXIMATE LOCATIONS OF FACILITY PADS AND TENT PADS ARE SHOWN ON SHEETS C2.1 TO C2.2.
2. APPROXIMATE LOCATIONS OF RETAINING WALLS TO BE IDENTIFIED BY THE C.O. IN THE FIELD AFTER ALL SPUR LOCATIONS ARE STAKED AND APPROVED.
3. AFTER IDENTIFICATION BY THE C.O., ALL RETAINING WALLS SHALL BE STAKED IN THE FIELD AND APPROVED BY THE C.O. PRIOR TO CONSTRUCTION.
4. RETAINING WALL HEIGHT SHALL BE MINIMIZED AND SHALL NOT EXCEED 36".
5. RETAINING WALLS ARE INTENDED TO BE INSTALLED AT THE FRONT AND/OR SIDES OF SPURS, FACILITY PADS, AND/OR TENT PADS IN WHICH A CUT/FILL SLOPE OF 3:1 CANNOT MATCH EXISTING GRADE WITHIN 10' AND/OR DISTURBANCE OF MATURE VEGETATION AND/OR ADJOINING SPURS WOULD RESULT WITHOUT A RETAINING WALL.

GENERAL CAMPGROUND PATH NOTES

1. CAMPGROUND PATH CAN SHIFT TO AVOID EXISTING OR UNFORESEEN SITE CONDITIONS. APPROVAL BY CONTRACTING OFFICER IS REQUIRED.
2. ENTIRE PATH SURFACE MUST MEET FOREST SERVICE ACCESSIBILITY REQUIREMENTS.
3. PROVIDE A 5'x5' LEVEL LANDING FOR EVERY 50 FEET OF HORIZONTAL DISTANCE WHERE THE SLOPE OF THE PATH EXCEEDS 8%.
4. ALL TREES 6" IN DIAMETER SHALL BE CUT IF THEY ARE WITHIN 3 FEET OF PATH CENTERLINE, BOTH SIDES OF PATH TYPICAL.
5. BRUSH EXTENDING INTO CLEARING LIMITS THAT IS OVER 12" IN HEIGHT AND ½" IN DIAMETER SHALL BE CUT FLUSH WITH THE MAIN STEM AT THE BRANCH FORK OR AT GROUND LIVE IF FORKED MAIN STEM IS NON-EXISTENT.
6. CONTRACTOR TO INSTALL DRAINAGE DIP AS REQUIRED BY SITE DRAINAGE CONDITIONS OR AS DIRECTED BY CONTRACTING OFFICER.

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

DETAILS

ARCHIVE NO.

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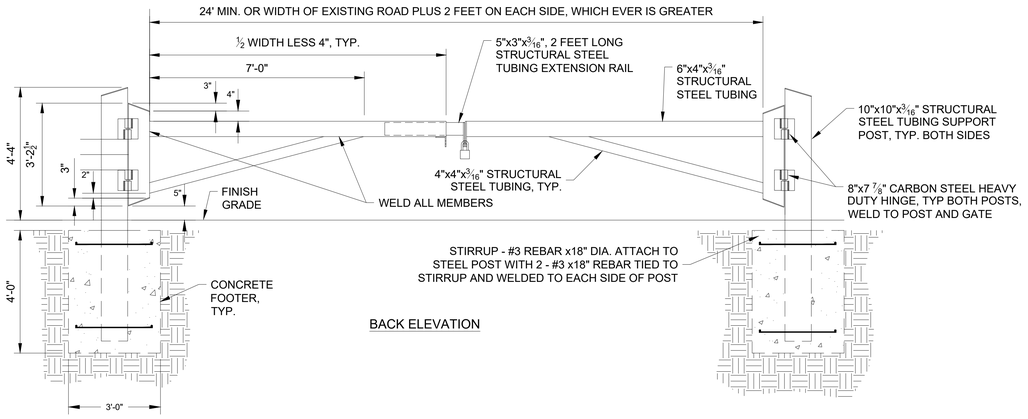
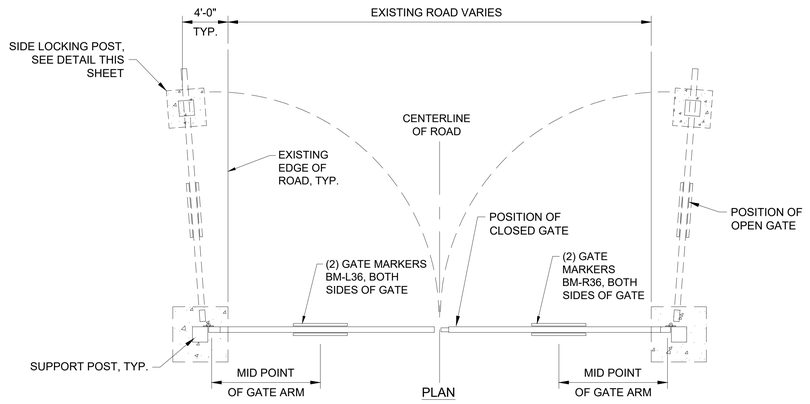
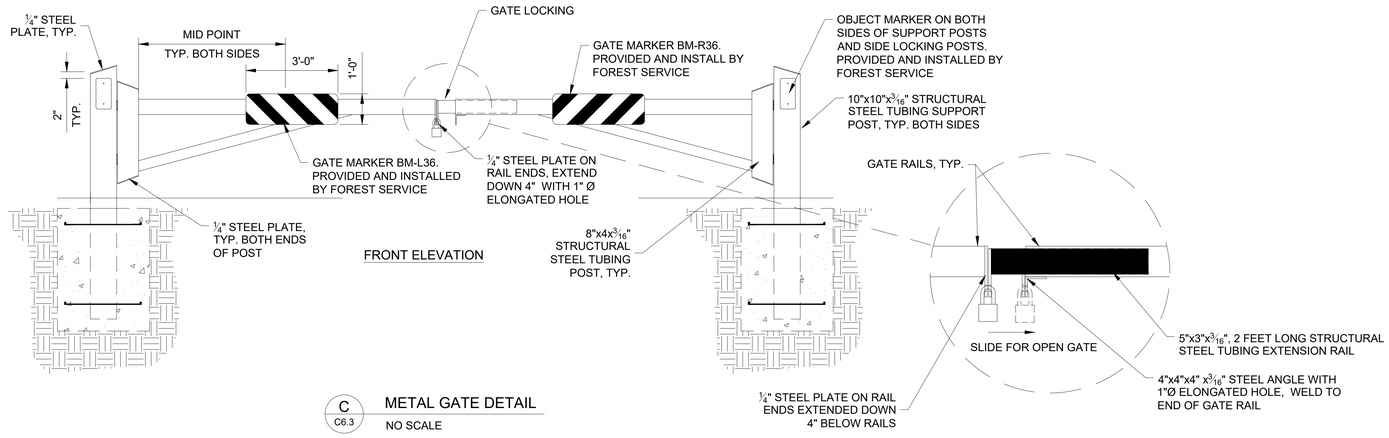


BACK ELEVATION



PLAN



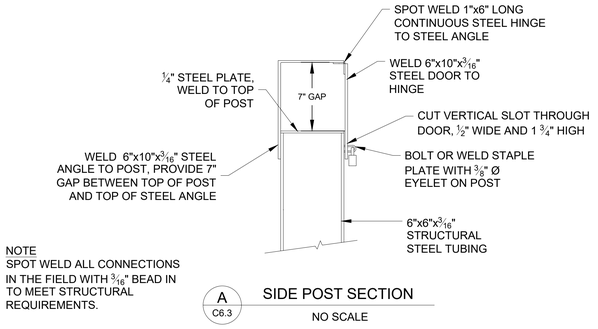
FRONT ELEVATION

C C6.3 METAL GATE DETAIL

NO SCALE

NOTES

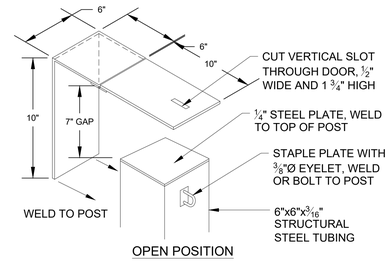
1. LOCATION OF GATES TO BE APPROVED BY CONTRACTING OFFICER PRIOR TO INSTALLATION.
2. FIELD LOCATE HOLDING POSTS TO SUPPORT GATE IN OPEN POSITION, COORDINATE WITH C.O..
3. IF HOLES FOR POSTS ARE OVER-EXCAVATED, THE CONTRACTOR SHALL FILL THE OVER-EXCAVATED HOLES WITH CONCRETE AT NO ADDITIONAL EXPENSE TO THE GOVERNMENT.
4. THE CONTRACTOR SHALL FURNISH AND INSTALL OBJECT MARKERS. THESE SHALL CONFORM TO THE MANUAL OF UNIFORM TRAFFIC CONTROL DEVICES. BOLT OBJECT MARKERS TO STEEL BACKING PLATES. FASTENERS FOR OBJECT MARKER INSTALLATIONS SHALL BE VANDAL-PROOF.
5. CONCRETE SHALL BE IN ACCORDANCE WITH THE SPECIFICATIONS.
6. ALL MEMBERS OF THE GATE ASSEMBLY SHALL BE FABRICATED FROM A-36 STANDARD STEEL SECTIONS. FABRICATED MEMBERS SHALL RECEIVE ONE SHOP COAT OF BENJAMIN MOORE RETARD PRIMER AND TWO COATS OF FOREST SERVICE BROWN PAINT OR EQUAL AFTER FABRICATION. ALL HARDWARE AND FASTENERS SHALL BE ZINC PLATED.
7. FASTENERS FOR OBJECT MARKER INSTALLATIONS SHALL BE SUPPLIED BY THE CONTRACTOR AND SHALL BE VANDAL-PROOF.
8. INSTALL GATE SECTIONS TO LEAVE REQUIRED SPACE FOR LOCKABLE LATCH TO FUNCTION. FIELD ADJUST GATE DIMENSIONS AS NECESSARY TO EXISTING CONDITIONS.



NOTE
SPOT WELD ALL CONNECTIONS IN THE FIELD WITH 3/16" BEAD IN TO MEET STRUCTURAL REQUIREMENTS.

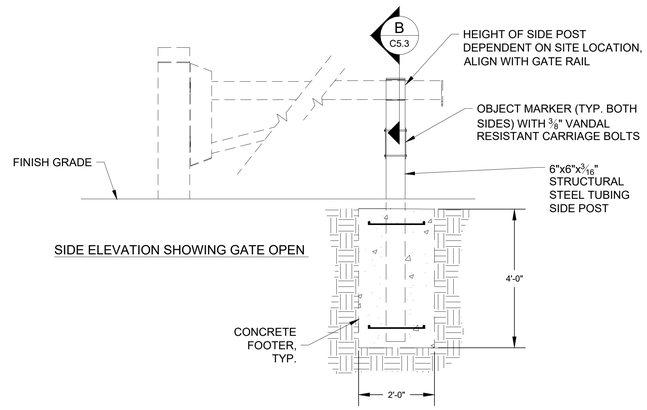
A C6.3 SIDE POST SECTION

NO SCALE



OPEN POSITION

SIDE POST LOCKING DEVICE DETAIL - ISOMETRIC



SIDE ELEVATION SHOWING GATE OPEN

A C6.3 SIDE POST DETAIL

NO SCALE

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
100% CDs

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HELL CANYON RANGER DISTRICT

DRAWING TITLE

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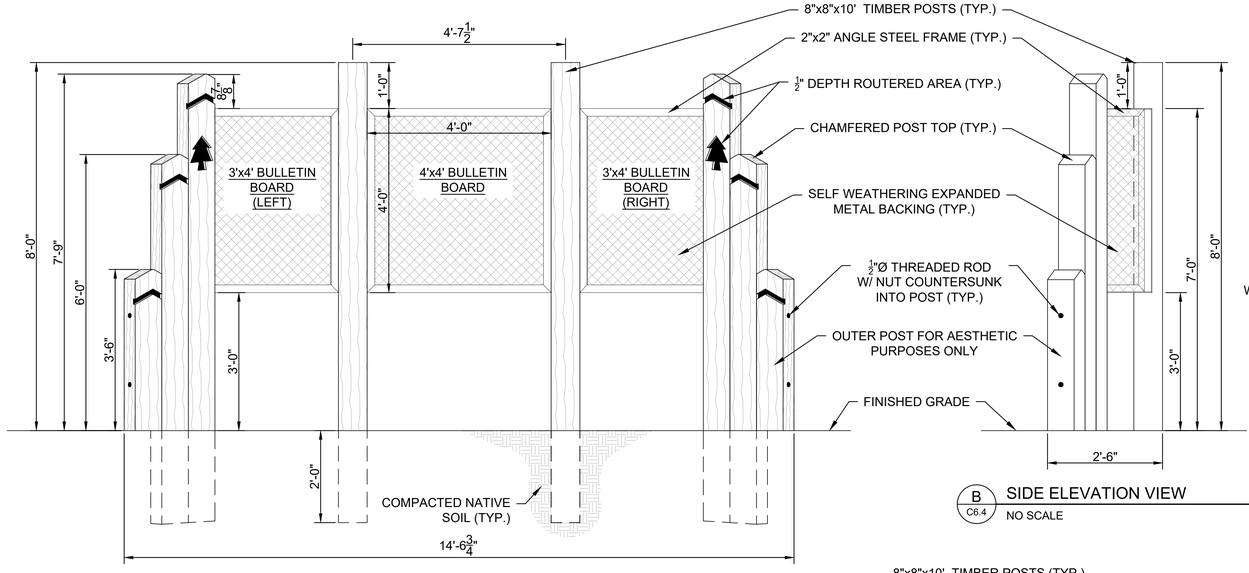
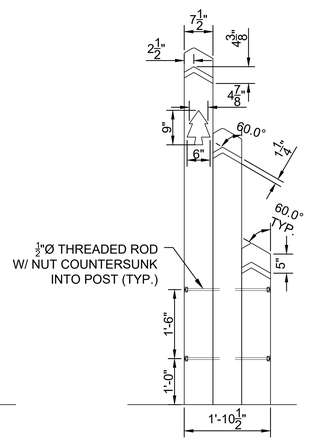
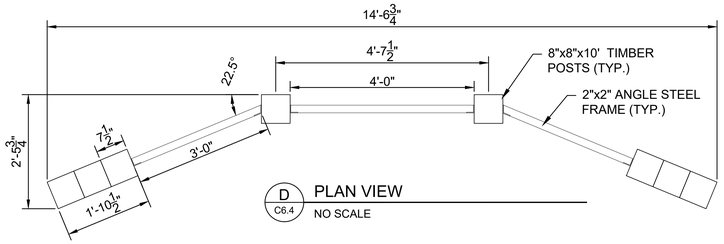


A C6.4 ELEVATION VIEW
NO SCALE

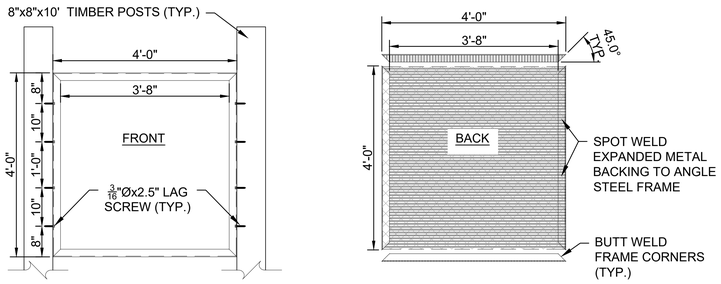


B C6.4 SIDE ELEVATION VIEW
NO SCALE

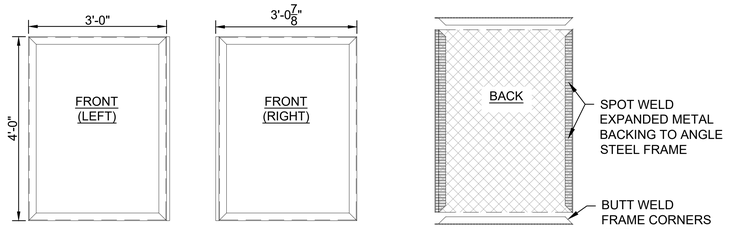
C C6.4 ROUTER & CHAMFER DETAILS
NO SCALE



D C6.4 PLAN VIEW
NO SCALE



E C6.4 4'x4' BULLETIN BOARD DETAILS
NO SCALE



F C6.4 3'x4' BULLETIN BOARD DETAILS
NO SCALE

NOTES:
1. ALL LUMBER SHALL BE ROUGH SAWN (R.S.) AND SHOWN AS NOMINAL DIMENSIONS, UNLESS OTHERWISE NOTED.
2. ALL WOOD SURFACES SHALL BE STAINED USING "OLYMPIC'S SEMI-TRANSPARENT STAIN IN FEDERAL STANDARD BROWN COLOR" OR APPROVED EQUAL.
3. BULLETIN BOARD SHALL HAVE 2"x2" SELF-WEATHERING ANGLED STEEL FRAME W/ SELF-WEATHERING EXPANDED METAL BACKING.
4. ALL FASTENER HARDWARE SHALL BE GALVANIZED.

KIOSK DETAIL SHOWN FOR REFERENCE ONLY. NOT PART OF CONTRACT

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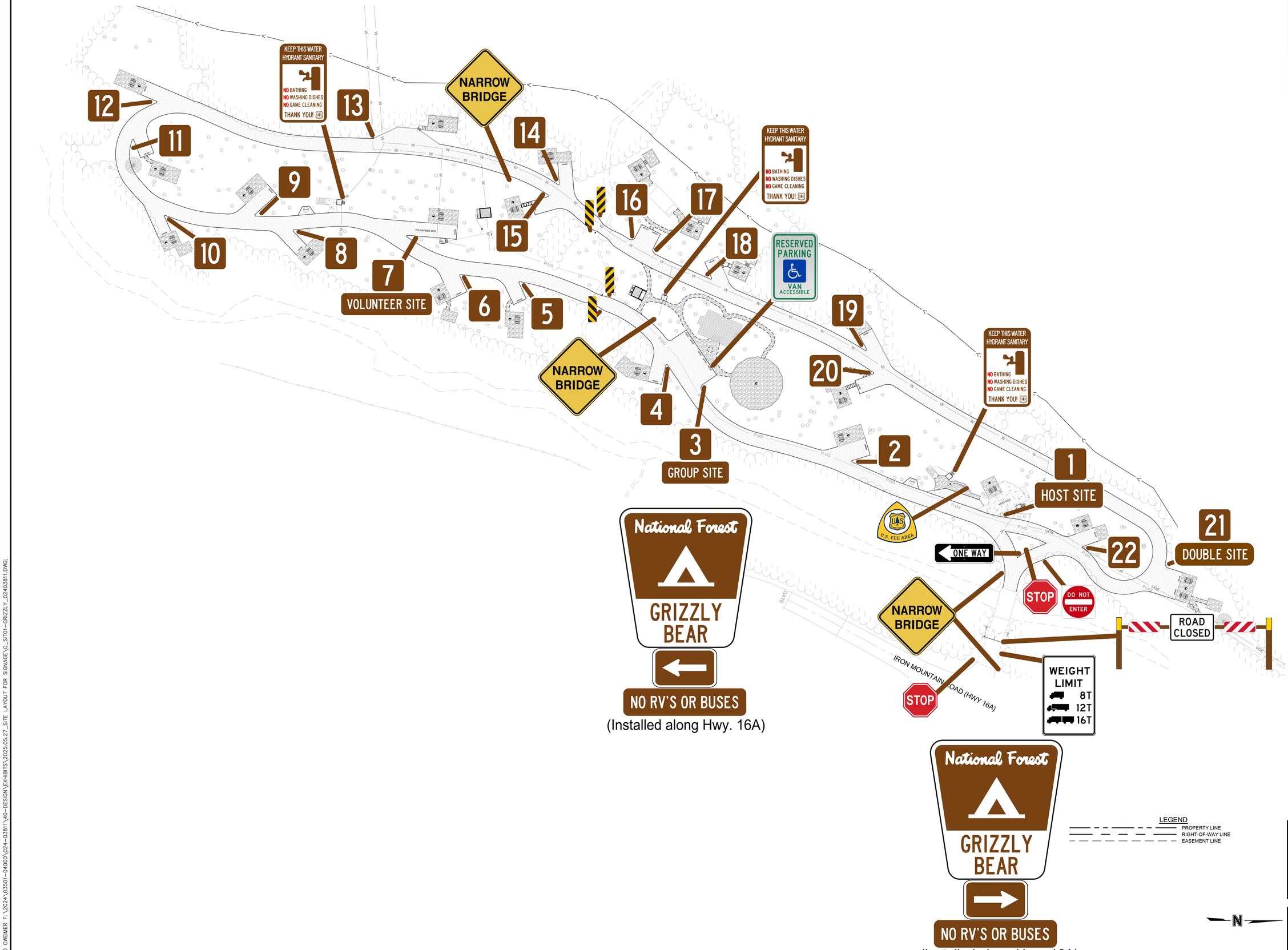

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SIGNAGE
PLAN

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GRIZZLY BEAR CAMPGROUND - DIRECTIONAL SIGNS

SIGN NUMBER	QTY.	WORDING/TYPE	SIGN CODE	TEXT HEIGHT	SIZE
1	2		RS-1	-	24"x24"
2	1		RS5-1	-	24"x24"
3	1		R6-1L	-	12"x36"
4	22	Site Marker Signs (Numbered 1 to 22)	-	3"	5"x5"
5	2		-	3"	25"x10"
6	1		-	3"	28"x10"
7	1		R7-8 & R7-8aP	-	12"x18"
8	1		-	3"	29"x8"
9	2		OM-3R	-	6"x12"
10	2		OM-3L	-	6"x12"
11	1		-	3"	24"x8"
12	1		R11-2	-	48"x30"
13	2		-	4"	41"x10"
14	2		SQ1-A	5"	40"x38"
15	1		M6-1L	-	21"x15"
16	1		M6-1R	-	21"x15"
17	4		W5-2	-	24"x24"
18	1		FBM-L	-	24"x8"
19	1		FBM-R	-	24"x8"
20	2		OM2-2V	-	6"x12"
21	1	Weight Limit Sign - Weights to be determined	R12-5	-	24"x30"
22	1		FA-1	-	9"x9"
23	3		-	-	5"x8"

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CODE REQUIREMENTS:

- 2024 INTERNATIONAL BUILDING CODE IBC
- 2024 NFPA LIFE SAFETY CODE 101 LSC
- ABA ACCESSIBILITY GUIDELINES ABAAG
- USFS SUSTAINABILITY GUIDELINES ZONING DESIGNATION: USDA FOREST SERVICE NATIONAL FOREST PROPERTY

CONSTRUCTION TYPE: V-B
OCCUPANCY: COMMERCIAL -N/A

ACCESSIBILITY SIGNAGE:
- PER IBC 1112.2, SIGNS REQUIRED IN SECTION 403.4 OF THE INTERNATIONAL PLUMBING CODE IDENTIFYING TOILET ROOMS AND BATHING ROOMS SHALL BE VISUAL CHARACTERS, RAISED CHARACTERS AND BRAILLE COMPLYING WITH ICC A117.1. WHERE PICTOGRAMS ARE PROVIDED AS DESIGNATIONS FOR TOILET ROOMS AND BATHING ROOMS, THE PICTOGRAMS SHALL HAVE VISUAL CHARACTERS, RAISED CHARACTERS AND BRAILLE COMPLYING WITH ICC A117.1.

INT WALL & CEILING REQUIREMENTS @ ROOMS AND ENCLOSED SPACES:
- CLASS 'C' PER IBC TABLE 803.13,
- PER IBC 1210.2.1, TOILET, BATHING AND SHOWER ROOM FLOOR FINISH MATERIALS SHALL HAVE A SMOOTH, HARD, NONABSORBENT SURFACE. THE INTERSECTIONS OF SUCH FLOORS WITH WALLS SHALL HAVE A SMOOTH, HARD, NONABSORBENT VERTICAL BASE THAT EXTENDS UPWARD ONTO THE WALLS NOT LESS THAN 4 INCHES.
- PER IBC 1210.2.2, WALLS AND PARTITIONS WITHIN 2 FEET OF SERVICE SINKS, URINALS AND WATER CLOSETS SHALL HAVE A SMOOTH, HARD, NONABSORBENT SURFACE, TO A HEIGHT OF NOT LESS THAN 4 FEET ABOVE THE FLOOR, AND EXCEPT FOR STRUCTURAL ELEMENTS, THE MATERIALS USED IN SUCH WALLS SHALL BE OF A TYPE THAT IS NOT ADVERSELY AFFECTED BY MOISTURE.

ROOF COVERING CLASSIFICATION:
- CLASS 'C' PER IBC TABLE 1505.1

STRUCTURAL DESIGN LOADS:
- 2024 INTERNATIONAL BUILDING CODE, ASCE 7-10
- DEFLECTION LIMITS PER IBC TABLE 1604.3,
- RISK CATEGORY II -STANDARD,
- LIVE LOADS PER IBC TABLE 1607.1:
- FLOOR 100 psf UNIFORM
- ROOF 20 psf UNIFORM, 300 # CONCENTRATED

OPTIONS:

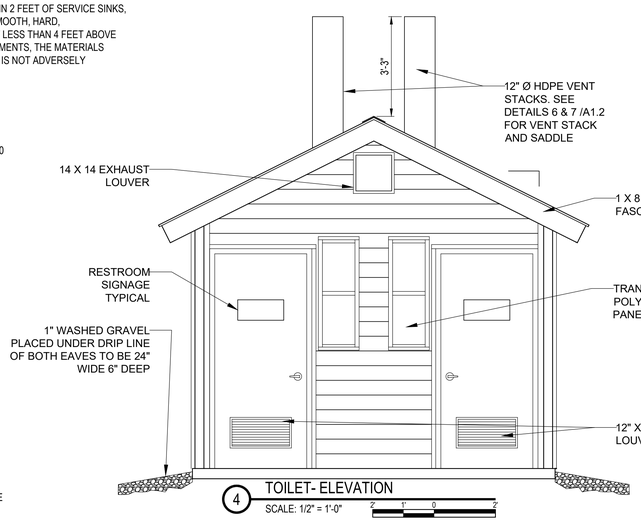
SEE CMEP FOR OPTIONS 1 THROUGH 3.

OPTION 4
CONTRACTOR SHALL INSTALL SINGLE POLYCARBONATE PANEL WINDOW UNIT INTO TWO (2) EXISTING SINGLE TOILET BUILDINGS ON SITE. LOCATE WINDOW UNIT IN SIMILAR LOCATION RELATIVE TO WHERE VENT STACK IS LOCATED IN DOUBLE TOILET BLDG (A1.1). COORDINATE AND VERIFY LOCATION OF WINDOW UNIT WITH COR.

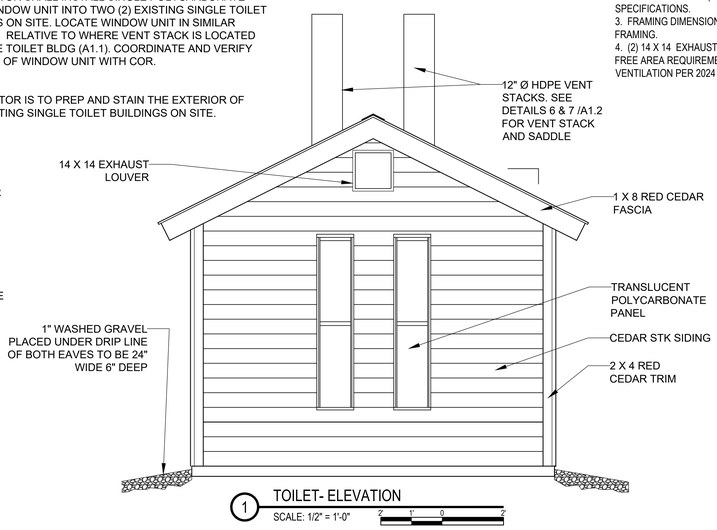
OPTION 5
CONTRACTOR IS TO PREP AND STAIN THE EXTERIOR OF TWO EXISTING SINGLE TOILET BUILDINGS ON SITE.

GENERAL NOTES:

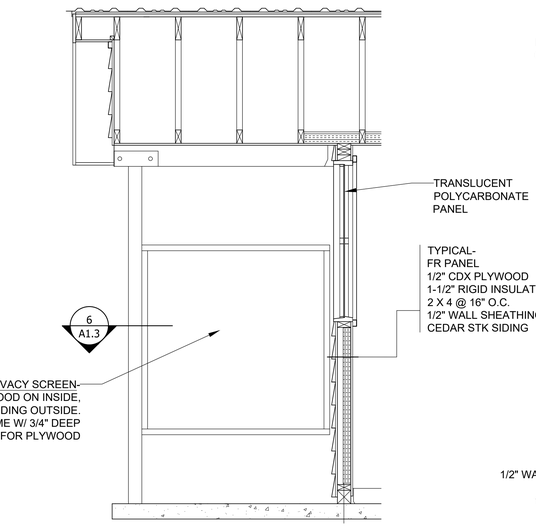
1. VAULT DETAILS HAVE BEEN OMITTED FOR CLARITY. SEE STRUCTURAL FOR VAULT DETAILS.
2. SIDING WILL BE 5/4 X 8 CEDAR LAP RABBITED SELECT TIGHT KNOT (STK) STAINED PER SPECIFICATIONS.
3. FRAMING DIMENSIONS FROM OUTSIDE FACE OF FRAMING.
4. (2) 14 X 14 EXHAUST LOUVERS PROVIDES 1/150 NET FREE AREA REQUIREMENT OF ATTIC ROOF VENTILATION PER 2024 IBC 1202.2.



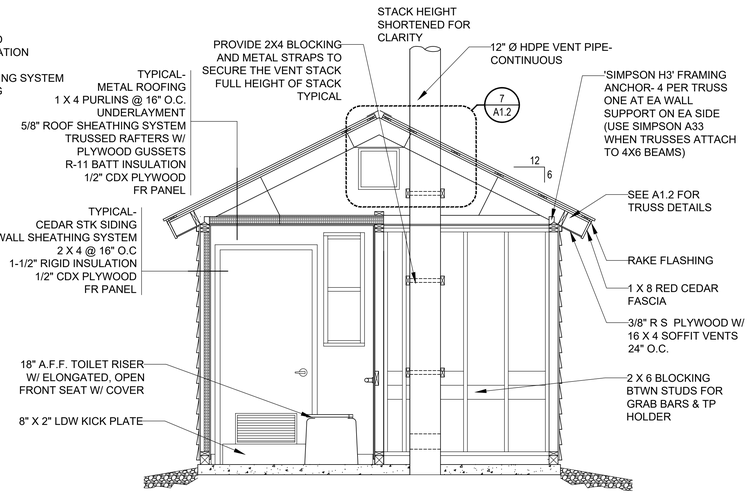
4 TOILET- ELEVATION
SCALE: 1/2" = 1'-0"



1 TOILET- ELEVATION
SCALE: 1/2" = 1'-0"

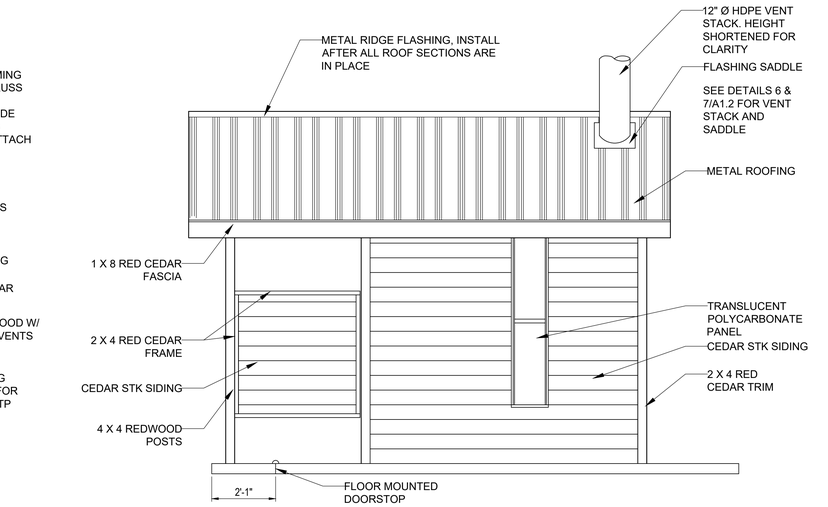


7 TOILET- SECTION
SCALE: 3/4" = 1'-0"



5 TOILET- SECTION
SCALE: 1/2" = 1'-0"

SEE 1/A1.2 FOR TRUSS DETAILS



2 TOILET- ELEVATION
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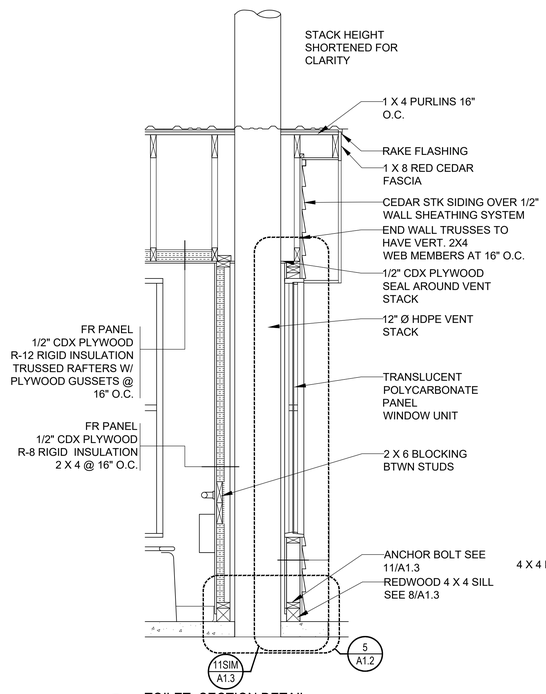
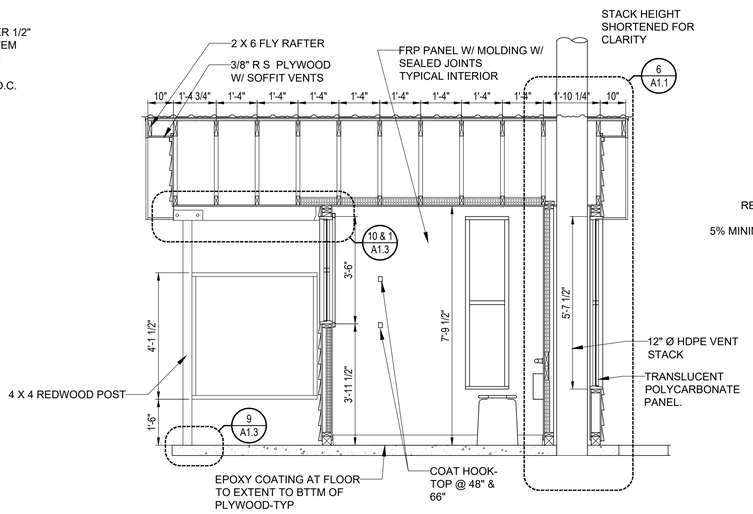
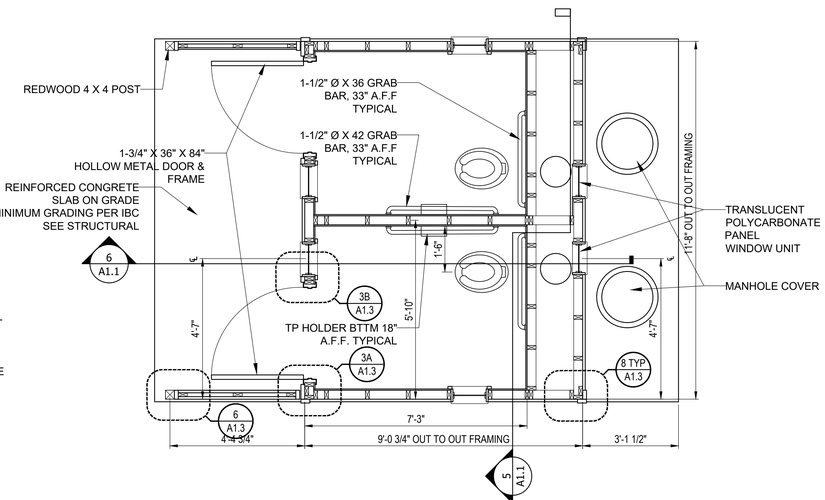


8 TOILET- SECTION DETAIL
SCALE: 3/4" = 1'-0"



6 TOILET- SECTION
SCALE: 1/2" = 1'-0"



3 TOILET- FLOOR PLAN
SCALE: 1/2" = 1'-0"

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

DOUBLE TOILET- FLOOR PLAN, ELEVATIONS, MISC

ARCHIVE NO.

DESIGNER

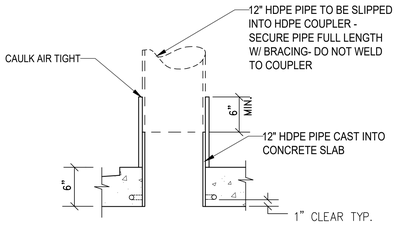
CHECKED

DATE 05/28/2025

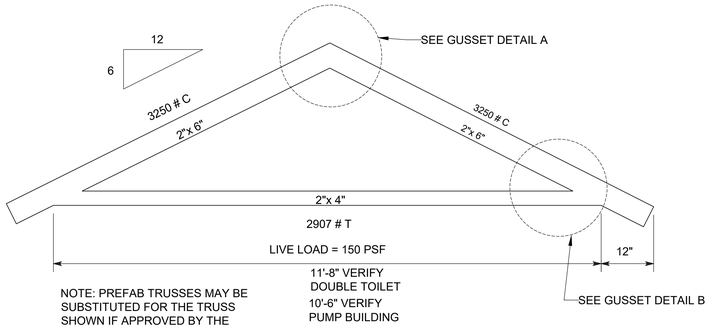
SUB SHEET NO. A1.1

SHEET 22 OF 48

5/28/25 11:27 TODD C:\TODD\1A-KHA WORK\201-023 BLACK HILLS USDA CAMPGROUNDS\DWGS\DWGS\GRIZZLY_A1.1.DWG;

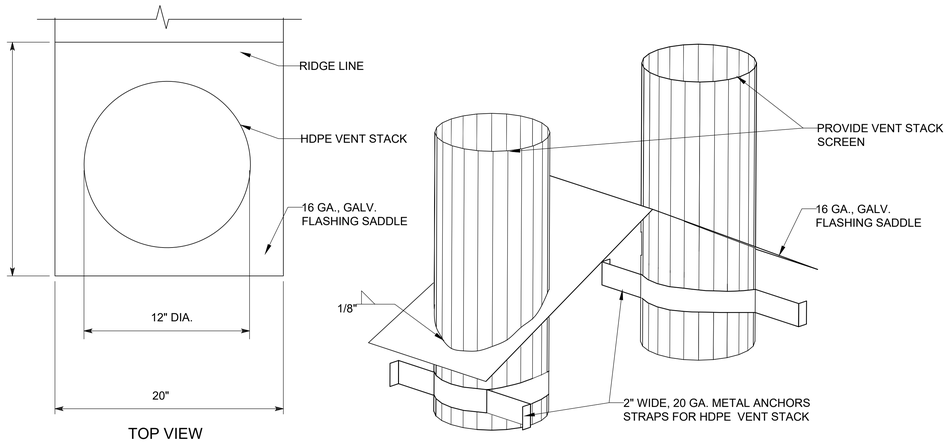


5 VENT PIPE DETAIL
NOT TO SCALE

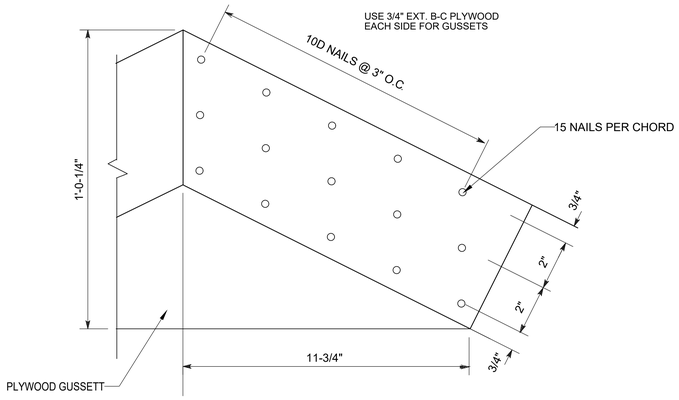


NOTE: PREFAB TRUSSES MAY BE SUBSTITUTED FOR THE TRUSS SHOWN IF APPROVED BY THE CONTRACTING OFFICER

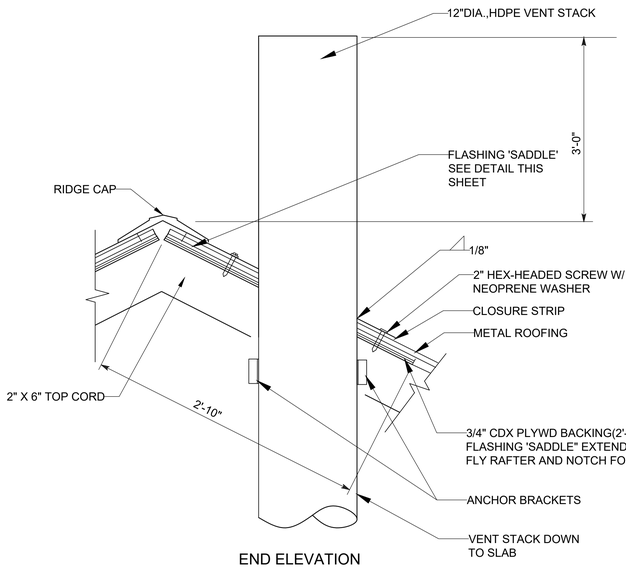
1 TRUSS DIAGRAM
NOT TO SCALE

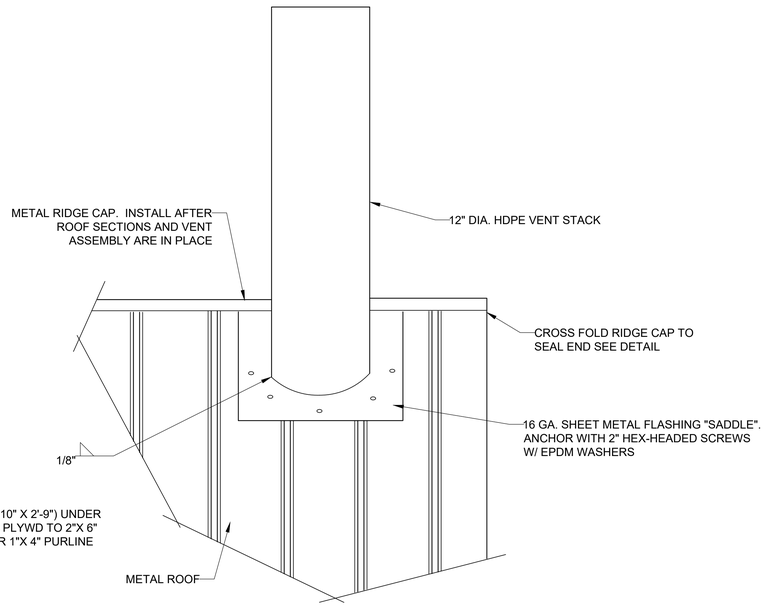


6 VENT STACK ASSEMBLY ISOMETRIC
NOT TO SCALE



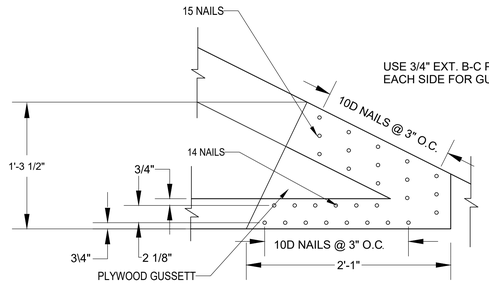
2 GUSSET DETAIL A
NOT TO SCALE



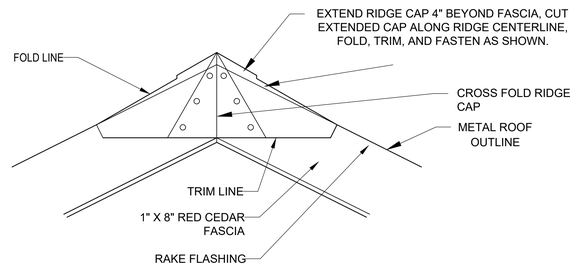


7 VENT STACK ASSEMBLY DETAILS
SCALE: 1-1/2" = 1'-0"

8" 4" 0" 8"



3 GUSSET DETAIL B
NOT TO SCALE



4 RIDGE CAP END DETAIL
NOT TO SCALE

5/28/25 11:27 TODD C:\TODD\1A-KHA WORK\201-023 BLACK HILLS USDA CAMPGROUNDS\DWGS\GRIZZLY_A1.2.DWG

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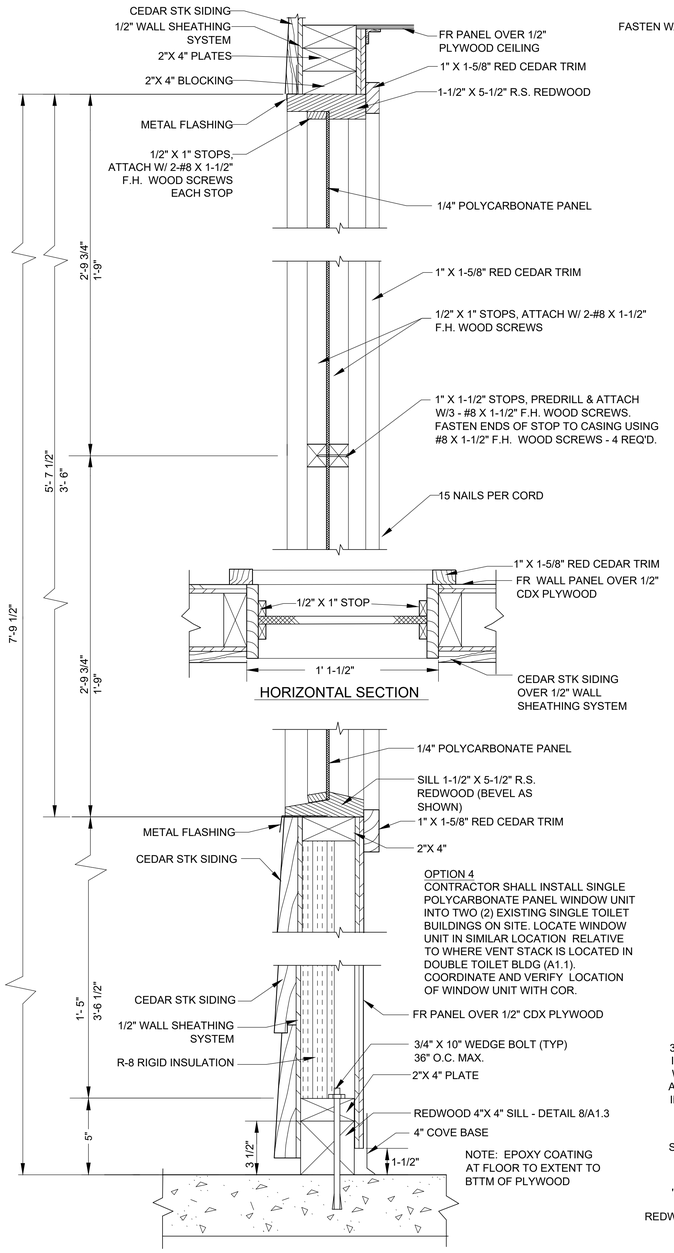
PROJECT NAME
GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
100% CDs
BLACK HILLS NATIONAL FOREST
HELL CANYON RANGER DISTRICT

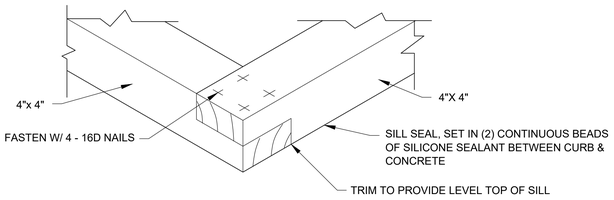
DRAWING TITLE
DOUBLE TOILET- DETAILS, MISC

ARCHIVE NO.

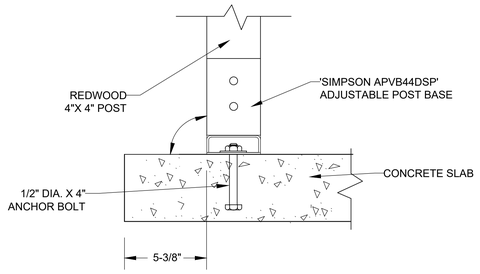
DESIGNER	SUB SHEET NO.
CHECKED	A1.2
DATE 05/28/2025	SHEET 23 OF 48



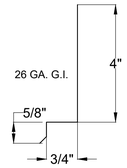
11 POLYCARBONATE PANEL WINDOW DETAILS
NOT TO SCALE



8 SILL DETAIL (TYP)
NOT TO SCALE



9 POST ATTACHMENT DETAIL
NOT TO SCALE



5 WINDOW FLASHING (TYP)
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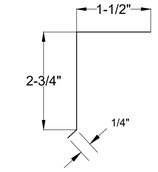
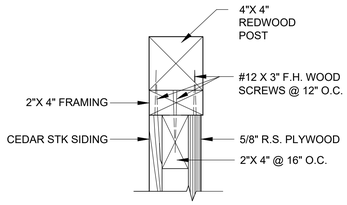
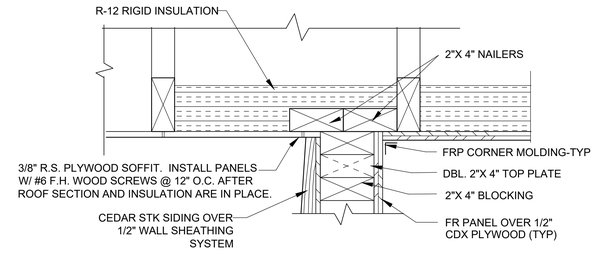


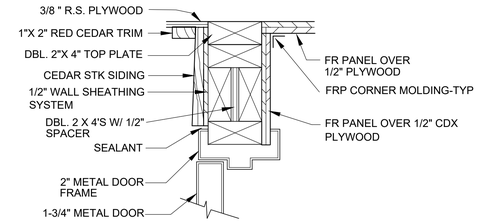
7 RAKE FLASHING (TYP)
NOT TO SCALE



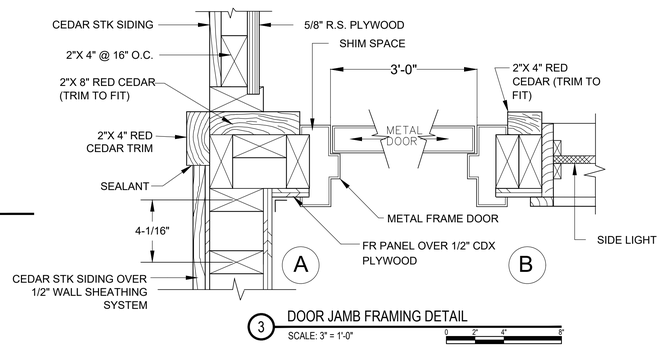
6 PRIVACY SCREEN AT POST DETAIL
SCALE: 3" = 1'-0"



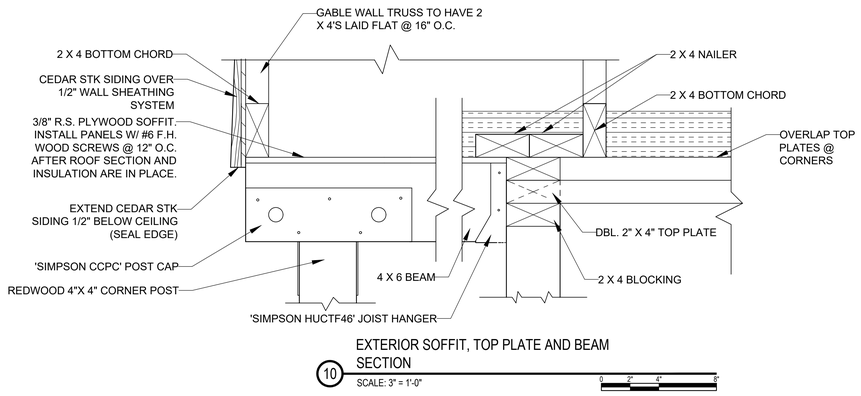
1 TOP PLATE SECTION
NOT TO SCALE



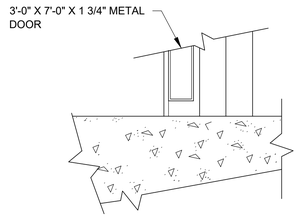
2 DOOR HEAD
SCALE: 3" = 1'-0"



3 DOOR JAMB FRAMING DETAIL
SCALE: 3" = 1'-0"



10 EXTERIOR SOFFIT, TOP PLATE AND BEAM SECTION
SCALE: 3" = 1'-0"



4 DOOR SILL DETAIL
NOT TO SCALE

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PROJECT NAME
GRIZZLY BEAR CAMPGROUND RECONSTRUCTION 100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE
DOUBLE TOILET-DETAILS, MISC

ARCHIVE NO.

DESIGNER
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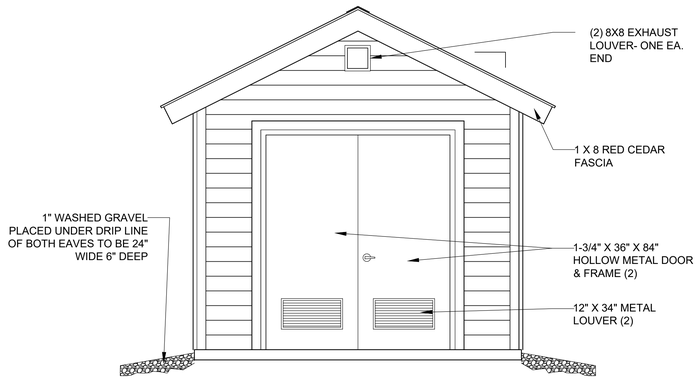
DATE 05/28/2025
SUB SHEET NO. A1.3
SHEET 24 OF 48

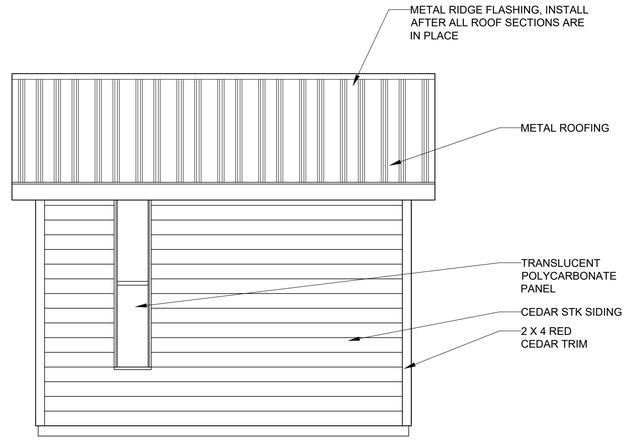
5/28/25 11:27 TODD C:\1TODD\1A-KHA WORK\201-023 BLACK HILLS USDA CAMPGROUNDS\DWGS\GRIZZLY_A1.2.DWG;

GENERAL NOTES:

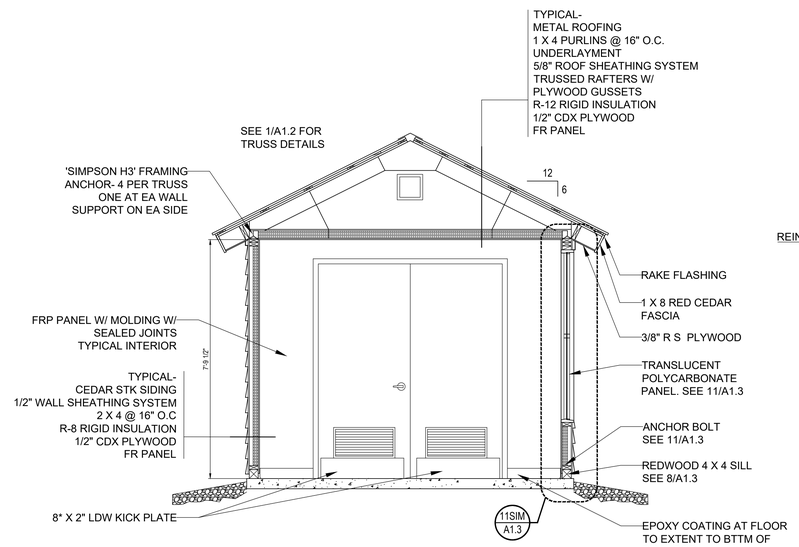
1. SIDING WILL BE 5/4 X 8 CEDAR LAP RABBITED SELECT TIGHT KNOT (STK) STAINED PER SPECIFICATIONS.
2. FRAMING DIMENSIONS FROM OUTSIDE FACE OF FRAMING.
3. (2) 8X8 EXHAUST LOUVERS PROVIDES 1/150 NET FREE AREA REQUIREMENT OF ATTIC ROOF VENTILATION PER 2024 IBC 1202.2.



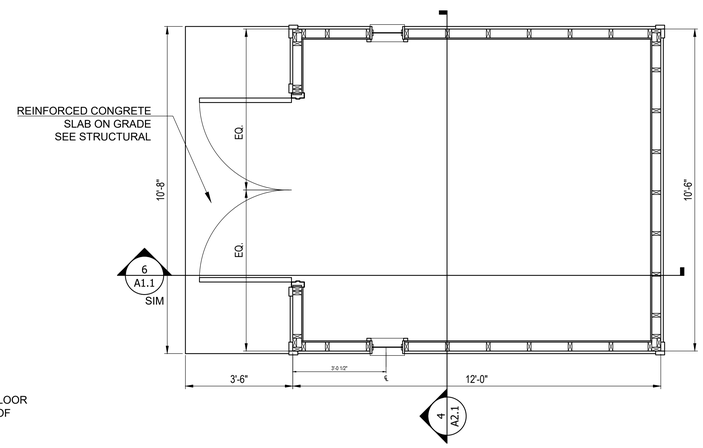
1 PUMP HOUSE- ELEVATION
SCALE: 1/2" = 1'-0"



2 PUMP HOUSE- ELEVATION
SCALE: 1/2" = 1'-0"



4 PUMP HOUSE SECTION
SCALE: 3/4" = 1'-0"



3 PUMP HOUSE- FLOOR PLAN
SCALE: 1/2" = 1'-0"

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
100% CDs

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

PUMP HOUSE- FLOOR PLAN, ELEVATIONS, SECTION

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
CHECKED	A2.1
DATE 05/28/2025	SHEET 25 OF 48

5/27/25 17:08 TODD C:\TODD\1A-KHA WORK\201-023 BLACK HILLS USDA CAMPGROUNDS\DWGS\GRIZZLY_A2.1.DWG

GENERAL CONCRETE NOTES

1. REFER TO ACI 318, BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE FOR CONCRETE CONSTRUCTION REQUIREMENTS
2. PROVIDE CONCRETE WITH A MINIMUM COMPRESSIVE STRENGTH OF 4000 PSI AT 28 DAYS. SUBMIT ENGINEERED CONCRETE MIX DESIGNS, INCLUDING BACKUP DATA FOR EACH TYPE OF CONCRETE PROPOSED FOR USE PRIOR TO THE START OF CONSTRUCTION. IDENTIFY WHERE EACH DIFFERENT CONCRETE MIX IS INTENDED TO BE USED.
3. PROVIDE AIR ENTRAINMENT PER ACI 318 FOR ALL EXTERIOR CONCRETE. LIMIT WATER CEMENT RATIO FOR ALL CONCRETE TO 0.45. FOR INTERIOR SLAB ON GROUND, LIMIT CEMENT CONTENT TO 620 POUNDS PER CUBIC YARD.
4. PROVIDE FOUR TEST CYLINDERS FOR EVERY 50 CUBIC YARDS, OR LESS (EACH DAY) AND FURNISH TEST REPORTS FOR 7, 14, AND 28 DAY COMPRESSION TESTS.
5. WATER MAY BE ADDED INITIALLY TO CONCRETE WHICH ARRIVES AT THE SITE IF AFTER ADDITION OF THE WATER THE MAXIMUM WATER/CEMENT RATIO IS NOT EXCEEDED. RETEMPERING AFTER INITIAL SLUMP ADJUSTMENT IS NOT PERMITTED.
6. FABRICATE AND PLACE REINFORCING STEEL IN ACCORDANCE WITH ACI 318 BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE, ACI SP-66 DETAILING MANUAL, ACI 301 SPECIFICATIONS FOR STRUCTURAL CONCRETE FOR BUILDINGS.
7. PROVIDE REINFORCING STEEL CONFORMING TO ASTM A615 GRADE 60. SUBMIT REINFORCING STEEL SHOP DRAWINGS FOR APPROVAL PRIOR TO THE START OF CONSTRUCTION.
8. PROVIDE ACI CLASS B REINFORCING BAR SPLICES BUT NOT LESS THAN 12 INCHES OF LAP.
9. PROVIDE CORNER REINFORCING BARS AT ALL EXTERIOR CORNERS OF CONCRETE WORK. CORNER BARS TO MATCH HORIZONTAL REINFORCING.
10. PROVIDE TWO #4 REBAR AT 4'-0" LONG (MINIMUM) AT ALL RE-ENTRANT CORNERS OF SLABS, FOOTINGS, AND WALLS. PLACE REBAR 2" CLEAR FROM THE CORNER AT 45°. SEE TYPICAL RE-ENTRANT CORNER DETAIL.
11. PROVIDE #4 REBAR REINFORCEMENT SPACED AT 12" O.C. CONFORMING TO ASTM A615 IN ALL SLABS ON GROUND UNLESS OTHERWISE NOTED.
12. PROVIDE THE FOLLOWING CONCRETE COVER IN ACCORDANCE WITH ACI 318:

CONCRETE CAST AGAINST EARTH OR FILL	3 IN
CONCRETE EXPOSED TO WEATHER OR IN CONTACT WITH THE GROUND	
NO 5 BAR AND SMALLER	1 1/2 IN
NO 6 BAR AND LARGER	2 IN
CONCRETE NOT EXPOSED TO WEATHER OR IN CONTACT WITH THE GROUND	
SLABS AND WALLS	3/4 IN
BEAMS AND COLUMNS	1 1/2 IN

13. PROVIDE ANCHOR BOLTS THAT CONFORM TO ASTM F1554 GRADE 55 SUPPLEMENT S1 WITH HEAVY HEX NUTS AND FLAT WASHERS, UNLESS NOTED OTHERWISE. HOT DIP GALVANIZE ALL ANCHOR BOLT ASSEMBLIES.
14. SET CAST-IN-PLACE ANCHOR BOLTS WITH TEMPLATES TO ELEVATIONS SHOWN ON DRAWINGS PRIOR TO CONCRETE BEING PLACED. "WET-STICKING" ANCHORS IS STRICTLY PROHIBITED.
15. VERIFY THAT ALL OPENINGS, INSERTS, ANCHOR BOLTS, EMBEDDED PIPING, REBAR, DOWELS, CONDUITS, GROUNDING WIRES, DUCTWORK, ETC. ARE IN PLACE BEFORE PLACING CONCRETE. "WET-STICKING" EMBEDMENTS IS STRICTLY PROHIBITED.
16. PROVIDE REINFORCING STEEL FREE OF OIL, MUD, HARDENED CONCRETE, LOOSE SCALE, AND RUST.
17. BRACE, TIE, AND SUPPORT REINFORCING WITH CHAIRS IN ACCORDANCE WITH ACI 315.
18. DO NOT WELD REINFORCING STEEL UNLESS INDICATED ON THE DRAWINGS.
19. PROVIDE 3/4" X 45° CHAMFER ON ALL EXPOSED EDGES.
20. PROVIDE 1/2" BITUMINOUS FIBER BOARD AT ALL SLAB ON GROUND EXPANSION JOINTS UNLESS NOTED OTHERWISE. TOOL ALL CONCRETE EDGES AND WITH THEM SEMI-RIGID EPOXY JOINT FILLER WITH 100% SOLIDS AND A TYPE A SHORE DUROMETER HARDNESS OF 80 ACCORDING TO ASTM D2240.
21. PROVIDE CONTRACTION JOINTS IN ALL SLABS ON GROUND WITH A MINIMUM DEPTH OF 1/3 OF THE CONCRETE THICKNESS AT THE LOCATIONS SHOWN ON THE DRAWING. TOOL JOINTS AFTER INITIAL FLOAT OR SAW-CUT WITHIN 8 HOURS OF PLACEMENT.
22. DO NOT USE CONCRETE CONSTRUCTION JOINTS UNLESS SHOWN ON DRAWINGS OR APPROVED IN WRITING BY THE ENGINEER.
23. USE CONTINUOUS WATERSTOP AT ALL HORIZONTAL AND VERTICAL CONCRETE JOINTS WHERE WATERSTOP IS INDICATED ON THE DRAWINGS. USE FACTORY MADE SPLICES AT INTERSECTING WALLS.
24. FOR BONDING NEW CONCRETE TO EXISTING CONCRETE, USE A TWO COMPONENT EPOXY BONDING ADHESIVE COMPLYING WITH ASTM C881 SUITABLE FOR THE APPLICATION TEMPERATURE. ROUGHEN AND CLEAN THE EXISTING CONCRETE AND APPLY IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS.
25. DO NOT PLACE CONCRETE INTO ANY FORM OR EXCAVATION CONTAINING TRASH, FREE WATER, ICE, OR FROZEN GROUND.
26. USE APPROPRIATE METHODS TO MINIMIZE FREE-FALL DISTANCE OF CONCRETE. DO NOT USE VIBRATORS TO TRANSPORT CONCRETE. DO NOT ALLOW CONCRETE TO FLOW HORIZONTALLY WITHIN FORMS OVER A DISTANCE EXCEEDING 5 FEET.
27. FINISH SLABS, PADS, AND RAMPS TO WITHIN 1/8 INCH OF ELEVATIONS SHOWN ON DRAWINGS.
28. PROVIDE A TROWELED FINISH ON TYPICAL INTERIOR SURFACES. PROVIDE A NON-SLIP BROOM FINISH ON TYPICAL EXTERIOR SURFACES. SCREED AND FLOAT CONCRETE AND USE A STIFF BRUSH OR BROOM TO PRODUCE A PROFILE OF 3/64 INCH IN ONE DIRECTION, PERPENDICULAR TO TRAFFIC IF APPLICABLE
29. PROVIDE A CLEAR, NON-YELLOWING, WATER BASED CURING AND SEALING COMPOUND ON FLOOR SLABS AND EXPOSED SURFACES UNLESS NOTED OTHERWISE.
30. REFER TO ACI 305 FOR REQUIREMENTS FOR PLACING CONCRETE IN HOT WEATHER AND TO ACI 306 FOR REQUIREMENTS FOR PLACING CONCRETE IN COLD WEATHER.
31. USE NON-SHRINK, NON-METALLIC, FACTORY PREMIXED CEMENTITIOUS GROUT WITH NO DRYING, SHRINKAGE, OR SETTLEMENT AT ANY AGE. COMPRESSIVE STRENGTH PER ASTM C109 OF NOT LESS THAN 5000 PSI AT 7 DAYS AND 7500 PSI AT 28 DAYS WHEN PLACED IN FLOWABLE CONSISTENCY. MIX AND PLACE PER MANUFACTURER'S INSTRUCTIONS.

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NO.	REVISION / ISSUE	DATE
6		
5		
4		
3		
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1		

PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

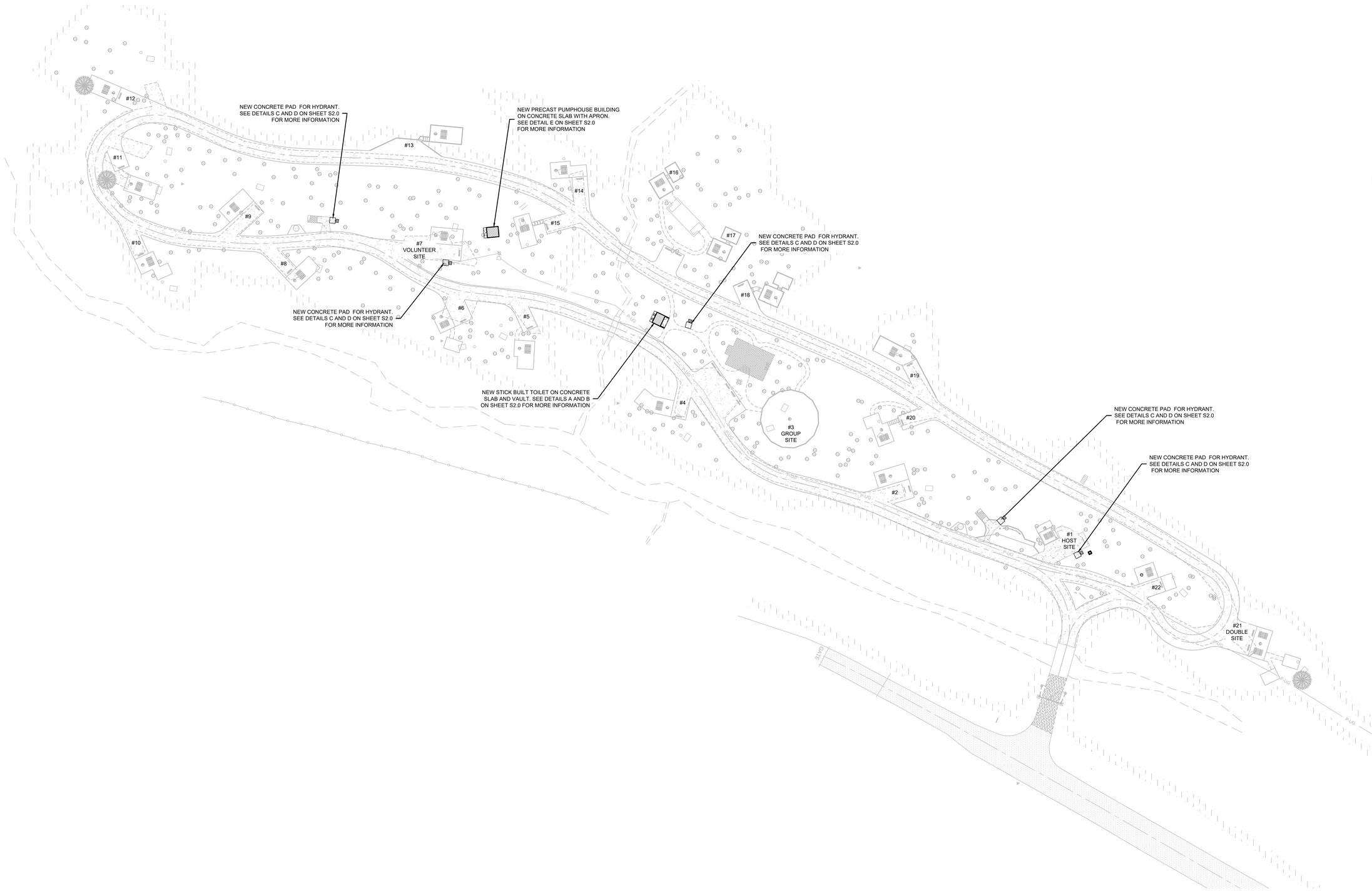
DRAWING TITLE

GENERAL CONCRETE NOTES

ARCHIVE NO.

DESIGNER TT	SUB SHEET NO. S0.1
CHECKED SR	
DATE 05.28.2025	SHEET 26 OF 48

5/27/25 18:43 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\STRUCTURAL\STRUCTURAL\GRIZZLY\S0.1.DWG



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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

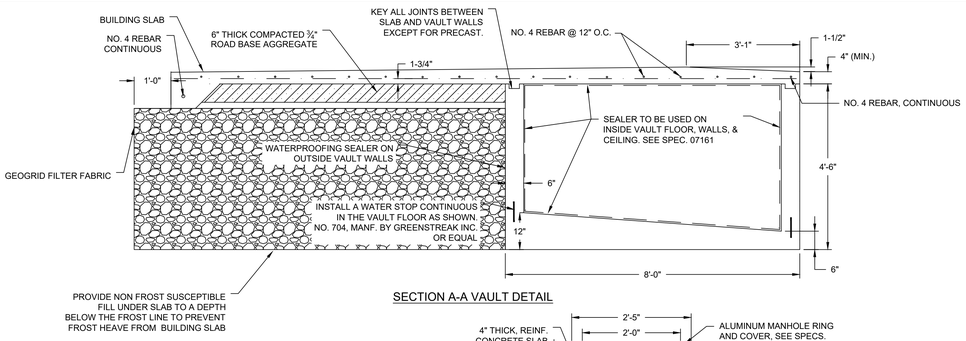
DRAWING TITLE

STRUCTURAL SITE PLAN

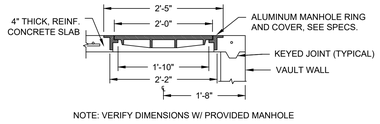
ARCHIVE NO.

DESIGNER	SUB SHEET NO.
TT	S1.1
CHECKED: SR	
DATE: 05.28.2025	SHEET 27 OF 48

5/28/25 01:48 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\STRUCTURAL\STRUCTURAL\GRIZZLY\S1.1.DWG

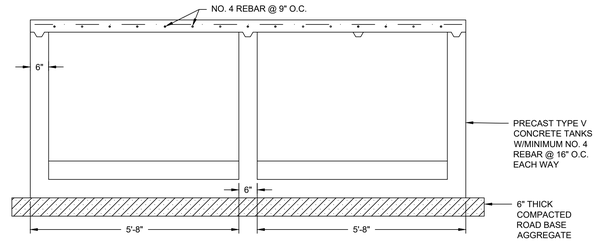


SECTION A-A VAULT DETAIL

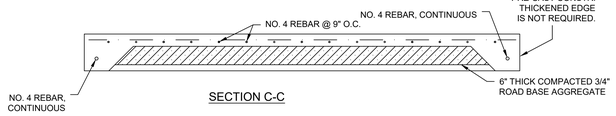


NOTE: VERIFY DIMENSIONS W/ PROVIDED MANHOLE

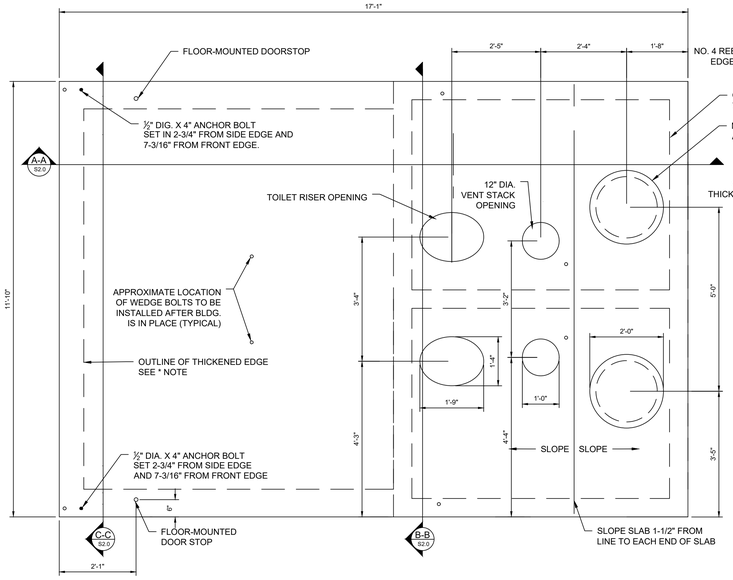
MANHOLE RING DETAIL



SECTION B-B

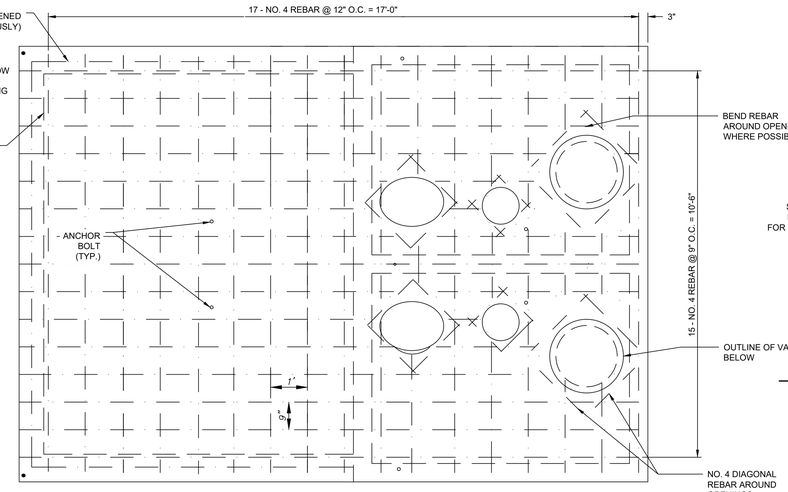


SECTION C-C



A S2.0 RESTROOM FOUNDATION PLAN

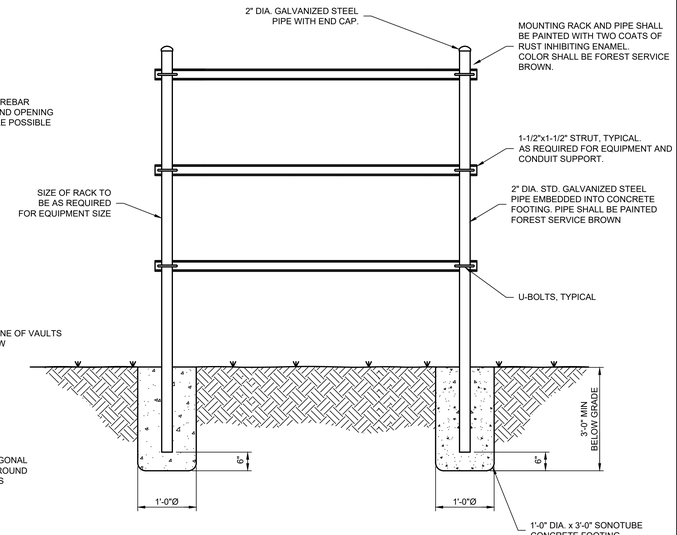
NO SCALE



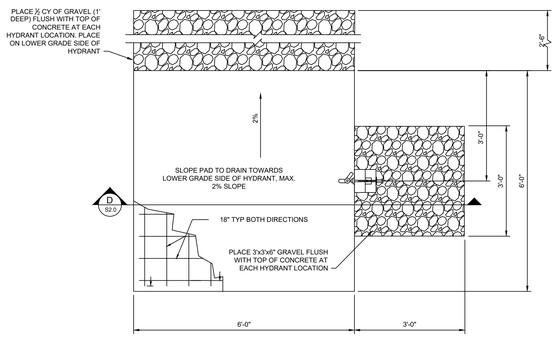
NOTE:
A PRECAST CONCRETE VAULT ENGINEERED BY A MANUFACTURER MAY BE SUBSTITUTED FOR THE CAST-IN-PLACE VAULT SHOWN IF APPROVED BY THE CONTRACTING OFFICER. IF PRE-CAST OPTION IS APPROVED, THE BUILDING SLAB (VAULT LID) SHALL BE ONE PIECE WITHOUT JOINTS.

B S2.0 REBAR PLAN

NO SCALE

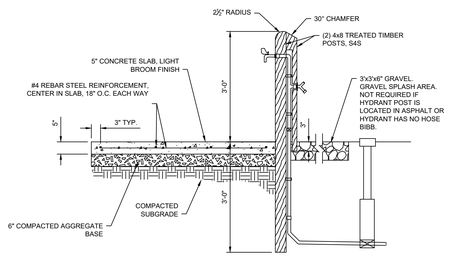


F S2.0 SECTION AT ELECTRICAL EQUIPMENT RACK



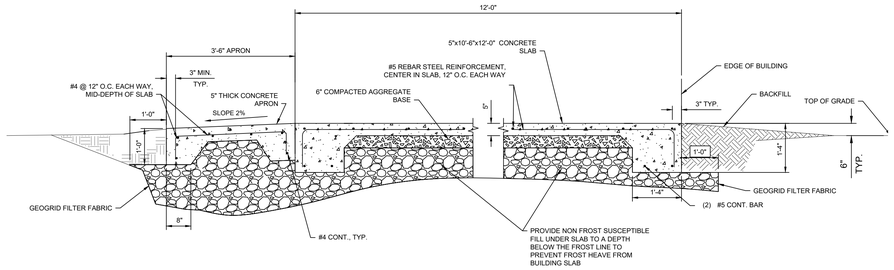
C S2.0 PAD AT HYDRANT DETAIL

NO SCALE



D S2.0 SECTION AT HYDRANT PAD

NO SCALE



E S2.0 SECTION AT PUMPHOUSE

NO SCALE

5/28/25 16:57 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\STRUCTURAL\STRUCTURAL\GRIZZLY\S2.0.DWG

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

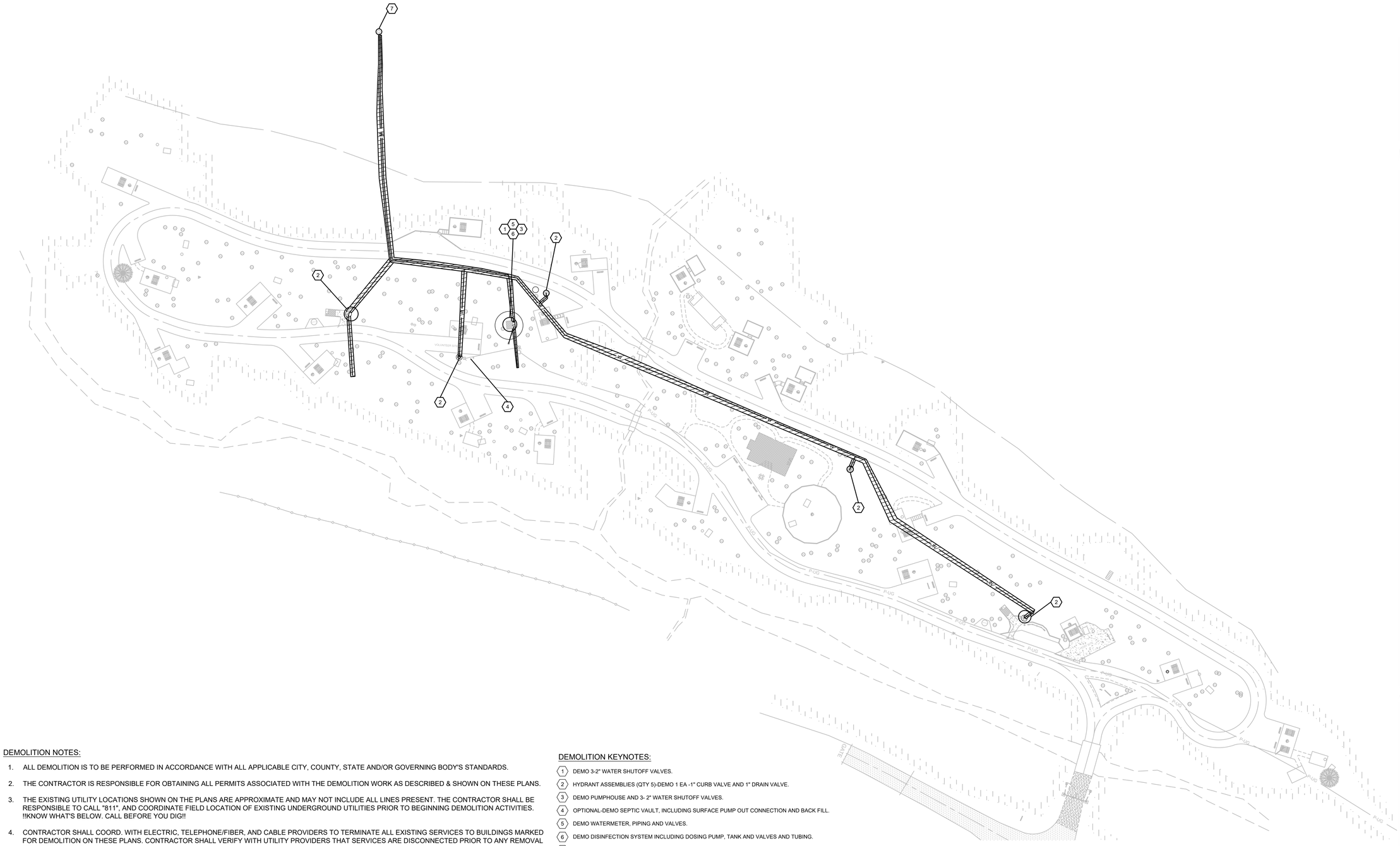
HELL CANYON RANGER DISTRICT

DRAWING TITLE

TYPICAL CONCRETE DETAILS

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
TT	S2.0
CHECKED	
JR	
DATE	SHEET
05.28.2025	28 OF 48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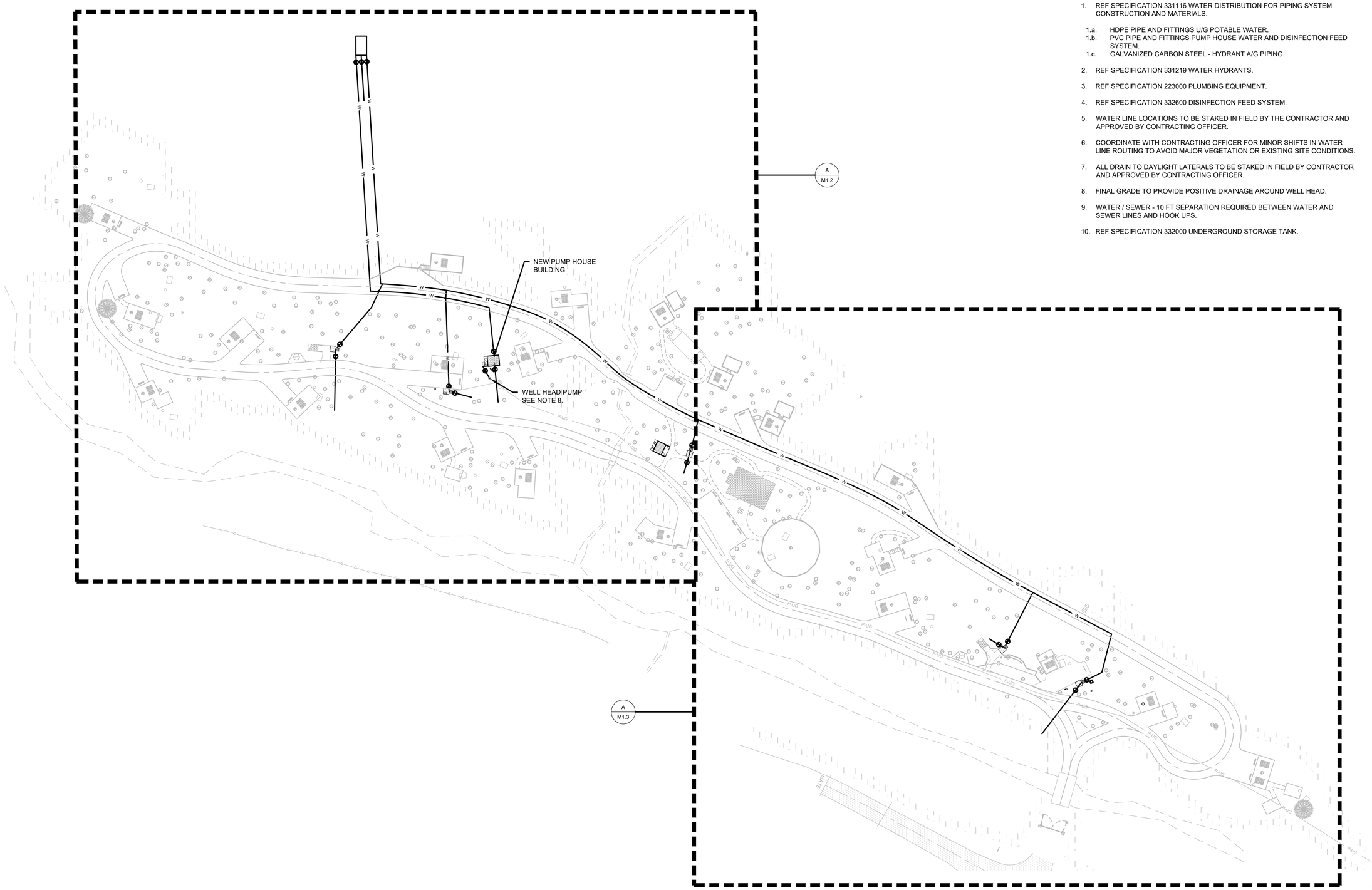
VOLUNTEER
PUG
PUG
PUG



GENERAL NOTES:

1. REF SPECIFICATION 331116 WATER DISTRIBUTION FOR PIPING SYSTEM CONSTRUCTION AND MATERIALS.
   - 1.a. HDPE PIPE AND FITTINGS U/G POTABLE WATER.
   - 1.b. PVC PIPE AND FITTINGS PUMP HOUSE WATER AND DISINFECTION FEED SYSTEM.
   - 1.c. GALVANIZED CARBON STEEL - HYDRANT A/G PIPING.
2. REF SPECIFICATION 331219 WATER HYDRANTS.
3. REF SPECIFICATION 223000 PLUMBING EQUIPMENT.
4. REF SPECIFICATION 332600 DISINFECTION FEED SYSTEM.
5. WATER LINE LOCATIONS TO BE STAKED IN FIELD BY THE CONTRACTOR AND APPROVED BY CONTRACTING OFFICER.
6. COORDINATE WITH CONTRACTING OFFICER FOR MINOR SHIFTS IN WATER LINE ROUTING TO AVOID MAJOR VEGETATION OR EXISTING SITE CONDITIONS.
7. ALL DRAIN TO DAYLIGHT LATERALS TO BE STAKED IN FIELD BY CONTRACTOR AND APPROVED BY CONTRACTING OFFICER.
8. FINAL GRADE TO PROVIDE POSITIVE DRAINAGE AROUND WELL HEAD.
9. WATER / SEWER - 10 FT SEPARATION REQUIRED BETWEEN WATER AND SEWER LINES AND HOOK UPS.
10. REF SPECIFICATION 332000 UNDERGROUND STORAGE TANK.

N

0' 20' 40' 80'
SCALE IN FEET

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

MECHANICAL/PLUMBING SITE PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
JR	M1.1
CHECKED: SR	
DATE: 05.28.2025	SHEET: 30 OF 48

5/28/25 01:48 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\MECH\WORKING CAD FILES\GRIZZLY\M1.1.DWG;

STORAGE TANK
N: 5770334.00
E: 1149138.39
EL: 4529

DRAIN
N: 577036.95
E: 1149185.79
EL: 4526

EXAMPLE SHUTOFF VALVE
MARKER POST
(SEE NOTE 12)

GENERAL NOTES:

1. REF SPECIFICATION 331116 WATER DISTRIBUTION FOR PIPING SYSTEM CONSTRUCTION AND MATERIALS.
   A. HDPE PIPE AND FITTINGS U/G POTABLE WATER.
   B. PVC PIPE AND FITTINGS PUMP HOUSE WATER AND DISINFECTION FEED SYSTEM.
   C. GALVANIZED CARBON STEEL - HYDRANT A/G PIPING.
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3. REF SPECIFICATION 223000 PLUMBING EQUIPMENT.
4. REF SPECIFICATION 332600 DISINFECTION FEED SYSTEM.
5. WATER LINE LOCATIONS TO BE STAKED IN FIELD BY THE CONTRACTOR AND APPROVED BY CONTRACTING OFFICER.
6. COORDINATE WITH CONTRACTING OFFICER FOR MINOR SHIFTS IN WATER LINE ROUTING TO AVOID MAJOR VEGETATION OR EXISTING SITE CONDITIONS.
7. ALL DRAIN TO DAYLIGHT LATERALS TO BE STAKED IN FIELD BY CONTRACTOR AND APPROVED BY CONTRACTING OFFICER.
8. UNLESS OTHERWISE NOTED:
   A. ALL MAIN RUNS ARE 2" HDPE SDR11 PIPE MATERIAL.
   B. ALL LATERALS TO HYDRANTS ARE 1" HDPE SDR11 PIPE MATERIAL.
9. SITE COORDINATES/ELEVATIONS ARE ESTIMATED AND SHOULD BE VERIFIED BY CONTRACTOR.
10. SEE SHEET M5.2 FOR HYDRANT DETAILS.
11. REF SPECIFICATION 332000 UNDERGROUND STORAGE TANK.
12. ALL FILL VALVES AND DRAIN VALVES TO BE MARKED WITH BROWN CARSONITE POST WITH 8" COLORED STICKER ON TOP. FILL VALVES TO BE MARKED RED AND DRAIN VALVES TO BE MARKED BLUE.

LEGEND:

- WATER PIPE
- HYDRANT
- COORDINATE CALLOUT WITH ELEVATION
- SEPTIC VAULT
- R — WATER SHUTOFF VALVE W/ CARSONITE POST (RED)
- B — WATER SHUTOFF VALVE W/ CARSONITE POST (BLUE)

N: 577056.09
E: 1149376.37
EL: 4537

N: 577122.54
E: 1149383.06
EL: 4535

N: 577167.09
E: 1149401.01
EL: 4535

H2 HYDRANT
N: 577007.52
E: 1149444.49
EL: 4529

NEW PUMP HOUSE

WELLHEAD PUMP

#7 VOLUNTEER SITE

LOW POINT DRAIN
N: 577176.58
E: 1149499.46
EL: 4526

DRAIN
N: 577140.56
E: 1149494.40
EL: 4526

H3 HYDRANT
N: 577125.46
E: 1149488.48
EL: 4527

DRAIN
N: 577006.43
E: 1149507.50
EL: 4520

H2 HYDRANT
N: 577376.64
E: 1149550.72
EL: 4523

DRAIN
N: 577369.82
E: 1149573.50
EL: 4520

#4 #5 #6 #8 #9 #10 #11 #12 #13 #14 #15 #16 #17

P-UG

MATCHLINE - SEE SHEET M1.3 FOR CONTINUATION

A / M1.2 ENLARGED SITE PLAN
1" = 20'

N

0' 10' 20' 40'
SCALE IN FEET

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PROJECT NAME
GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

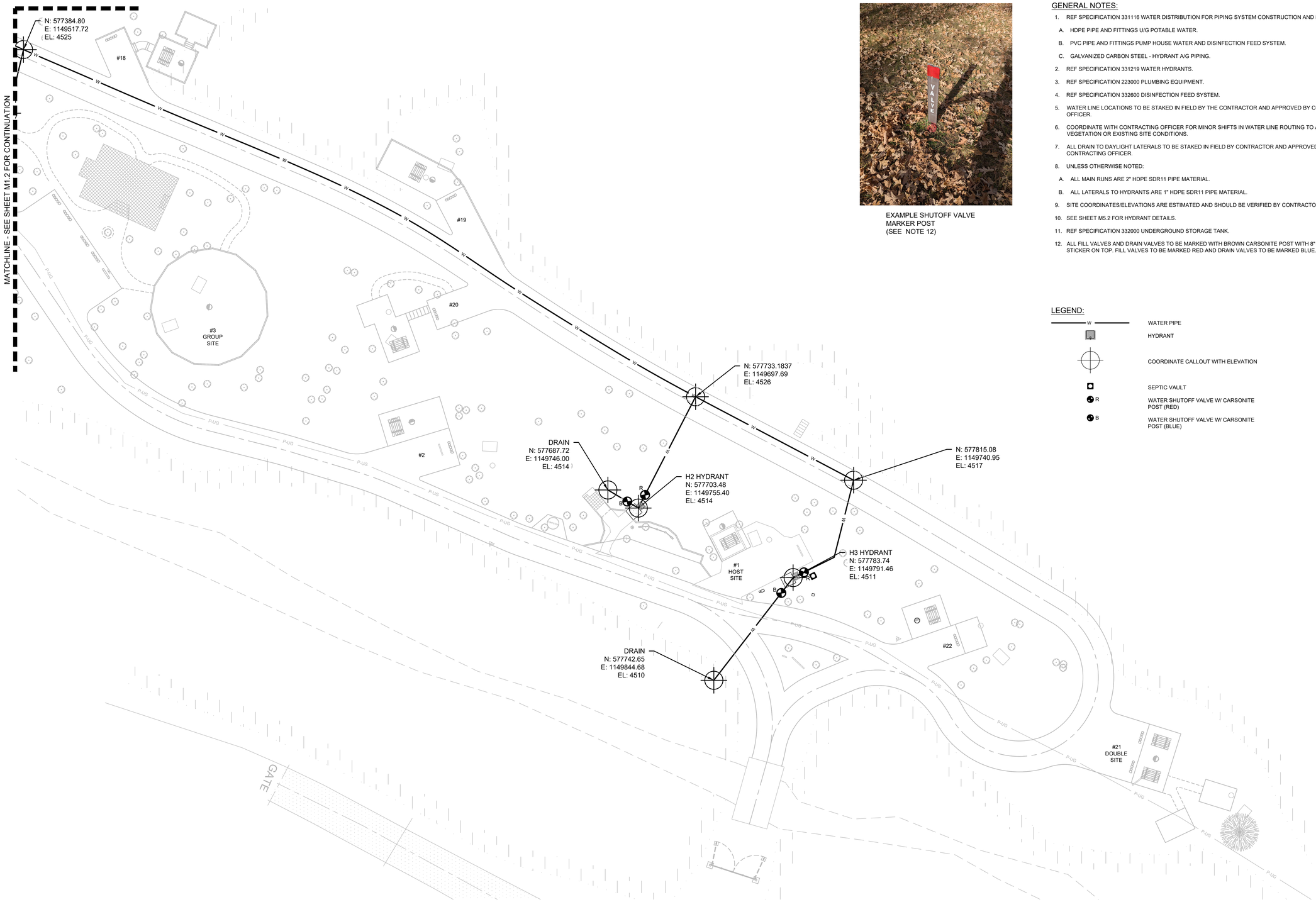
HELL CANYON RANGER DISTRICT

DRAWING TITLE
MECHANICAL/PLUMBING ENLARGED SITE PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
JR	M1.2
CHECKED: SR	
DATE: 05.28.2025	SHEET 31 OF 48

5/28/25 01:48 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\MECH\WORKING CAD FILES\GRIZZLY\M1.2.DWG



EXAMPLE SHUTOFF VALVE
MARKER POST
(SEE NOTE 12)

GENERAL NOTES:

1. REF SPECIFICATION 331116 WATER DISTRIBUTION FOR PIPING SYSTEM CONSTRUCTION AND MATERIALS.
   - A. HDPE PIPE AND FITTINGS U/G POTABLE WATER.
   - B. PVC PIPE AND FITTINGS PUMP HOUSE WATER AND DISINFECTION FEED SYSTEM.
   - C. GALVANIZED CARBON STEEL - HYDRANT A/G PIPING.
2. REF SPECIFICATION 331219 WATER HYDRANTS.
3. REF SPECIFICATION 223000 PLUMBING EQUIPMENT.
4. REF SPECIFICATION 332600 DISINFECTION FEED SYSTEM.
5. WATER LINE LOCATIONS TO BE STAKED IN FIELD BY THE CONTRACTOR AND APPROVED BY CONTRACTING OFFICER.
6. COORDINATE WITH CONTRACTING OFFICER FOR MINOR SHIFTS IN WATER LINE ROUTING TO AVOID MAJOR VEGETATION OR EXISTING SITE CONDITIONS.
7. ALL DRAIN TO DAYLIGHT LATERALS TO BE STAKED IN FIELD BY CONTRACTOR AND APPROVED BY CONTRACTING OFFICER.
8. UNLESS OTHERWISE NOTED:
   - A. ALL MAIN RUNS ARE 2" HDPE SDR11 PIPE MATERIAL.
   - B. ALL LATERALS TO HYDRANTS ARE 1" HDPE SDR11 PIPE MATERIAL.
9. SITE COORDINATES/ELEVATIONS ARE ESTIMATED AND SHOULD BE VERIFIED BY CONTRACTOR.
10. SEE SHEET M5.2 FOR HYDRANT DETAILS.
11. REF SPECIFICATION 332000 UNDERGROUND STORAGE TANK.
12. ALL FILL VALVES AND DRAIN VALVES TO BE MARKED WITH BROWN CARSONITE POST WITH 8" COLORED STICKER ON TOP. FILL VALVES TO BE MARKED RED AND DRAIN VALVES TO BE MARKED BLUE.

LEGEND:

- WATER PIPE
- HYDRANT
- COORDINATE CALLOUT WITH ELEVATION
- SEPTIC VAULT
- R — WATER SHUTOFF VALVE W/ CARSONITE POST (RED)
- B — WATER SHUTOFF VALVE W/ CARSONITE POST (BLUE)

A / M1.3 — ENLARGED SITE PLAN
1" = 20'

N

0' 10' 20' 40'
SCALE IN FEET

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

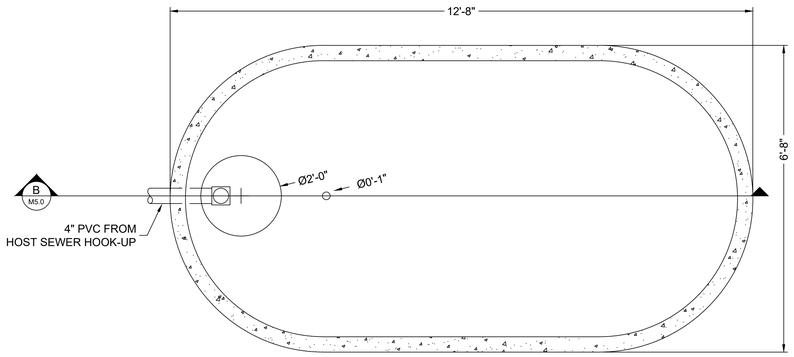
DRAWING TITLE

MECHANICAL/PLUMBING ENLARGED SITE PLAN

ARCHIVE NO.

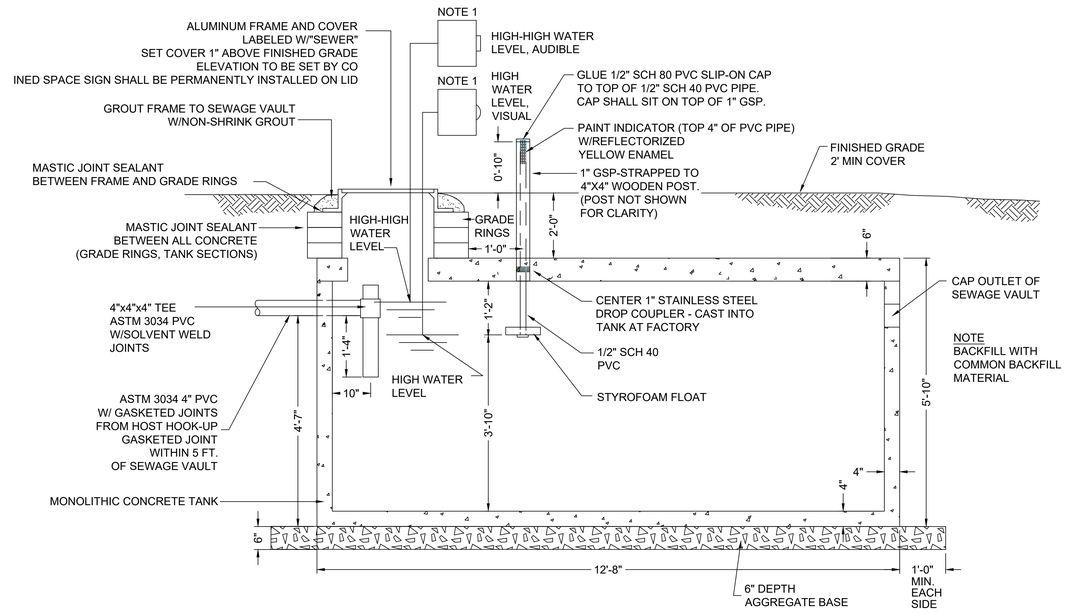
DESIGNER JR	SUB SHEET NO. M1.3
CHECKED SR	
DATE 05.28.2025	SHEET 32 OF 48

5/28/25 01:48 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\MECH\WORKING CAD FILES\GRIZZLY\M1.2.DWG;



A M5.0 2000 GALLON PRE CAST CONCRETE SEWAGE VAULT PLAN

NO SCALE

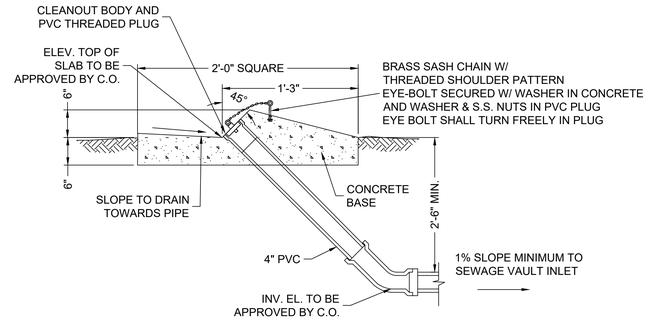


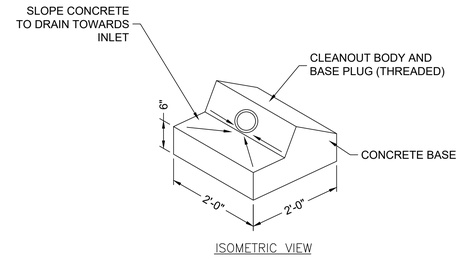
B M5.0 2000 GALLON PRE CAST CONCRETE SEWAGE VAULT SECTION

NO SCALE

NOTES

1. INDICATORS SHALL BE IN FULL VIEW OF HOST SITE.
   WITH BOTH VISUAL AND AUDIBLE ALARMS.
   - FOR HIGH WATER LEVEL (VISUAL ALARM)
   - FOR HIGH-HIGH WATER LEVEL (AUDIBLE ALARM).
   - EXISTING MECH FLOAT ASSY SHALL BE ABANDONED IN PLACE.

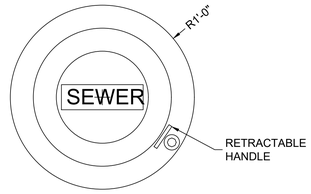




ISOMETRIC VIEW

C M5.0 HOST SEWER HOOK-UP

NO SCALE



D M5.0 SEWAGE VAULT COVER DETAIL

NO SCALE

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NO.	REVISION / ISSUE	DATE
6		
5		
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3		
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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

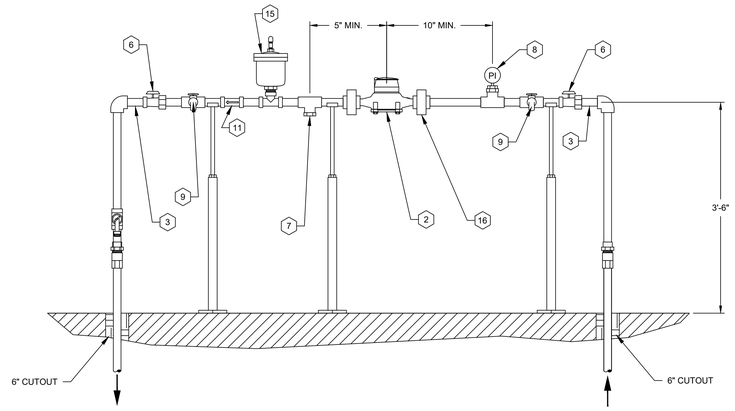
DRAWING TITLE

MECHANICAL DETAILS EXISTING SEWAGE VAULT

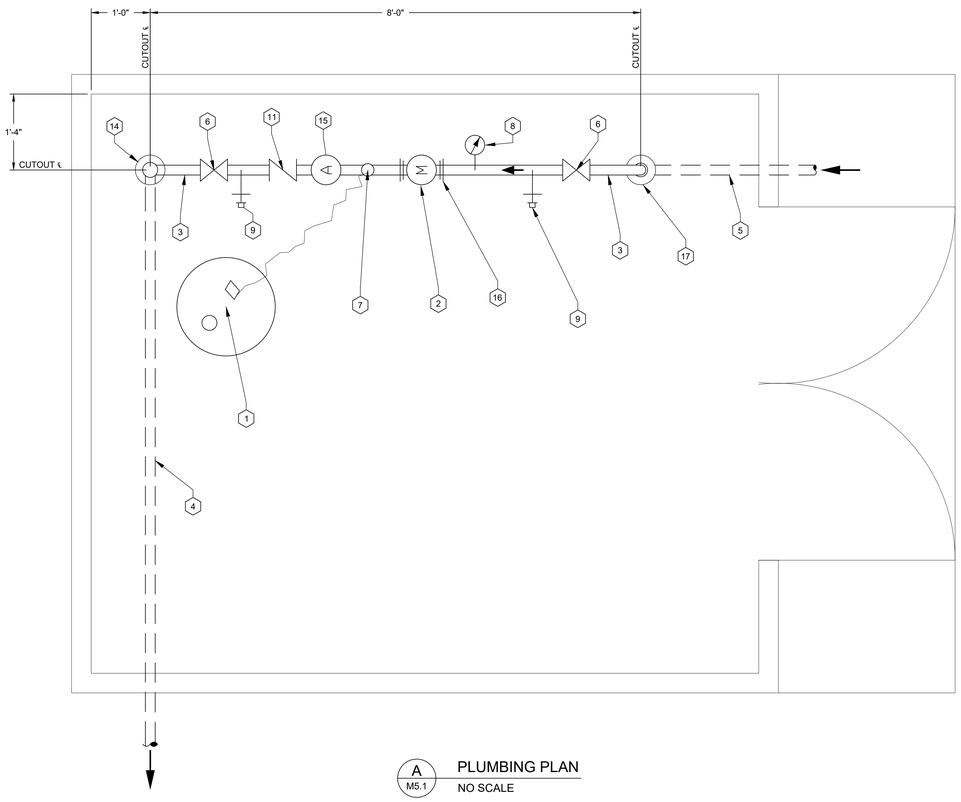
ARCHIVE NO.

DESIGNER	SUB SHEET NO.
JR	M5.0
CHECKED: SR	
DATE: 05.28.2025	SHEET: 33 OF 48

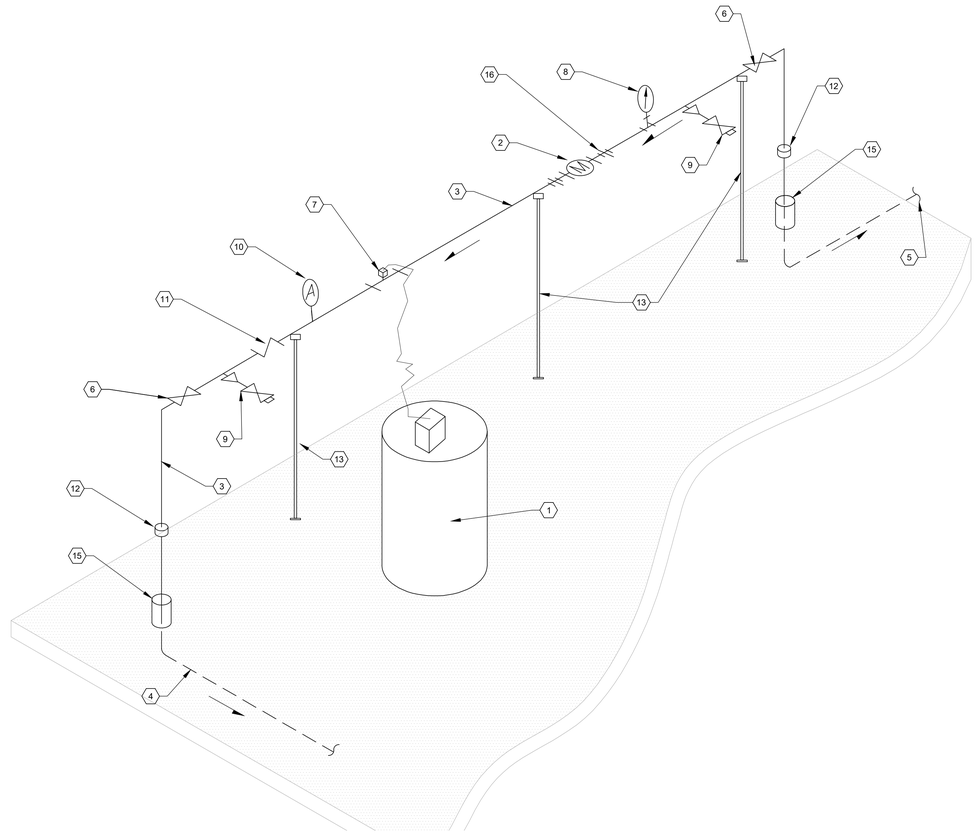
5/27/25 21:58 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\MECH\WORKING CAD FILES\GRIZZLY\M5.0_HOST_TANK_DETAIL.DWG;



C M5.1 PLUMBING DETAIL
NO SCALE



A M5.1 PLUMBING PLAN
NO SCALE



B M5.1 PLUMBING ISOMETRIC
NO SCALE

KEYED NOTES: ⟨XX⟩

1. CHLORINE SOLUTION TANK WITH CHEMICAL METERING PUMP
2. 1" WATER METER
3. 1" SCH 80 PVC
4. 1" HDPE TO STORAGE TANK
5. 1" HDPE FROM WELL
6. 1" PVC BALL VALVE
7. CHEMICAL INJECTOR FITTING TO BE INSTALLED IN VERTICAL POSITION INTO BOTTOM OF PIPE
8. PRESSURE GAGE
9. SMOOTH NOZZLE SAMPLE SPIGOT
10. AIR RELIEF VALVE
11. BRONZE CHECK VALVE
12. HDPE TO PVC COUPLER
13. PIPE SUPPORT
14. 6" DIA. HOLE IN FLOOR SLAB, FILL ANNULAR SPACE WITH FOAM, TYP.
15. 6" PIPE SLEEVE THROUGH CONCRETE SLAB (TYP.) FILL ANNULAR SPACE WITH FOAM
16. UNION, TYP.
17. 6" DIA HOLE IN FLOOR SLAB FOR 1" HDPE PIPE AND 1" ELECTRICAL CONDUIT, FILL ANNULAR SPACE WITH FOAM, TYP.

KEY:

SLOPE PIPES TO DRAIN IN DIRECTION OF ARROWS, TYP.

DIRECTION OF FLOW

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

MECHANICAL DETAILS PUMPHOUSE WATER METER AND DISINFECTION PIPING

ARCHIVE NO.

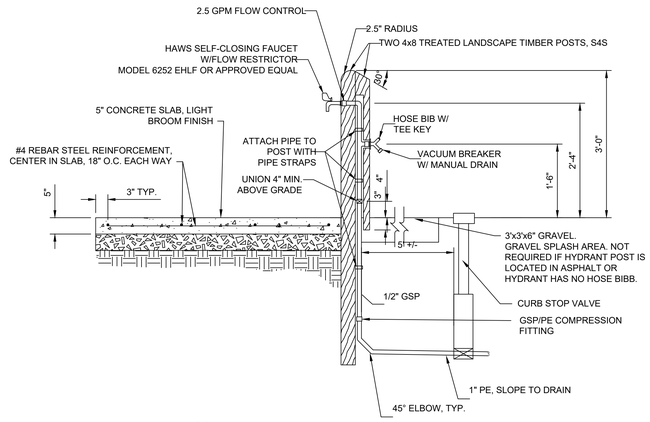
DESIGNER	SUB SHEET NO.
JR	M5.1
CHECKED: SR	
DATE: 05.28.2025	SHEET 34 OF 48

5/27/25 19:09 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\MECH\WORKING CAD FILES\GRIZZLY\M5.1_PUMPHOUSE_DETAILDWG;

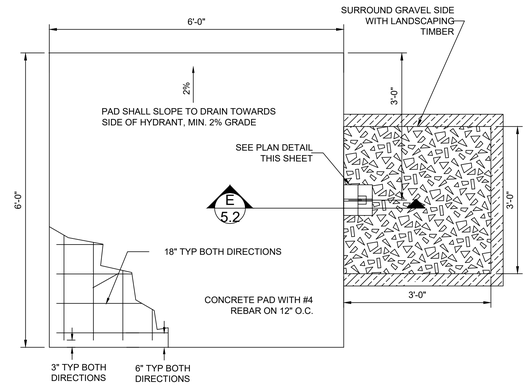
HYDRANT FIXTURES: (XX)

(A) 1/2" SELF-CLOSING FAUCET WITH FLOW RESTRICTOR, MODEL 6252 EHLF OR APPROVED EQUAL.

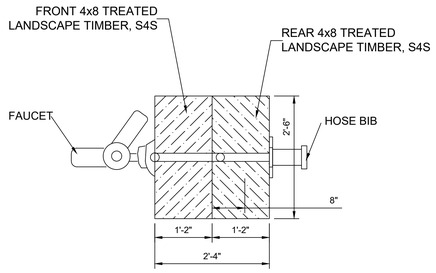
(B) HOSE BIBB WITH REMOVABLE TEE KEY AND VACUUM BREAKER WITH MANUAL DRAIN.



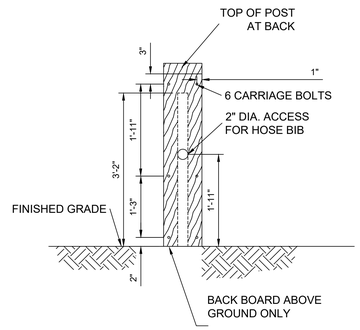
A 5.2 SECTION OF CAMPGROUND HYDRANT



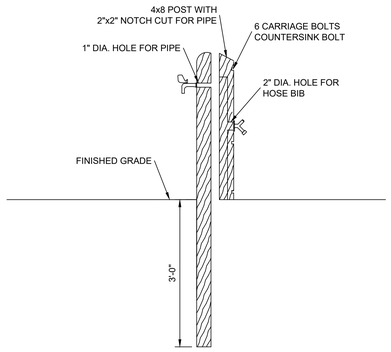
B 5.2 CONCRETE HYDRANT PAD



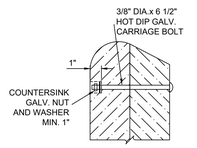
C 5.2 CAMPGROUND HYDRANT PLAN DETAIL



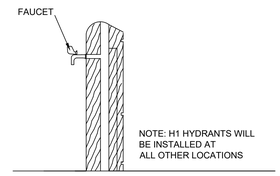
D 5.2 ROUTED PIPE CASE



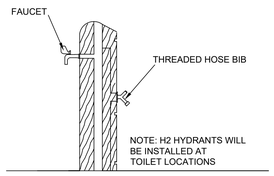
E 5.2 CAMPGROUND HYDRANT SIDE VIEW



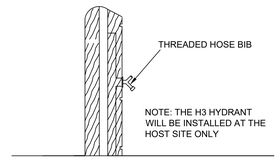
F 5.2 POST BOLT DETAIL



G 5.2 H1 HYDRANT DETAIL



H 5.2 H2 HYDRANT DETAIL



I 5.2 H3 HYDRANT DETAIL

TYPE	DESCRIPTION	LOCATION	QUANTITY
H2	HAWS SELF CLOSING FAUCET, THREADED HOSE BIBB AND VACUUM BREAKER	ALL TOILETS	3
H3	HOSE BIBB WITH TEE KEY AND VACUUM BREAKER	HOST & VOLUNTEER SITES	2

HYDRANT NOTES:

THREE DIFFERENT TYPES OF HYDRANTS WILL BE USED.

H1 HAWS SELF CLOSING FAUCET ONLY.
H2 HAWS SELF CLOSING FAUCET AND THREADED HOSE. BIB WITH TEE KEY INCLUDING VACUUM BREAKER.
H3 HOSE BIB WITH TEE KEY AND VACUUM BREAKER.

H2 HYDRANTS WILL BE PLACED AT ALL TOILET LOCATIONS, H3 HYDRANTS WILL BE PLACED ONLY AT THE HOST SITE, AND THE H1 HYDRANTS WILL BE USED AT ALL OTHER HYDRANT LOCATIONS.

EXACT LOCATIONS OF HYDRANTS WILL BE STAKED IN THE FIELD.

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6		
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4		
3		
2		
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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

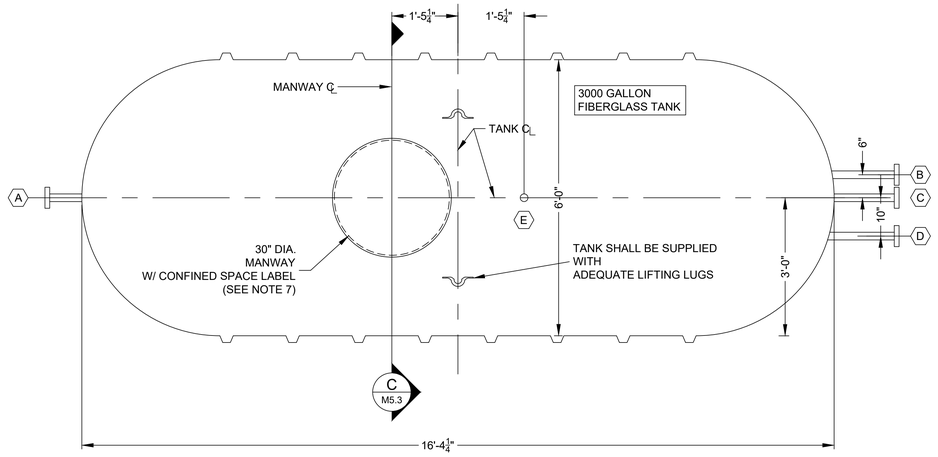
DRAWING TITLE

MECHANICAL DETAILS HYDRANT DETAILS

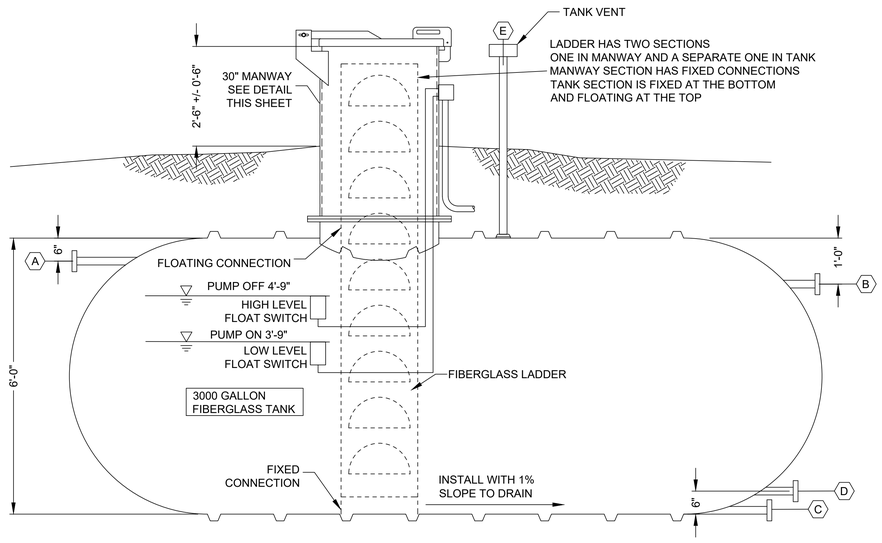
ARCHIVE NO.

DESIGNER	SUB SHEET NO.
JR	M5.2
CHECKED: SR	
DATE: 05.28.2025	SHEET 35 OF 48

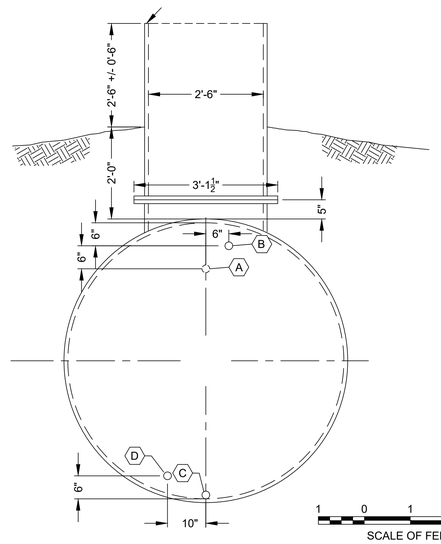
5/28/25 - 7:21 - BRIAN C:\USERS\BRIAN\DESKTOP\024-0148 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\MECH WORKING CAD FILES\GRIZZLY\M5.2_HYDRANT_DETAILS.DWG



A M5.3 PLAN VIEW
3/4" = 1'-0"



B M5.3 ELEVATION VIEW
3/4" = 1'-0"



C M5.3 ELEVATION SECTION
3/4" = 1'-0"

NOZZLE TABLE

NOZZLE	SIZE	DESCRIPTION
A FROM TREATMENT BUILDING (INLET)	2"	FLANGED AND GUSSETED HORIZONTAL NOZZLE
B OVERFLOW	2"	FLANGED AND GUSSETED HORIZONTAL NOZZLE
C TANK DRAIN	2"	FLANGED AND GUSSETED TANGENTIAL NOZZLE WITH BOTTOM SUPPORT
D DISTRIBUTION LINE (OUTLET)	2"	FLANGED AND GUSSETED HORIZONTAL NOZZLE
E VENT	2"	FEMALE NPT SERVICE FITTING

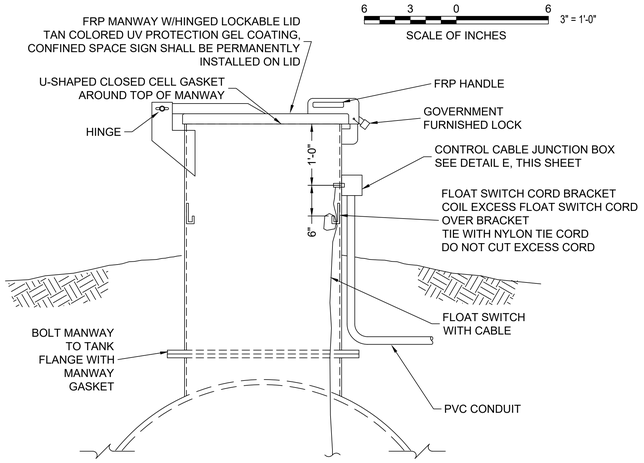
STORAGE TANK REQUIREMENTS

1. MINIMUM USEABLE VOLUME = 3000 GALLONS
2. INTERNAL DESIGN PRESSURE = 5 PSI, VENT TO ATMOSPHERE
3. EXTERNAL DESIGN PRESSURE: THE TANK SHALL BE DESIGNED TO WITHSTAND BURIAL IN SEVEN FEET OF WATER SATURATED SOIL OVERBURDEN.
4. LADDER SHALL BE ATTACHED ON THE BOTTOM BY A PIVOT AND ALLOWED TO FLOAT IN A BRACKET NEAR THE TOP.
5. USE STAINLESS STEEL HARDWARE (BOLTS, NUTS, WASHERS, ETC.) FOR CONNECTIONS TO TANK AND MANWAY.
6. HOLES DRILLED THROUGH MANWAY SHALL BE MADE WATERPROOF WITH NSF APPROVED CAULKING OR RUBBER SEALS.
7. MANWAY LID SHALL BE LABLED "DANGER CONFINED SPACE ENTER BY PERMIT ONLY".

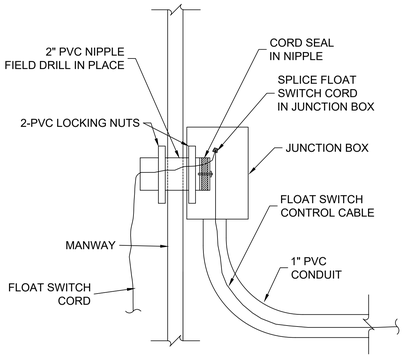
1 0 1 2 3 3/4" = 1'-0"
SCALE OF FEET

1 0 1 2 1"=1'-0"
SCALE OF FEET

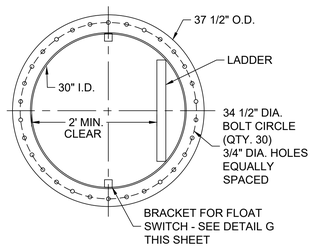
6 3 0 6 3" = 1'-0"
SCALE OF INCHES



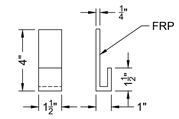
D M5.3 MANWAY DETAIL
1" = 1'-0"



E M5.3 JUNCTION BOX DETAIL
3" = 1'-0"



F M5.3 MANWAY FLANGE DETAIL
1" = 1'-0"



NOTE:
WELD BRACKETS TO
WALL OF FIBERGLASS
MANWAY

G M5.3 BRACKET DETAIL
3" = 1'-0"

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6		
5		
4		
3		
2		
1		

PROJECT NAME
GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
BLACK HILLS NATIONAL FOREST
HELL CANYON RANGER DISTRICT

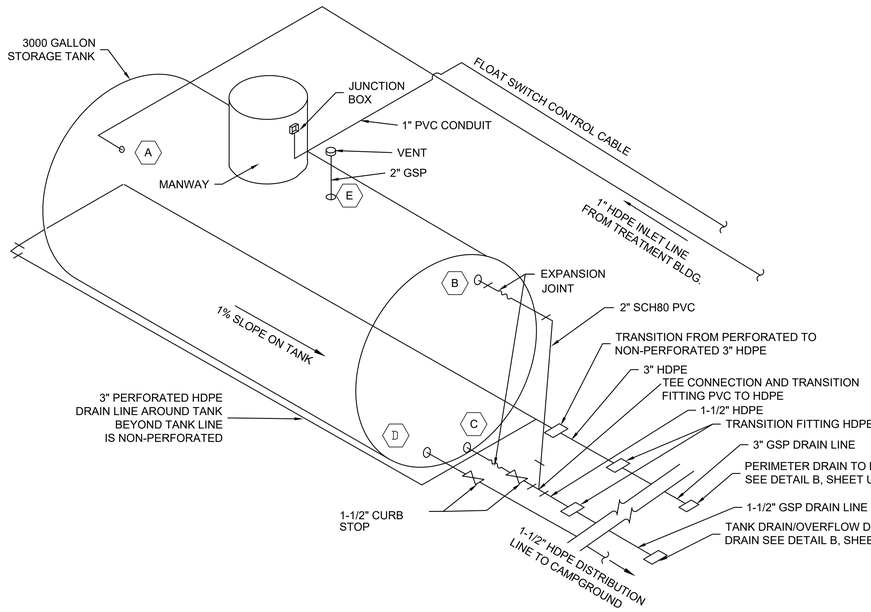
DRAWING TITLE
MECHANICAL DETAILS WATER STORAGE TANK DETAIL

ARCHIVE NO.

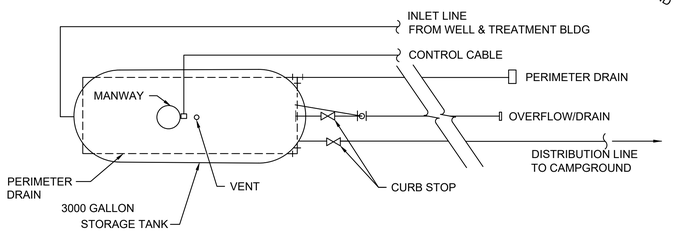
DESIGNER	SUB SHEET NO.
JR	M5.3
CHECKED: SR	
DATE: 05.28.2025	SHEET 36 OF 48

5/27/25 19:10 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\MECH\WORKING CAD FILES\GRIZZLY\M5.3_STORAGE_TANK_DETAIL.DWG;

TANK CONNECTION	TYPE	FITTINGS FOR TANK CONNECTION
(A) FROM TREATMENT BUILDING (INLET)	2" FLANGE	2" FLANGE x 1" HDPE FUSION FITTING + 2" x 1" REDUCTION COMPRESSION COUPLING TO 1" HDPE
(B) OVERFLOW	2" FLANGE	2" FLANGE x 2" FLANGE EXP JOINT + 2" FLANGED x FIPT SCH 80 PVC TO 1-1/2" SCH 80 PVC
(C) TANK DRAIN	2" FLANGE	2" FLANGE x 2" FLANGE EXP JOINT + 2" FLANGED x FIPT SCH 80 PVC TO 1-1/2" SCH 80 PVC
(D) DISTRIBUTION LINE (OUTLET)	2" FLANGE	2" FLANGE x 2" HDPE FUSION FITTING TO 1-1/2" HDPE
(E) VENT	2" NPT	THREAD 2" GSP INTO TANK CONNECTION



A M5.4 ISOMETRIC VIEW
NTS

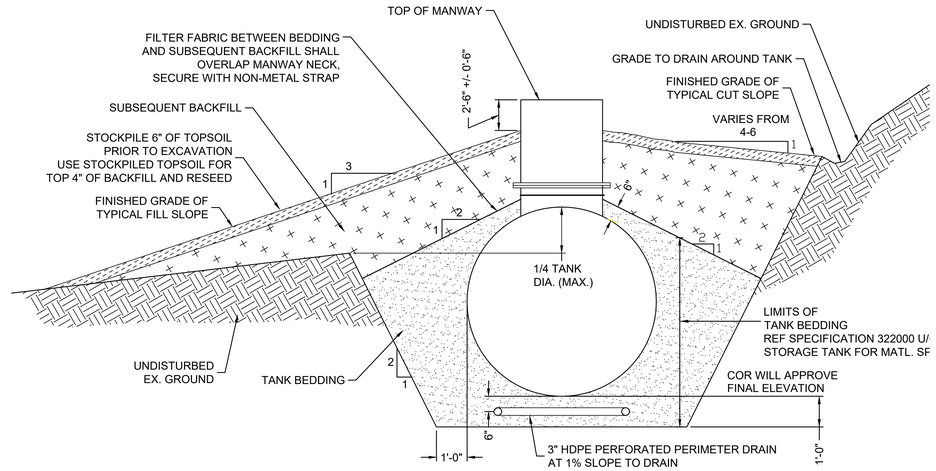


NOTES

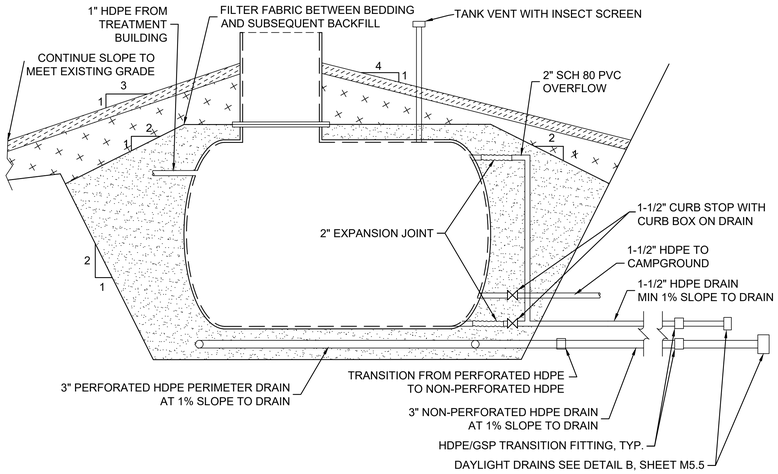
1. ALL PIPING SHALL BE SLOPED TO DRAIN
2. CONTRACTING OFFICER WILL APPROVE TANK LOCATION AND ELEVATION

PIPING LEGEND

- CURB STOP
- UNION
- 90 ELBOW
- TEE



B M5.4 ELEVATION VIEW
NTS



C M5.4 ELEVATION VIEW
NTS

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PROJECT NAME
GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

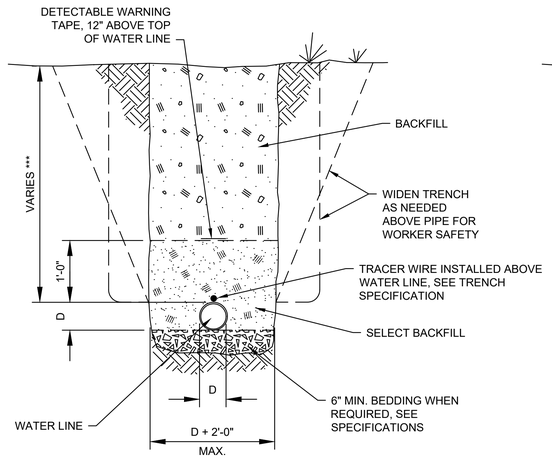
HELL CANYON RANGER DISTRICT

DRAWING TITLE
MECHANICAL DETAILS STORAGE TANK INSTALLATION DETAILS

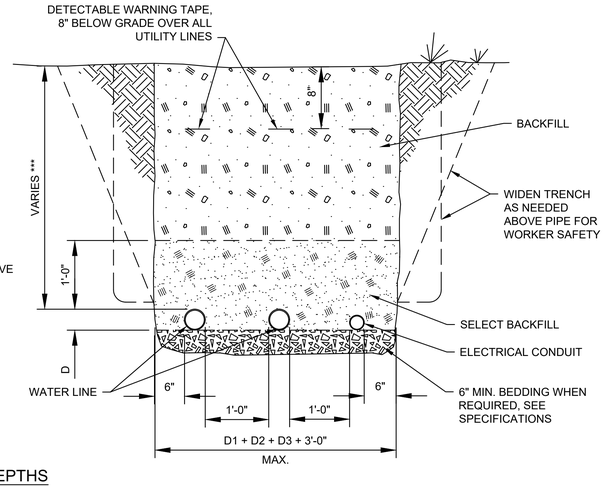
ARCHIVE NO.

DESIGNER	SUB SHEET NO.
JR	M5.4
CHECKED: SR	
DATE: 05.28.2025	SHEET 37 OF 48

5/27/25 19:11 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\MECH\WORKING CAD FILES\GRIZZLY\M5.4_STORAGE_TANK_INSTALLATION.DWG;



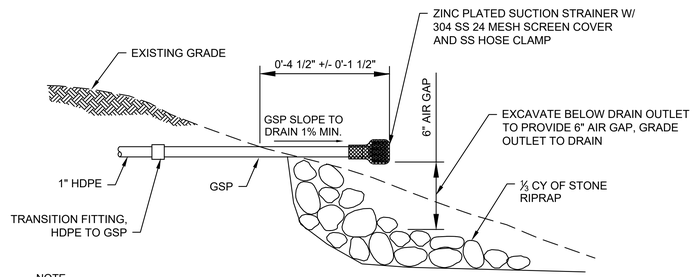
A M5.5 TYPICAL SINGLE UTILITY TRENCH
NO SCALE



UTILITY LINE DEPTHS

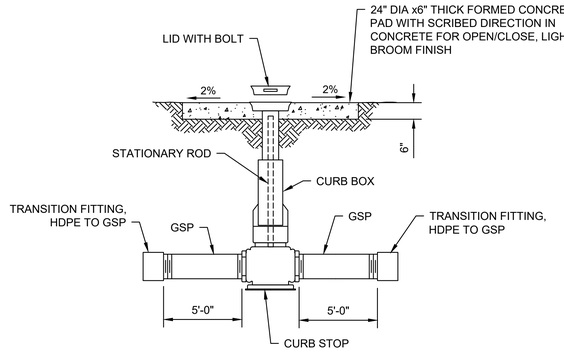
*** 2'-0" MIN. COVER OVER WATER LINE
2'-0" MIN. COVER OVER ELECTRICAL

B M5.5 TYPICAL MULTIPLE UTILITY TRENCH
NO SCALE



NOTE
THE LAST 10 FT. OF DRAIN LINE SHALL BE GALVANIZED STEEL PIPE

C U3.1 DAYLIGHT DRAIN DETAIL
NO SCALE



D M5.5 CURB STOP (VALVES 2" AND SMALLER)
NO SCALE

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

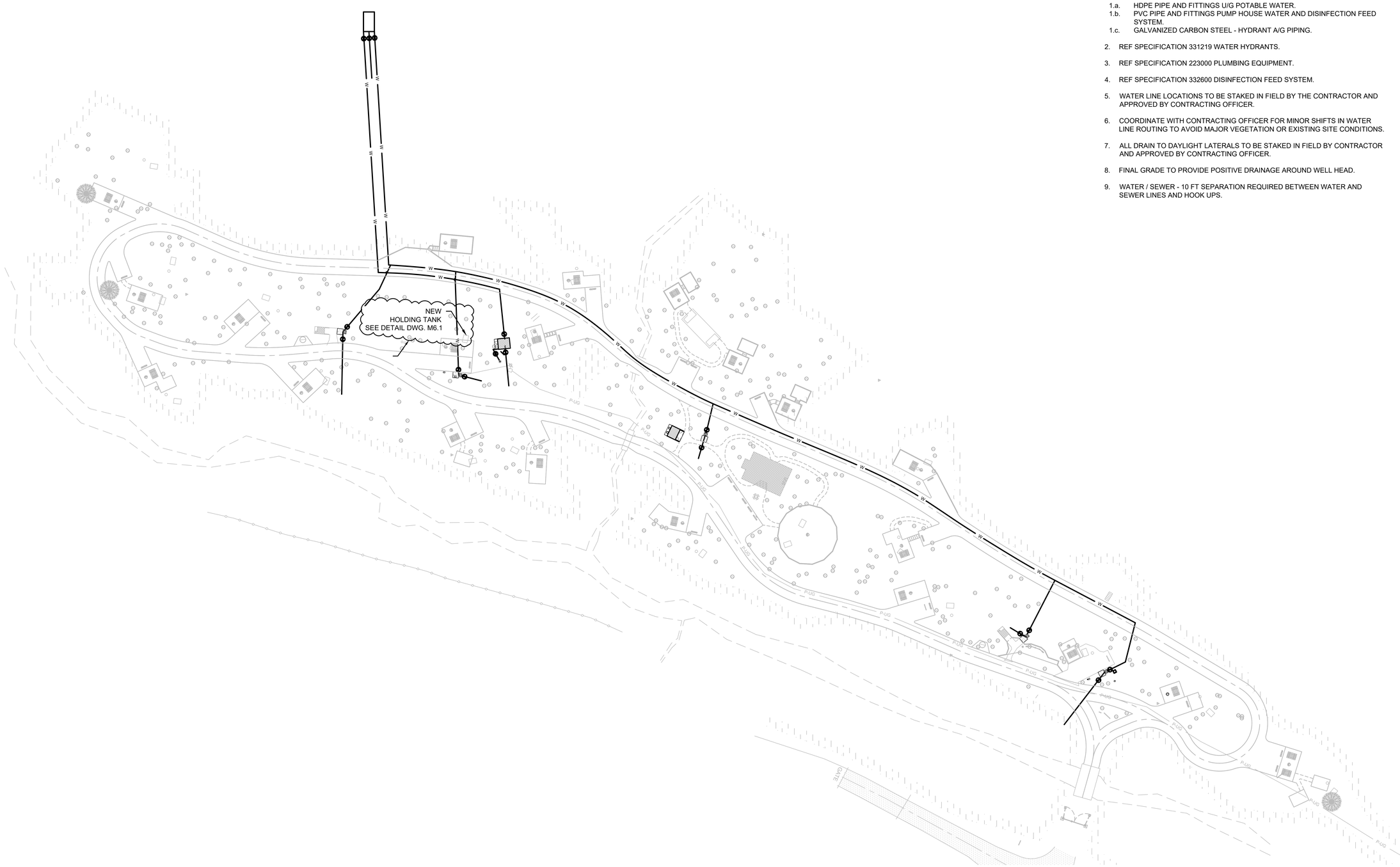
DRAWING TITLE

MECHANICAL DETAILS MISCELLANEOUS UTILITY DETAILS

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
JR	M5.5
CHECKED: SR	
DATE: 05.28.2025	SHEET 38 OF 48

5/27/25 19:11 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\MECH\WORKING CAD FILES\GRIZZLY\M5.5_MISCELLANEOUS UTILITY DETAILS.DWG;



GENERAL NOTES:

1. REF SPECIFICATION 331116 WATER DISTRIBUTION FOR PIPING SYSTEM CONSTRUCTION AND MATERIALS.

   1.a. HDPE PIPE AND FITTINGS U/G POTABLE WATER.
   1.b. PVC PIPE AND FITTINGS PUMP HOUSE WATER AND DISINFECTION FEED SYSTEM.
   1.c. GALVANIZED CARBON STEEL - HYDRANT A/G PIPING.

2. REF SPECIFICATION 331219 WATER HYDRANTS.
3. REF SPECIFICATION 223000 PLUMBING EQUIPMENT.
4. REF SPECIFICATION 332600 DISINFECTION FEED SYSTEM.
5. WATER LINE LOCATIONS TO BE STAKED IN FIELD BY THE CONTRACTOR AND APPROVED BY CONTRACTING OFFICER.
6. COORDINATE WITH CONTRACTING OFFICER FOR MINOR SHIFTS IN WATER LINE ROUTING TO AVOID MAJOR VEGETATION OR EXISTING SITE CONDITIONS.
7. ALL DRAIN TO DAYLIGHT LATERALS TO BE STAKED IN FIELD BY CONTRACTOR AND APPROVED BY CONTRACTING OFFICER.
8. FINAL GRADE TO PROVIDE POSITIVE DRAINAGE AROUND WELL HEAD.
9. WATER / SEWER - 10 FT SEPARATION REQUIRED BETWEEN WATER AND SEWER LINES AND HOOK UPS.

GRIZZLY BEAR MECHANICAL BID OPTIONS	
OPTION #	DESCRIPTION
1	REMOVE AND REPLACE SEWAGE VAULT

OPTIONS LISTED ABOVE SHALL BE COORDINATED WITH ALL OTHER DISCIPLINE SHEETS. REFERENCE CIVIL SHEET C7.0.

N

0' 20' 40' 80'
SCALE IN FEET

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

MECHANICAL/PLUMBING SITE PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
JR	M6.0
CHECKED: SR	
DATE: 05.28.2025	SHEET 39 OF 48

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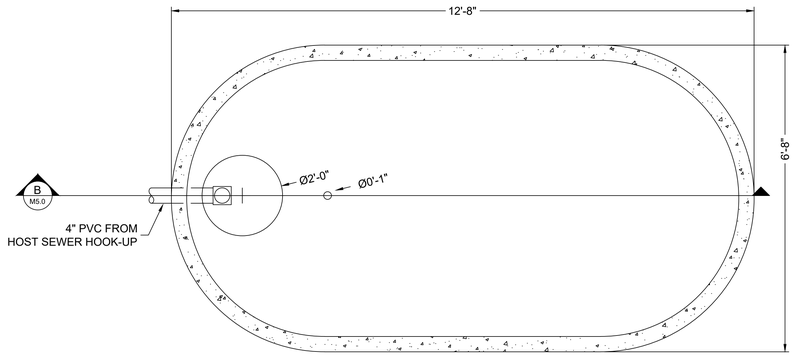
NOTES
1. INDICATORS SHALL BE IN FULL VIEW OF HOST SITE. WITH BOTH VISUAL AND AUDIBLE ALARMS.
   - FOR HIGH WATER LEVEL (VISUAL ALARM)
   - FOR HIGH-HIGH WATER LEVEL (AUDIBLE ALARM).
   - EXISTING MECH FLOAT ASSY SHALL BE ABANDONED IN PLACE.



A M5.0 2000 GALLON PRE CAST CONCRETE SEWAGE VAULT PLAN
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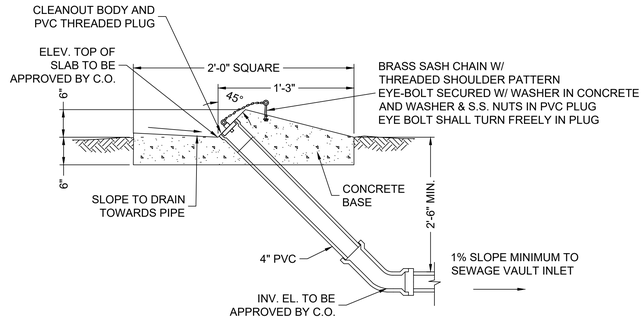
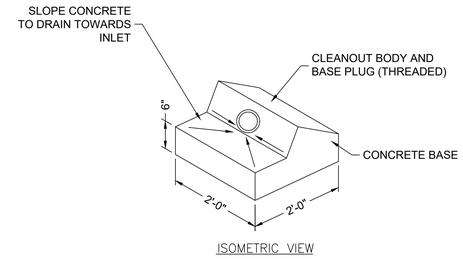
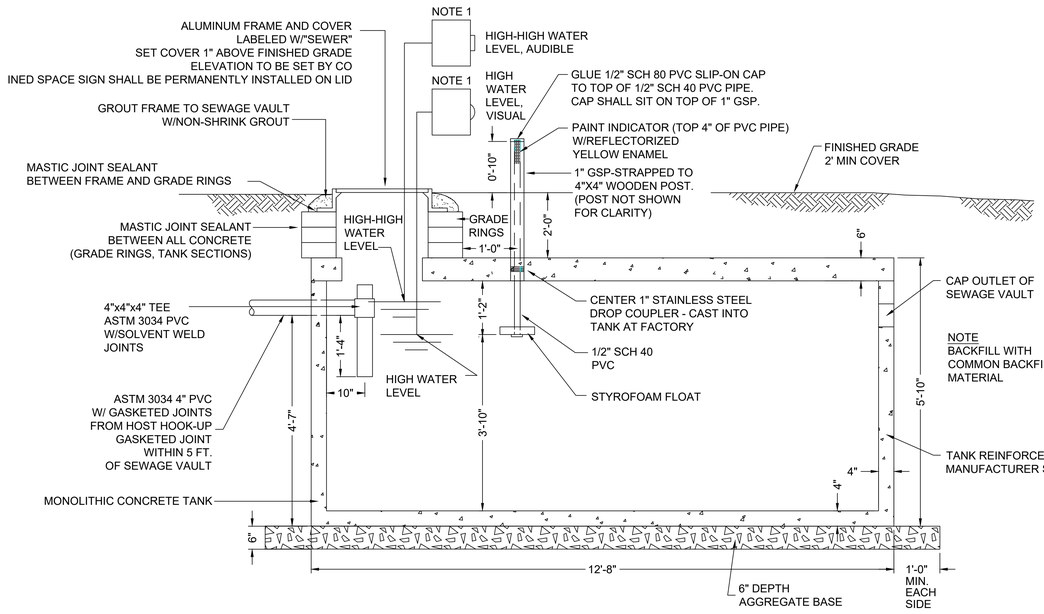




ISOMETRIC VIEW

C M5.0 HOST SEWER HOOK-UP
NO SCALE



B M5.0 2000 GALLON PRE CAST CONCRETE SEWAGE VAULT SECTION
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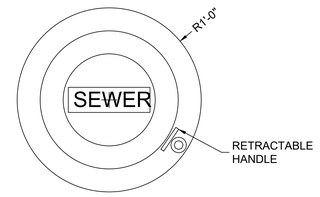

D M5.0 SEWAGE VAULT COVER DETAIL
NO SCALE

OPTION 1

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PROJECT NAME
GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
BLACK HILLS NATIONAL FOREST
HELL CANYON RANGER DISTRICT

DRAWING TITLE
OPTIONAL MECHANICAL DETAILS NEW SEWAGE VAULT

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
JR	M6.1
CHECKED: SR	
DATE: 05.28.2025	SHEET 40 OF 48

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ABBREVIATIONS

A	AMPERE
AC	ANALYSIS CONTROL SWITCH
AFF	ABOVE FINISHED FLOOR
AIC	AMPERE INTERRUPTING CAPACITY
AIT	ANALYSIS INDICATION TEMPERATURE
AIP	ANALYSIS INDICATION PH
AL	ALUMINUM
ATS	AUTOMATIC TRANSFER SWITCH
AWG	AMERICAN WIRE GAUGE
A/C	ABOVE COUNTER
C	CONDUIT
C/W	COMPLETE WITH
CIR	CIRCUIT
CU	COPPER
DIV	DIVISION
EC	ELECTRICAL CONTRACTOR
EGC	EQUIPMENT GROUNDING CONDUCTOR
EMT	ELECTRIC METALLIC TUBING
EA	VOLTAGE ALARM
ES	VOLTAGE SWITCH
ER	EXISTING TO REMAIN
FFC	FLOW RATION CONTROL
FI	FLOW INDICATION SWITCH
FS	FLOW SWITCH
GEC	GROUNDING ELECTRODE CONDUCTOR
GEN	GENERATOR
GND	GROUND
HC	HAND CONTROL
HP	HORSEPOWER
HZ	HERTZ
IECC	INTERNAL ENERGY CONSERVATION CODE
IMC	INTERMEDIATE METALLIC CONDUIT
IN	INCHES
KC	TIME INDICATION SWITCH
KV	KILOVOLT
KVA	KILOVOLT-AMPS
KW	KILOWATT
LA	LEVEL ALARM
LB	POUND
LED	LIGHT EMITTING DIODE
LI	LEVEL INDICATION SWITCH
LIC	LEVEL INDICATION CONTROL
LPS	LIGHTNING PROTECTION SYSTEM
MAX	MAXIMUM
MC	MECHANICAL CONTRACTOR
MCC	MOTOR CONTROL CENTER
MIN	MINIMUM
MFR	MANUFACTURER
MV	MANUAL VALVE
NEC	NATIONAL ELECTRIC CODE
NEMA	NATIONAL ELECTRIC MANUFACTURERS ASSOCIATION
NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
NIC	NOT IN CONTRACT
NUM	NUMBER
NTS	NOT TO SCALE
OEM	ORIGINAL EQUIPMENT MANUFACTURER
OHD	OVERHEAD (ROLLUP) DOOR
OSHA	OCCUPATIONAL SAFETY AND HEALTH ADMINISTATION
P	POLE
PC	PRESSURE CONTROL
PH	PHASE
PI	PRESSURE INDICATION SWITCH
PRI	PRIMARY
PVC	POLYVINYL CHLORIDE
QI	QUALITY INDICATION SWITCH
QTY	QUANTITY
RL	RELOCATE
RLD	RELOCATED
RS	RIGID STEEL
RMC	RIGID METAL CONDUIT
RTU	ROOFTOP UNIT
SC	SPEED CONTROL
SCH	SCHEDULE
SEC	SECONDARY
SPD	SURGE PROTECTIVE DEVICE
SQFT	SQUARE FEET
TBD	TO BE DETERMINED
TI	TEMPERATURE INDICATION
TMP	TEMPORARY
TYP	TYPICAL
TSTAT	THERMOSTAT
UFC	UNITED FACILITIES CRITERIA
UG	UNDERGROUND
UL	UNDERWRITER LABORATORIES
UON	UNLESS OTHERWISE NOTED
UPS	UNINTERRUPTIBLE POWER SUPPLY
UH	UNIT HEATER
V	VOLTS
VAC	VOLTS ALTERNATING CURRENT
VC	VIBRATION CONTROL
VDC	VOLTS DIRECT CURRENT
VS	VIBRATION SWITCH
W	WATTS
W/	WITH
WP	WEATHERPROOF
XFMR	TRANSFORMER
XP	EXPLOSION PROOF

GENERAL NOTES

1. ALL WORK SHALL COMPLY WITH THE NATIONAL ELECTRICAL CODE (NFPA 70), U.S. FOREST SERVICE ENGINEERING MANUAL AND STANDARD SPECIFICATIONS AS WELL AS LOCAL ELECTRICAL UTILITY REQUIREMENTS AND ORDINANCES.
2. OBTAIN ALL NECESSARY PERMITS BEFORE COMMENCING WORK.
3. COORDINATE INSPECTIONS WITH THE AUTHORITY HAVING JURISDICTION (AHJ) AND THE CONTRACTING OFFICER.
4. INSTALL ELECTRICAL PROVISIONS FOR RV PEDESTALS AND CAMPGROUND FACILITIES AS INDICATED ON THE DRAWINGS.
5. ALL NEW ELECTRICAL EQUIPMENT, COMPONENTS, AND WIRING SHALL BE UL-LISTED, AND INSTALLED PER MANUFACTURER RECOMMENDATIONS.
6. PROVIDE WEATHERPROOF NEMA 3R ENCLOSURES AND CONNECTIONS FOR ALL OUTDOOR EQUIPMENT.
7. TEST ALL ELECTRICAL CIRCUITS AND EQUIPMENT FOR PROPER OPERATION AND GROUNDING BEFORE FINAL ACCEPTANCE. SUBMIT TEST RESULTS TO THE U.S. FOREST SERVICE FOR REVIEW.
8. COORDINATE ALL UTILITY CONNECTIONS AND SERVICE UPGRADES WITH THE LOCAL UTILITY PROVIDER.
9. VERIFY EXACT LOCATIONS OF EXISTING UTILITIES IN THE FIELD PRIOR TO EXCAVATION.
10. PRIOR TO EXCAVATION, VERIFY THE LOCATION OF ALL EXISTING UNDERGROUND UTILITIES, INCLUDING WATER, SEWER, AND GAS LINES, THROUGH THE APPROPRIATE UTILITY LOCATING SERVICE.
11. ELECTRICAL CONTRACTOR SHALL BORE UNDER ANY EXISTING PAVED SURFACES.
12. WHERE UNDERGROUND ELECTRICAL LINES INTERSECT WITH SIGNIFICANT TREE ROOTS, USE TRENCHLESS METHODS (E.G., BORING) OR HAND-DIGGING TO PRESERVE TREE HEALTH. REPORT ANY UNAVOIDABLE ROOT DAMAGE TO THE CONTRACTING OFFICER.
13. ELECTRICAL CONTRACTOR IS RESPONSIBLE FOR RECALCULATING VOLTAGE-DROP FOR ALTERNATE CIRCUIT ROUTES AND SUBMITTING REVISED CALCULATIONS FOR APPROVAL IF ADDITIONAL CONDUCTOR LENGTHS ARE REQUIRED.
14. RESTORE ALL DISTURBED AREAS, INCLUDING VEGETATION AND PAVING, TO THEIR ORIGINAL CONDITION UPON COMPLETION OF WORK. EXCAVATIONS SHALL BE BACKFILLED WITH CLEAN, COMPACTED SOIL FREE OF DEBRIS AND SHARP OBJECTS AND ROCKS.

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PROJECT NAME

GRIZZLY
CAMPGROUND
60% DESIGN
DEVELOPMENT

BLACK HILLS
NATIONAL FOREST

HELL CANYON RANGER DISTRICT

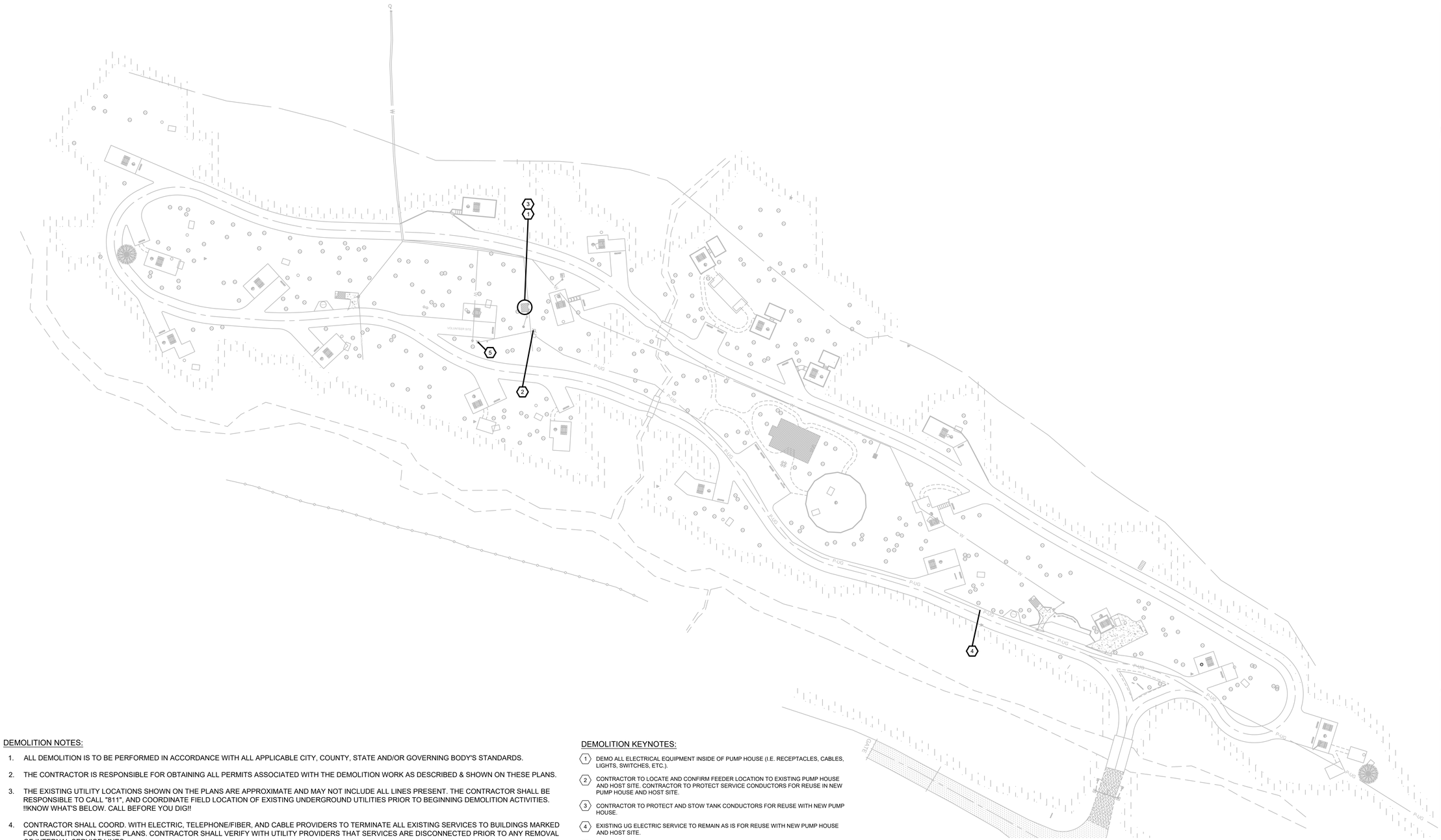
DRAWING TITLE

GENERAL NOTES,
ABBREVIATIONS

ARCHIVE NO.

DESIGNER DGK	SUB SHEET NO. E0.1
CHECKED DGK	
DATE 05.28.2025	SHEET 41 OF 48

5/27/25 19:18 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\ELECTRICAL\WORKING CAD FILES\GRIZZLY\E0.1_GEN_NOTES.DWG;



DEMOLITION NOTES:

1. ALL DEMOLITION IS TO BE PERFORMED IN ACCORDANCE WITH ALL APPLICABLE CITY, COUNTY, STATE AND/OR GOVERNING BODY'S STANDARDS.
2. THE CONTRACTOR IS RESPONSIBLE FOR OBTAINING ALL PERMITS ASSOCIATED WITH THE DEMOLITION WORK AS DESCRIBED & SHOWN ON THESE PLANS.
3. THE EXISTING UTILITY LOCATIONS SHOWN ON THE PLANS ARE APPROXIMATE AND MAY NOT INCLUDE ALL LINES PRESENT. THE CONTRACTOR SHALL BE RESPONSIBLE TO CALL "811", AND COORDINATE FIELD LOCATION OF EXISTING UNDERGROUND UTILITIES PRIOR TO BEGINNING DEMOLITION ACTIVITIES. !!KNOW WHAT'S BELOW. CALL BEFORE YOU DIG!!
4. CONTRACTOR SHALL COORD. WITH ELECTRIC, TELEPHONE/FIBER, AND CABLE PROVIDERS TO TERMINATE ALL EXISTING SERVICES TO BUILDINGS MARKED FOR DEMOLITION ON THESE PLANS. CONTRACTOR SHALL VERIFY WITH UTILITY PROVIDERS THAT SERVICES ARE DISCONNECTED PRIOR TO ANY REMOVAL OF INTERNAL SERVICE LINES.
5. THE DEMOLITION PLAN AND EXISTING CONDITIONS SHOWN ARE APPROXIMATE. CONTRACTOR SHALL INSPECT AND FIELD VERIFY THE EXISTING SITE CONDITIONS TO BECOME ACQUAINTED WITH ALL EXISTING CONDITIONS WHICH EFFECT ANY AND ALL WORK ASSOCIATED WITH THE DEMOLITION PLANS.
6. ALL EXISTING BUILDING RELATED FEATURES NOT SHOWN ON THIS PLAN SHALL BE COORD. WITH ALL OTHER PLANS ASSOCIATED WITH THIS PROJECT. ALL CONCRETE OR ASPHALT PAVEMENT, CURB & GUTTER, RETAINING WALLS AND MISCELLANEOUS STRUCTURES INCLUDING BUT NOT LIMITED TO WATER HYDRANTS, FIRE RINGS, PICNIC TABLES, SITE NUMBERING POSTS, WHEELSTOPS, KIOSKS, INFO BOARDS, AND MISCELLANEOUS DEBRIS SHALL BE DEMOLISHED AND REMOVED FROM THE SITE AND SHALL BE DISPOSED OF OFF-SITE AT CONTRACTORS EXPENSE TO AN APPROVED LANDFILL, UNLESS OTHERWISE NOTED.
7. IN THE EVENT THE CONTRACTOR ENCOUNTERS ON THE SITE MATERIAL REASONABLY BELIEVED TO BE ASBESTOS, LEAD-BASED PAINT, OR ANY OTHER HAZARDOUS MATERIAL, CONTRACTOR SHALL IMMEDIATELY NOTIFY THE CONTRACTING OFFICER
8. DEBRIS SHALL NOT BE BURIED ON SITE. ALL UNSUITABLE MATERIAL AND DEBRIS SHALL BE REMOVED FROM THE SITE AND PROPERLY DISPOSED OFF-SITE IN ACCORDANCE WITH ALL CITY, STATE, COUNTY & FEDERAL LAWS AND ORDINANCES. NO ON-SITE BURNING WILL BE PERMITTED.
9. IF EXISTING UN-CAPPED WATER WELLS ARE DISCOVERED ON-SITE, CONTRACTOR SHALL IMMEDIATELY NOTIFY THE CONTRACTING OFFICER.
10. CONTRACTOR SHALL COORDINATE WITH FOREST SERVICE TO PUMP SEPTIC VAULTS OF ALL WASTEWATER AND SLUDGE. CONTRACTOR SHALL REMOVE EXISTING VAULTS AND ANY ASSOCIATED APPURTENANCES IN ACCORDANCE WITH CITY, COUNTY, STATE, AND FEDERAL LAWS AND ORDINANCES.
11. ANY AREA DISTURBED BY DEMOLITION THAT WILL NOT BE DISTURBED IN PROPOSED GRADING ACTIVITIES SHOWN ON THESE PLANS SHALL HAVE TOPSOIL PLACED, SEEDED AND MULCHED ACCORDING TO THE CONTRACT SPECIFICATIONS.
12. OWNER HAS FIRST RIGHT OF REFUSAL OF ALL MATERIALS REMOVED FROM THE SITE.

DEMOLITION KEYNOTES:

1. DEMO ALL ELECTRICAL EQUIPMENT INSIDE OF PUMP HOUSE (I.E. RECEPTACLES, CABLES, LIGHTS, SWITCHES, ETC.).
2. CONTRACTOR TO LOCATE AND CONFIRM FEEDER LOCATION TO EXISTING PUMP HOUSE AND HOST SITE. CONTRACTOR TO PROTECT SERVICE CONDUCTORS FOR REUSE IN NEW PUMP HOUSE AND HOST SITE.
3. CONTRACTOR TO PROTECT AND STOW TANK CONDUCTORS FOR REUSE WITH NEW PUMP HOUSE.
4. EXISTING UG ELECTRIC SERVICE TO REMAIN AS IS FOR REUSE WITH NEW PUMP HOUSE AND HOST SITE.
5. CONTRACTOR TO PROTECT AND REUSE CAMPSITE PEDESTAL WITH EXISTING ELECTRIC SERVICE.

DEMOLITION LEGEND

1. ABANDON IN PLACE.

N

0' 20' 40' 80'
SCALE IN FEET

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PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

ELECTRICAL SITE PLAN DEMOLITION PLAN

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
DGK	ED1.1
CHECKED: DGK	
DATE: 05.28.2025	SHEET 42 OF 48

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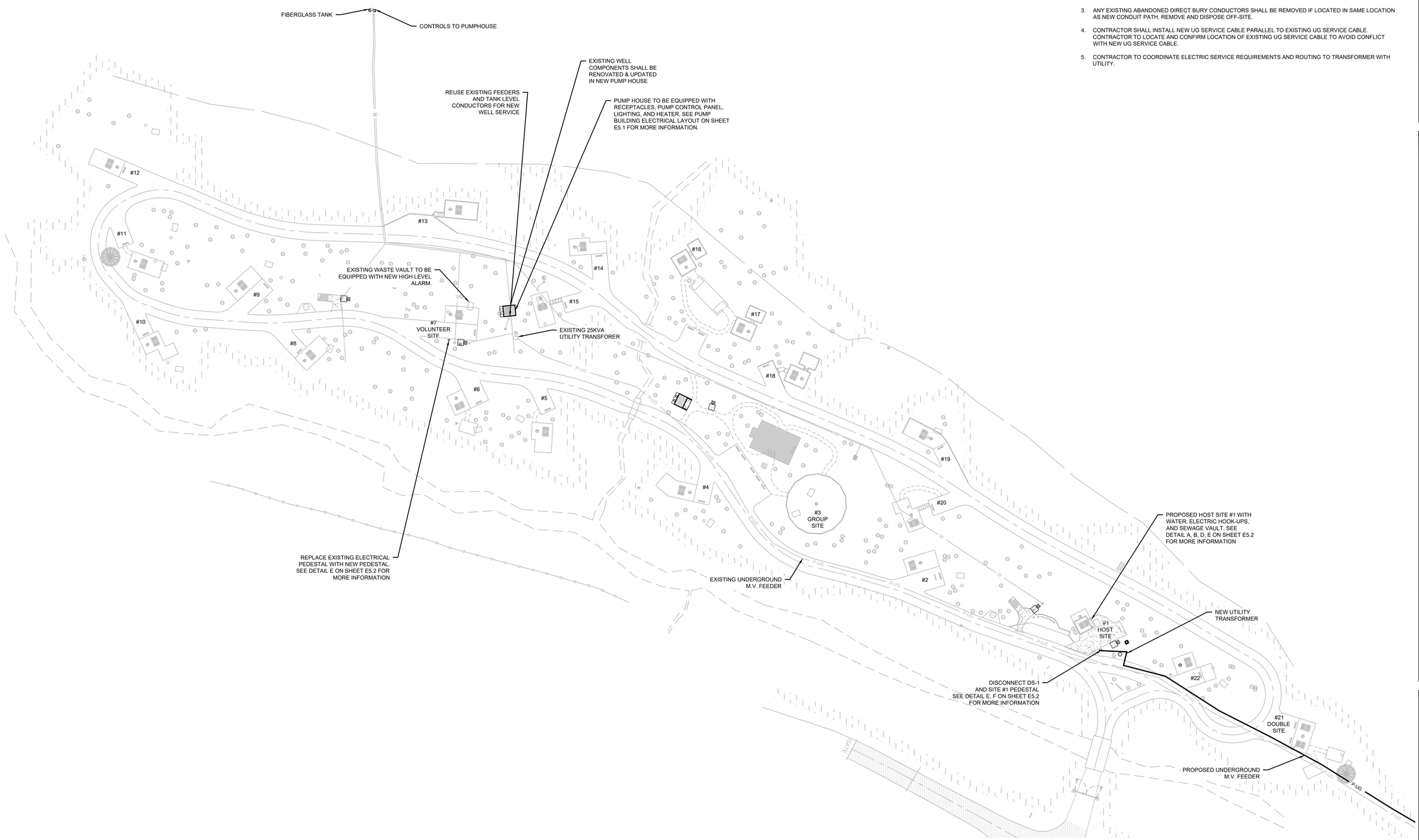
GENERAL NOTES:

1. CONTRACTOR SHALL BORE UNDER ALL GRAVEL ROADS.
2. CONTRACTOR SHALL LOCATE AND PROTECT ALL EXISTING UTILITIES DURING TRENCHING/BORING ACTIVITIES.
3. ANY EXISTING ABANDONED DIRECT BURY CONDUCTORS SHALL BE REMOVED IF LOCATED IN SAME LOCATION AS NEW CONDUIT PATH. REMOVE AND DISPOSE OFF-SITE.
4. CONTRACTOR SHALL INSTALL NEW UG SERVICE CABLE PARALLEL TO EXISTING UG SERVICE CABLE. CONTRACTOR TO LOCATE AND CONFIRM LOCATION OF EXISTING UG SERVICE CABLE TO AVOID CONFLICT WITH NEW UG SERVICE CABLE.
5. CONTRACTOR TO COORDINATE ELECTRIC SERVICE REQUIREMENTS AND ROUTING TO TRANSFORMER WITH UTILITY.



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BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

ELECTRICAL SITE PLAN

ARCHIVE NO.

DESIGNER: DGK	SUB SHEET NO. E1.1
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DATE: 05.28.2025	SHEET 43 OF 48

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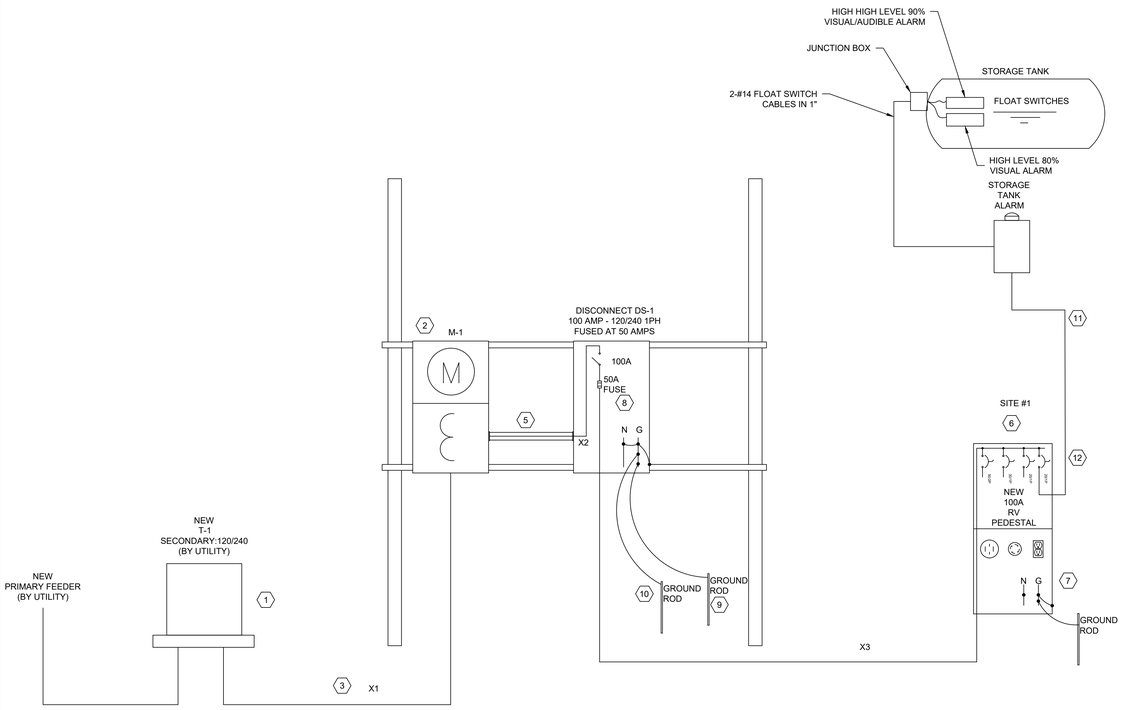

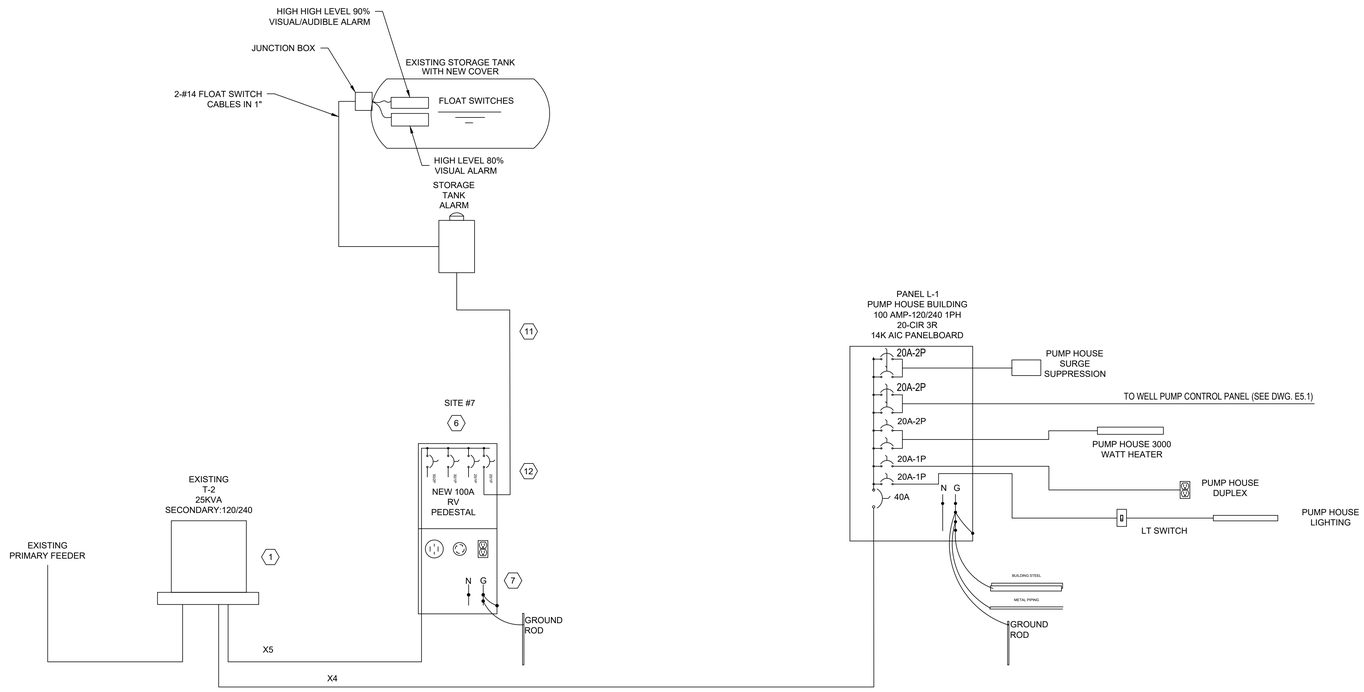

KEYED NOTES:

1. PAD MOUNTED TRANSFORMER BY UTILITY COMPANY
2. METER BY UTILITY COMPANY
3. SECONDARY CONDUCTORS, DIRECT BURIAL CABLE, ALUMINUM, 3-#2/0 AWG WITH #1 GROUND FROM TRANSFORMER TO METER M-1
4. DIRECT BURIAL CABLE, ALUMINUM, 3-#1 WITH #1 GROUND
5. ALUMINUM 3-#2/0 WITH #1 GROUND IN 1-1/2" CONDUIT FROM METER M-1 TO DISCONNECT DS-1. MOUNT DISCONNECT DS-1 ON METER STUB PROVIDED BY UTILITY COMPANY
6. RV POWER PEDESTAL, TYP.
7. GROUND ROD WITH #6 COPPER CONDUCTOR CONNECTED BACK TO GROUND LUG OF RV POWER PEDESTAL, WHERE ROCK IS ENCOUNTERED BURY ROD IN A TRENCH THAT IS AT LEAST 30 INCHES DEEP, TYP
8. BOND GROUND BUS AND NEUTRAL BUS TO PANEL
9. GROUNDING ELECTRODE CONDUCTOR TO GROUND ROD, SIZE IN ACCORDANCE WITH NEC
10. SUPPLEMENTAL GROUND ROD, LOCATE AT LEAST 6 FEET FROM GROUND ROD, SIZE IN ACCORDANCE WITH NEC
11. DIRECT BURIAL CABLE, COPPER, 2-#12 WITH #12 GROUND TO SEWAGE TANK ALARM
12. PURCHASE EXTRA BREAKER FOR SUMP TANK ALARM, ALARM PANEL, ADD CONDUIT AND WIRING TO TANK FLOATS
13. UPDATE WELL COMPONENTS TO INCLUDE LIGHTING RECEPTACLES AND PUMP CONTROLS IN PUMP HOUSE

GENERAL NOTES:

1. PROVIDE FULL AMPACITY SERVICE RATED BREAKERS FOR CIRCUIT BREAKERS PROTECTING RV POWER PEDESTAL.
2. FOR PANEL L1 PANEL SCHEDULE SEE SHEET E6.0.
3. FOR GRIZZLY SERVICE SHORT CIRCUIT SCHEDULE SEE SHEET E6.1.

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NO.	REVISION / ISSUE	DATE
6		
5		
4		
3		
2		
1		

PROJECT NAME

GRIZZLY BEAR
CAMPGROUND
RECONSTRUCTION

BLACK HILLS
NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

ELECTRICAL DETAILS
ONE-LINE DIAGRAM

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
DGK	E5.0
CHECKED: DGK	
DATE: 05.28.2025	SHEET 44 OF 48

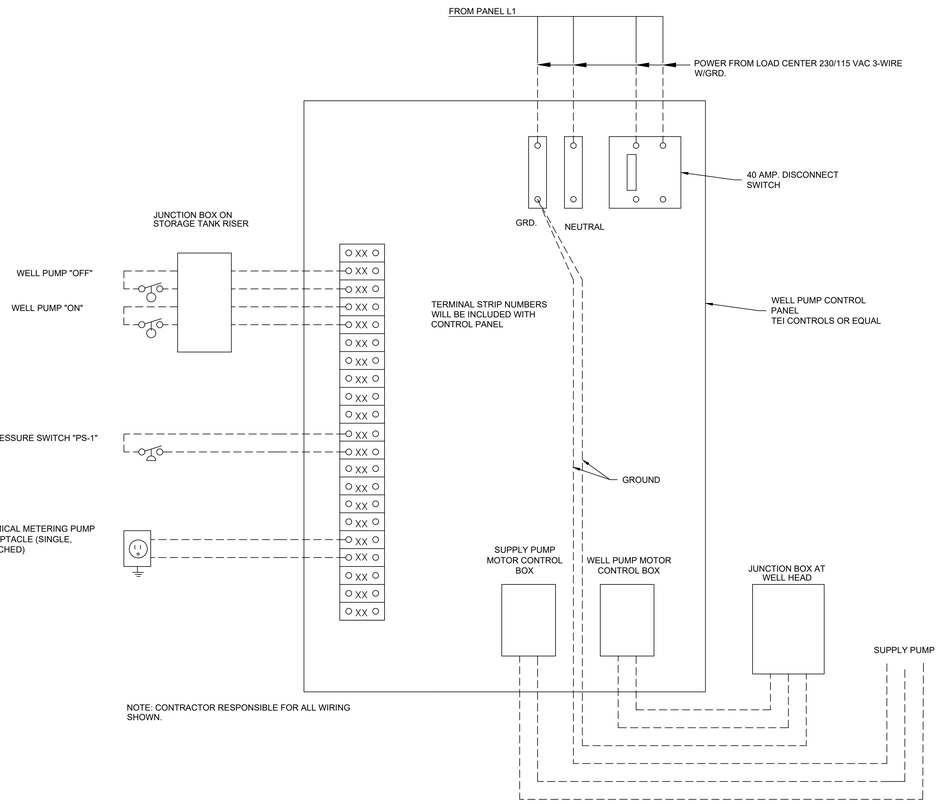
5/27/25 19:17 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\ELECTRICAL\WORKING CAD FILES\GRIZZLY\E5.0_ONE-LINE.DWG;

GENERAL NOTES:

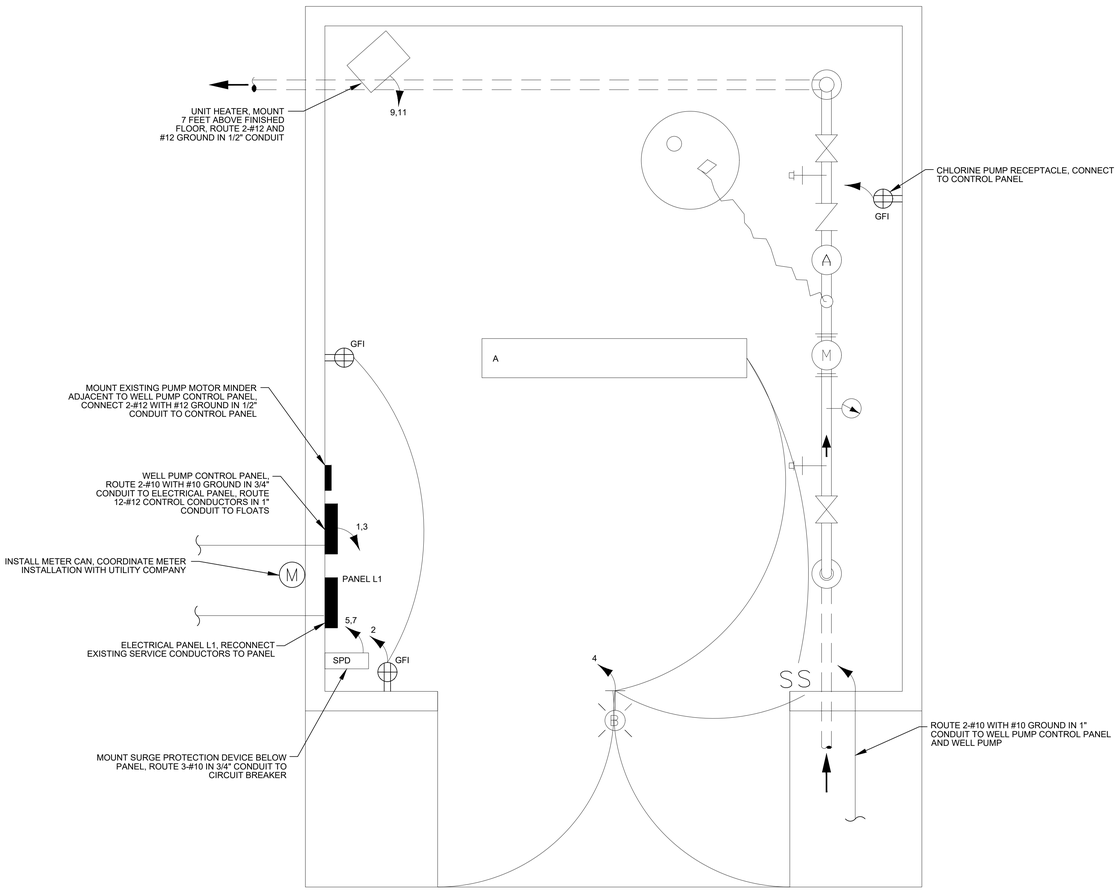
1. COORDINATE ELECTRICAL INSTALLATION AND ROUTING WITH EQUIPMENT AND PIPING IN PUMP BUILDING.
2. ROUTE POWER THROUGH OPENINGS IN CONCRETE FLOOR WITH 1" HDPE PIPE, FILL FLOOR OPENING ANNULAR SPACE WITH HIGH DENSITY POLYURETHANE FOAM AFTER INSTALLING CONDUIT. SEE M5.1 FOR FLOOR OPENINGS.
3. SURFACE MOUNT CONDUITS IN PUMP BUILDING.

LEGEND (NOT ALL SYMBOLS SHOWN IN LEGEND ARE USED ON PROJECT)

- OR — ELECTRICAL PANEL
- TVSS — TRANSIENT VOLTAGE SURGE SUPPRESSOR
- GFI — DUPLEX, GROUND FAULT CIRCUIT RECEPTACLE
- 120 VOLT DUPLEX RECEPTACLE
- S — 1-WAY SWITCH
- A2 — CIRCUIT, MINIMUM SIZE #12 AWG, # INDICATES PANEL AND CIRCUIT BREAKER
- J — JUNCTION BOX
- B / A — CEILING MOUNTED FIXTURE, LETTER INDICATES FIXTURE
- E / H / B — WALL MOUNTED FIXTURE, LETTER INDICATES FIXTURE
- X — CEILING MOUNTED FIXTURE, LETTER INDICATES FIXTURE



A / E5.1 GRIZZLY WELL PUMP CONTROL WIRING DETAIL



PUMP BUILDING ELECTRICAL LAYOUT
NOT TO SCALE

LUMINAIRE SCHEDULE

PLAN CODE	TYPE	MANUFACTURER	MODEL	LAMPS/ MODULES	LED COLOR TEMP	LUMENS	WATTS	NOMINAL SIZE W X D X L	VOLTS	REMARKS
A	ROUGH SERVICE	LITHONIA	VSL 2 32 MVOLT GEB10IS DL	32W T8	-	-	64	9 X 3 X 49	120	UL LISTED FOR DAMP LOCATIONS
B	WALL PACK	LITHONIA	TWR1 LED 1 40K MVOLT PE	1	4000K	2161	33	13 X 7.5 X 9	120	PROVIDE WITH PHOTOCELL

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5		
4		
3		
2		
1		

PROJECT NAME
GRIZZLY BEAR CAMPGROUND RECONSTRUCTION
BLACK HILLS NATIONAL FOREST
HELL CANYON RANGER DISTRICT

DRAWING TITLE
ELECTRICAL DETAILS PUMP HOUSE ELECTRICAL LAYOUT

ARCHIVE NO.

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DGK	E5.1
CHECKED: DGK	
DATE: 05.28.2025	SHEET 45 OF 48

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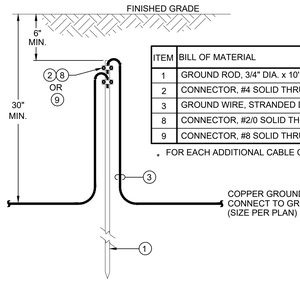
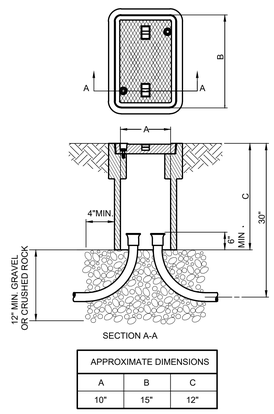


ITEM	BILL OF MATERIAL
1	GROUND ROD, 3/4" DIA. x 10' LONG, COPPER CLAD STEEL
2	CONNECTOR, #4 SOLID THRU #2/0 STR. CABLE TO GND. ROD, BURNDY #GAR-6426
3	GROUND WIRE, STRANDED DIRECT-BURRIED COPPER
8	CONNECTOR, #2/0 SOLID THRU 250KCMIL STR. CABLE TO GND. ROD, BURNDY #GAR-6429
9	CONNECTOR, #8 SOLID THRU #4 STR. CABLE TO GND. ROD, BURNDY #GAR-644C

. FOR EACH ADDITIONAL CABLE CONNECTION ADD 1 CONNECTOR.

A / E5.2 — GROUND ROD INSTALLATION

NOT TO SCALE



SECTION A-A

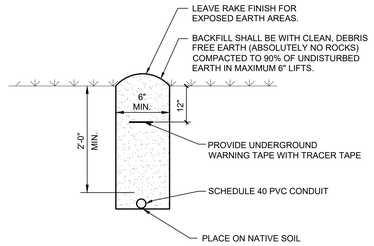
APPROXIMATE DIMENSIONS		
A	B	C
10"	15"	12"

B / E5.2 — GRADE PULL BOX DETAIL

NOT TO SCALE

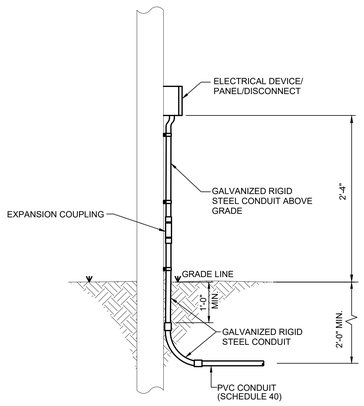
NOTES

1. ACCESS DUCTS, BENDS AND/OR SWEEPS, ARE SUBSIDIARY TO THE ITEM "PULL BOX, TYPE PB".
2. FILL MATERIAL SHALL BE GRAVEL OR CRUSHED ROCK COMPOSED OF CLEAN, HARD, DURABLE AND UNCOATED
3. PROLONGED EXPOSURE OF THE PULL BOX AND COVER TO SUNLIGHT, COMMON ICE MELTING CHEMICALS OR FERTILIZERS SHALL NOT CAUSE SIGNIFICANT DAMAGE TO THE PULL BOX OR COVER, OR IMPAIR ITS FUNCTION.
4. CONDUIT TERMINATIONS IN PULL BOX SHALL BE FITTED WITH BELL OR FLARED END.
5. CONCRETE SHALL BE CLASS 47B-3000.
6. PULL BOXES ARE IN GROUND, GRADE LEVEL ENCLOSURES INTENDED TO BE INSTALLED WHERE NON-DELIBERATE VEHICULAR TRAFFIC IS ANTICIPATED.
7. PULL BOX ILLUSTRATIONS ARE TYPICAL AND ARE NOT INTENDED TO LIMIT DESIGN. BOXES OF SQUARE OR CIRCULAR PLAN-SECTION AND OF EQUAL OR GREATER VOLUME ARE ACCEPTABLE. A MINIMUM DEPTH OF 18 INCHES, MUST BE MAINTAINED.
8. THE PULL BOX ENCLOSURES (BOX AND COVER) SHALL BE NON-METALLIC AND GREEN OR GRAY IN COLOR.
9. PULL BOX COVERS SHALL BE OF HEAVY-DUTY CONSTRUCTION TO WITHSTAND THE STRUCTURAL REQUIREMENTS OF THE SPECIFICATIONS. COVERS SHALL FIT SUFFICIENTLY TIGHT TO PREVENT THE ENTRANCE OF RUN-OFF WATER. COVERS SHALL BE EMBOSSED WITH AN ANTI-SLIP SURFACE. THE MANUFACTURER'S NAME AND THE LOGO "ELECTRIC". COVERS SHALL BE HELD SECURELY IN PLACE BY TWO OR MORE 3/8" (MIN. DIA.) STAINLESS STEEL HEX HEAD BOLTS WITH FLAT WASHER. BOLTS SHALL BE HELD CAPTIVE TO THE COVER. ALL COVERS SHALL BE EQUIPPED WITH A RECESSED "LIFT PIN" AND HAVE A MINIMUM 0.50 COEFFICIENT OF FRICTION SKID RESISTANT SURFACE.
10. EACH PULL BOX ASSEMBLY (BOX WITH HEAVY-DUTY COVER SECURELY ATTACHED) SHALL CONFORM TO ALL TEST PROVISIONS AND REQUIREMENTS FOR TIER 15 APPLICATIONS AS OUTLINED IN THE LATEST ISSUE OF THE ANSI/SCTE 77 2002 STANDARD TITLED "SPECIFICATION FOR UNDER GROUND ENCLOSURE INTEGRITY". INDEPENDENT THIRD-PARTY TEST REPORTS TOGETHER WITH CERTIFICATION THAT ALL TEST PROVISIONS OF THE ANSI/SCTE 77 2002 STANDARD HAVE BEEN MET WILL BE REQUIRED FOR EACH TYPE OF PULL BOX BEING FURNISHED. ALL TESTS SHALL BE DATED AND SIGNED BY A REGISTERED PROFESSIONAL ENGINEER AND SHALL BEAR THE LETTERHEAD OF THE TESTING AGENCY.



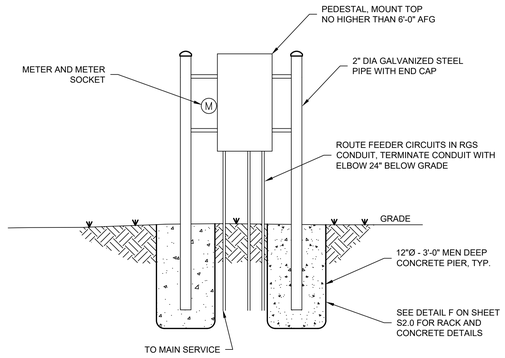
C / E5.2 — TRENCH DETAIL

NOT TO SCALE



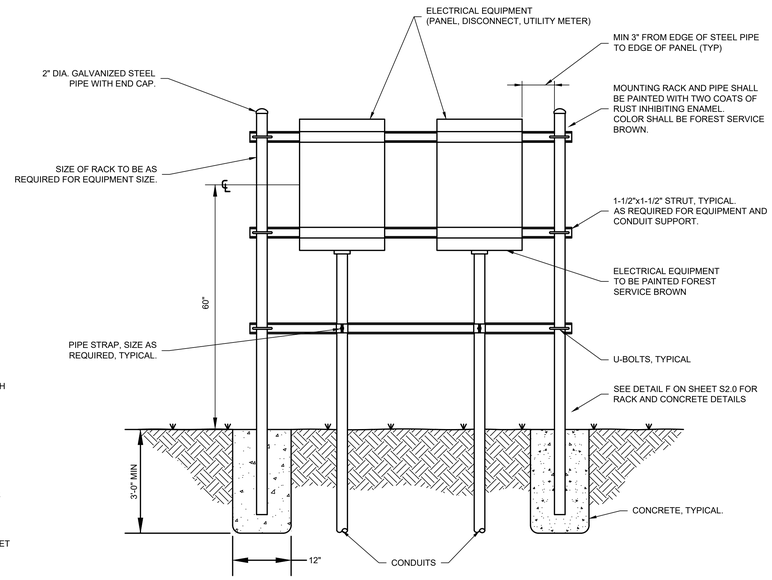
D / E5.2 — EXPOSED CONDUIT DETAIL

NOT TO SCALE



E / E5.2 — PEDESTAL MOUNTING DETAIL

NOT TO SCALE



F / E5.2 — EQUIPMENT RACK MOUNT DETAIL

NOT TO SCALE

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6		
5		
4		
3		
2		
1		

PROJECT NAME

GRIZZLY BEAR CAMPGROUND RECONSTRUCTION

BLACK HILLS NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

ELECTRICAL DETAILS

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
DGK	E5.2
CHECKED: DGK	
DATE: 05.28.2025	SHEET 46 OF 48

FED FROM:		UTILITY	PANEL SCHEDULE		L1		PANEL LOCATION:		PUMP BUILDING		
		MOUNTING METHOD: SURFACE			VOLTS: 240/120			BUS RATING: 100 Amp			
		ENCLOSURE: NEMA 3R			PHASE: 1			MAIN BREAKER: 40			
		FEED FROM: BOTTOM			WIRE: 3			AIC RATING: 14,000			
			LOAD	Brkr	PHASE		Brkr	LOAD			
#	Code	DESCRIPTION	VA	Size	A	B	Size	VA	DESCRIPTION	Code	#
1	6	WELL PUMP CONTROL PANEL	600	20/2	960		20/1	360	RECEPTACLES	2	2
3	6	WELL PUMP CONTROL PANEL	600			697	20/1	97	LIGHTING	1	4
5	6	SURGE PROTECTION	0	20/2	0						6
7	6	SURGE PROTECTION	0			0					8
9	4	ELECTRIC HEAT	1500	20/2	1500						10
11	4	ELECTRIC HEAT	1500			1500					12
13					0						14
15						0					16
17					0						18
19						0					20

CODE	DESCRIPTION	Load VA	DEM	TOTAL					
1	LIGHTING	97	1.25	121		TOTAL PHASE A:	**2,460**	VA	
2	REC. TO 10,000 VA	360	1	360		TOTAL PHASE B:	**2,197**	VA	
2.1	REC. OVER 10,000 VA	0	0.5	0					
3	MOTOR	0	1	0					
3.1	LARGEST MOTOR	0	1.25	0		1 ϕ CONNECTED:	**4,657**	VA	
4	HEATER	3,000	1.25	3,750		1 ϕ CODE DEMAND:	**5,431**	VA	
5	KITCHEN	0	1	0		Per ϕ Code Demand:	**22.6**	AMPS	
6	MISCELLANEOUS	1,200	1	1,200					
7	SUB PANEL	0	1	0					

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PROJECT NAME

GRIZZLY BEAR
CAMPGROUND
RECONSTRUCTION

BLACK HILLS
NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

ELECTRICAL SCHEDULES
PANEL SCHEDULE

ARCHIVE NO.

DESIGNER	SUB SHEET NO.
DGK	E6.0
CHECKED: DGK	
DATE: 05.28.2025	SHEET 47 OF 48

5/28/25 17:00 BRIAN C:\USERS\BRIAN\DESKTOP\024-0146 USDA FS BLACK HILLS CAMPGROUND REDESIGN (JV)\ELECTRICAL\WORKING CAD FILES\GRIZZLY\E6.0_PANEL_SCHEDULE.DWG;

GRIZZLY BEAR SERVICE TO SITE #1 PEDESTAL						
SHORT CIRCUIT SCHEDULE						
POINT	DESCRIPTION	WIRE SIZE	LENGTH OF RUN (FT)	RATED SHORT CIRCUIT BRACING	SHORT CIRCUIT (Isc SYM RMS)	LINE-LINE VOLTAGE
T1	UTILITY TRANSFORMER 50KVA 1.4% Z				14,800	240
X1	T-1 TO UTILITY METER M-1	2/0-ALU	30		11,361	240
X2	UTILITY METER TO DISCONNECT DS-1	2/0-ALU	6	65,000	10,711	240
X3	DISCONNECT DS-1 TO SITE #1 PEDESTAL	#1/0-ALU	15	10,000	9,005	240

GRIZZLY BEAR SERVICE TO PANEL L-1 & SITE #7 PEDESTAL						
SHORT CIRCUIT SCHEDULE						
POINT	DESCRIPTION	WIRE SIZE	LENGTH OF RUN (FT)	RATED SHORT CIRCUIT BRACING	SHORT CIRCUIT (Isc SYM RMS)	LINE-LINE VOLTAGE
T2	UTILITY TRANSFORMER 25KVA 1.4% Z				14,400	240
X4	T-2 TO PANEL L-1	2/0-ALU	30	14,000	11,142	240
X5	T-2 TO SITE #7 PEDESTAL	2/0-ALU	70	10,000	7,744	240

GENERAL NOTES:

1. FOR SHORT CIRCUIT POINT LOCATIONS SEE SHEET E5.0 ONE-LINE DIAGRAM.

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CAMPGROUND
RECONSTRUCTION

BLACK HILLS
NATIONAL FOREST

HELL CANYON RANGER DISTRICT

DRAWING TITLE

ELECTRICAL SCHEDULES
SHORT CIRCUIT

ARCHIVE NO.

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